

$\langle N_{part} \rangle$ and $\langle N_{coll} \rangle$ systematics study for centralities in d+Au and O+O Run 21 data

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September 9, 2025

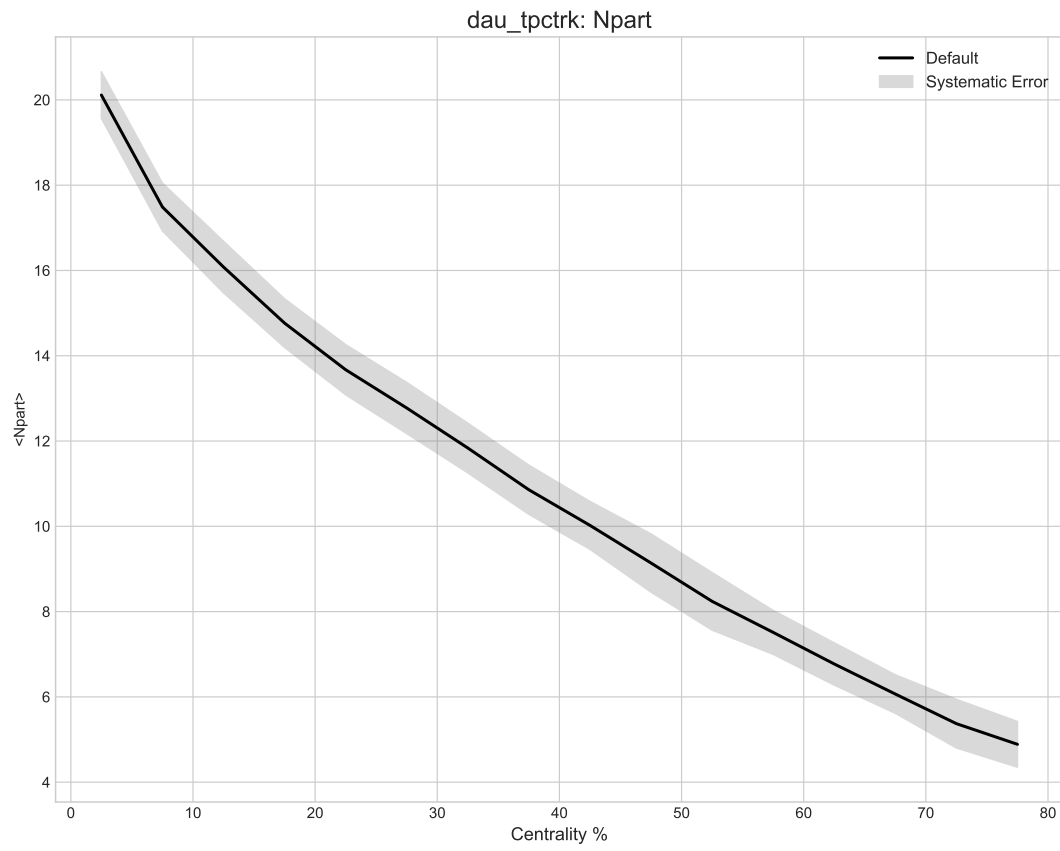
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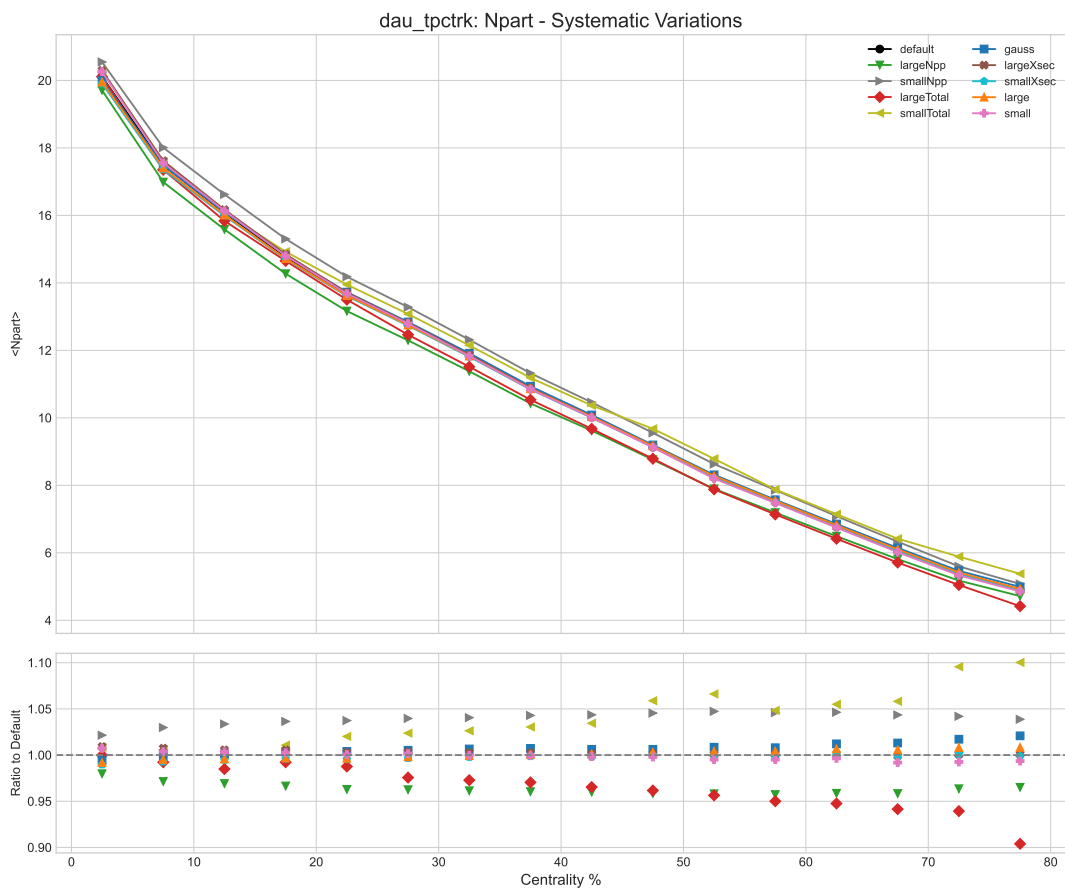
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d+Au TPC - Npart

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

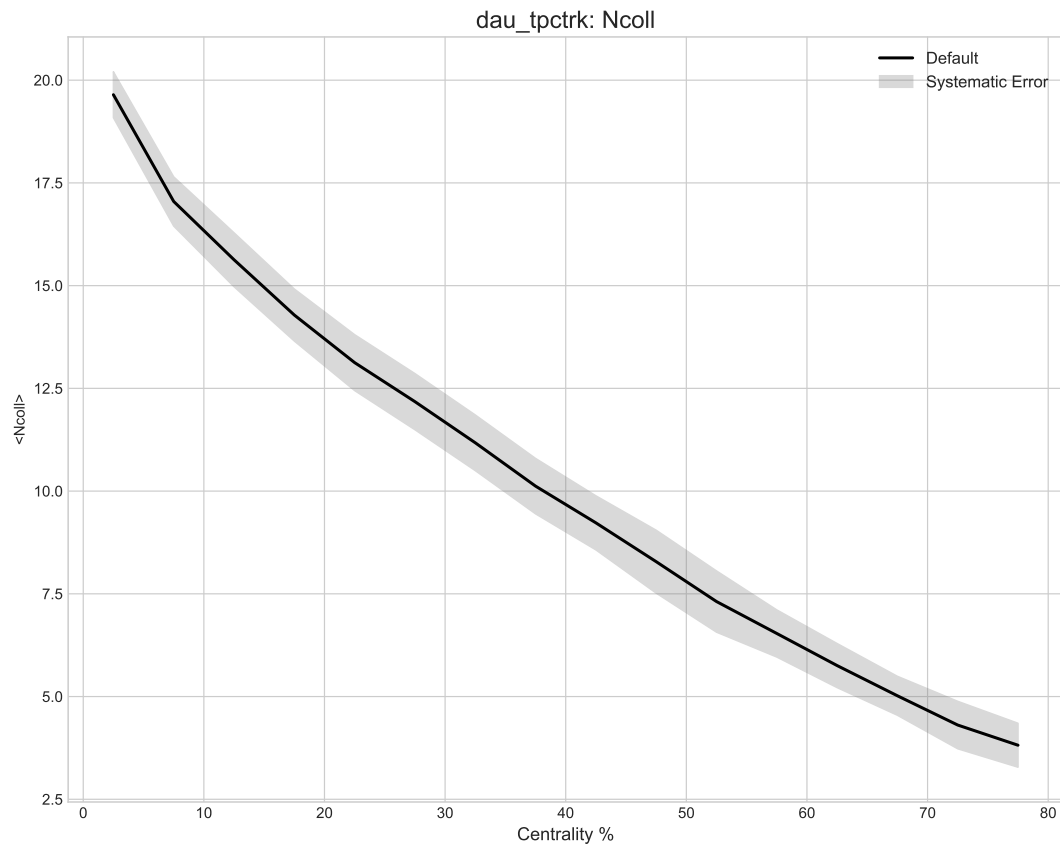
min_c	max_c	value	stat_err	syst_err	tot_err	gauss	large	largeNpp	largeTotal	largeXsec	small	smallNpp	smallTotal	smallXsec
0.0	5.0	20.1141	0.0074	0.5467	0.5467	19.9959	19.9536	19.7082	20.1100	20.2985	20.2597	20.5468	20.2814	19.9045
5.0	10.0	17.4876	0.0068	0.5605	0.5605	17.4789	17.4058	16.9871	17.3523	17.6176	17.5541	18.0093	17.5934	17.3591
10.0	15.0	16.0804	0.0066	0.6043	0.6044	16.1034	16.0132	15.5841	15.8371	16.1708	16.1362	16.6215	16.0895	15.9921
15.0	20.0	14.7651	0.0057	0.5705	0.5705	14.8145	14.7189	14.2710	14.6504	14.8472	14.8033	15.3026	14.9236	14.7031
20.0	25.0	13.6722	0.0066	0.5888	0.5888	13.7266	13.6250	13.1648	13.5028	13.7106	13.6808	14.1832	13.9486	13.6097
25.0	30.0	12.7745	0.0062	0.6004	0.6004	12.8406	12.7509	12.2959	12.4630	12.8176	12.7962	13.2811	13.0775	12.7307
30.0	35.0	11.8366	0.0058	0.5834	0.5834	11.9144	11.8315	11.3797	11.5149	11.8686	11.8323	12.3159	12.1480	11.8101
35.0	40.0	10.8568	0.0054	0.5769	0.5769	10.9351	10.8648	10.4267	10.5360	10.8816	10.8494	11.3227	11.1869	10.8432
40.0	45.0	10.0224	0.0063	0.5620	0.5621	10.0849	10.0340	9.6211	9.6752	10.0341	10.0056	10.4592	10.3673	10.0035
45.0	50.0	9.1393	0.0048	0.6830	0.6830	9.1960	9.1678	8.7589	8.7887	9.1505	9.1202	9.5550	9.6773	9.1416
50.0	55.0	8.2408	0.0055	0.6753	0.6753	8.3108	8.2762	7.8954	7.8814	8.2469	8.2005	8.6308	8.7861	8.2329
55.0	60.0	7.5127	0.0051	0.5154	0.5154	7.5725	7.5457	7.1916	7.1375	7.5156	7.4754	7.8589	7.8759	7.5079
60.0	65.0	6.7718	0.0047	0.4962	0.4962	6.8541	6.8181	6.4915	6.4167	6.7791	6.7478	7.0858	7.1443	6.7702
65.0	70.0	6.0667	0.0042	0.4524	0.4524	6.1459	6.1017	5.8139	5.7116	6.0614	6.0173	6.3308	6.4194	6.0582
70.0	75.0	5.3725	0.0037	0.5704	0.5704	5.4646	5.4165	5.1766	5.0464	5.3758	5.3323	5.5979	5.8864	5.3789
75.0	80.0	4.8878	0.0048	0.5370	0.5370	4.9897	4.9287	4.7171	4.4181	4.8911	4.8560	5.0771	5.3782	4.8831

Percentage difference from default											
min_c	max_c	tot_sys_err(%)	gauss(%)	large(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	small(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)
0.0	5.0	2.72	-0.59	-0.80	-2.02	-0.02	0.92	0.72	2.15	0.83	-1.04
5.0	10.0	3.20	-0.05	-0.47	-2.86	-0.77	0.74	0.38	2.98	0.60	-0.74
10.0	15.0	3.76	0.14	-0.42	-3.09	-1.51	0.56	0.35	3.37	0.06	-0.55
15.0	20.0	3.86	0.33	-0.31	-3.35	-0.78	0.56	0.26	3.64	1.07	-0.42
20.0	25.0	4.31	0.40	-0.34	-3.71	-1.24	0.28	0.06	3.74	2.02	-0.46
25.0	30.0	4.70	0.52	-0.19	-3.75	-2.44	0.34	0.17	3.97	2.37	-0.34
30.0	35.0	4.93	0.66	-0.04	-3.86	-2.72	0.27	-0.04	4.05	2.63	-0.22
35.0	40.0	5.31	0.72	0.07	-3.96	-2.95	0.23	-0.07	4.29	3.04	-0.13
40.0	45.0	5.61	0.62	0.12	-4.00	-3.46	0.12	-0.17	4.36	3.44	-0.19
45.0	50.0	7.47	0.62	0.31	-4.16	-3.84	0.12	-0.21	4.55	5.89	0.03
50.0	55.0	8.19	0.85	0.43	-4.19	-4.36	0.07	-0.49	4.73	6.62	-0.10
55.0	60.0	6.86	0.80	0.44	-4.28	-5.00	0.04	-0.50	4.61	4.83	-0.06
60.0	65.0	7.33	1.22	0.68	-4.14	-5.24	0.11	-0.36	4.64	5.50	-0.02
65.0	70.0	7.46	1.31	0.58	-4.17	-5.85	-0.09	-0.82	4.35	5.81	-0.14
70.0	75.0	10.62	1.72	0.82	-3.64	-6.07	0.06	-0.75	4.20	9.57	0.12
75.0	80.0	10.99	2.09	0.84	-3.49	-9.61	0.07	-0.65	3.87	10.03	-0.10

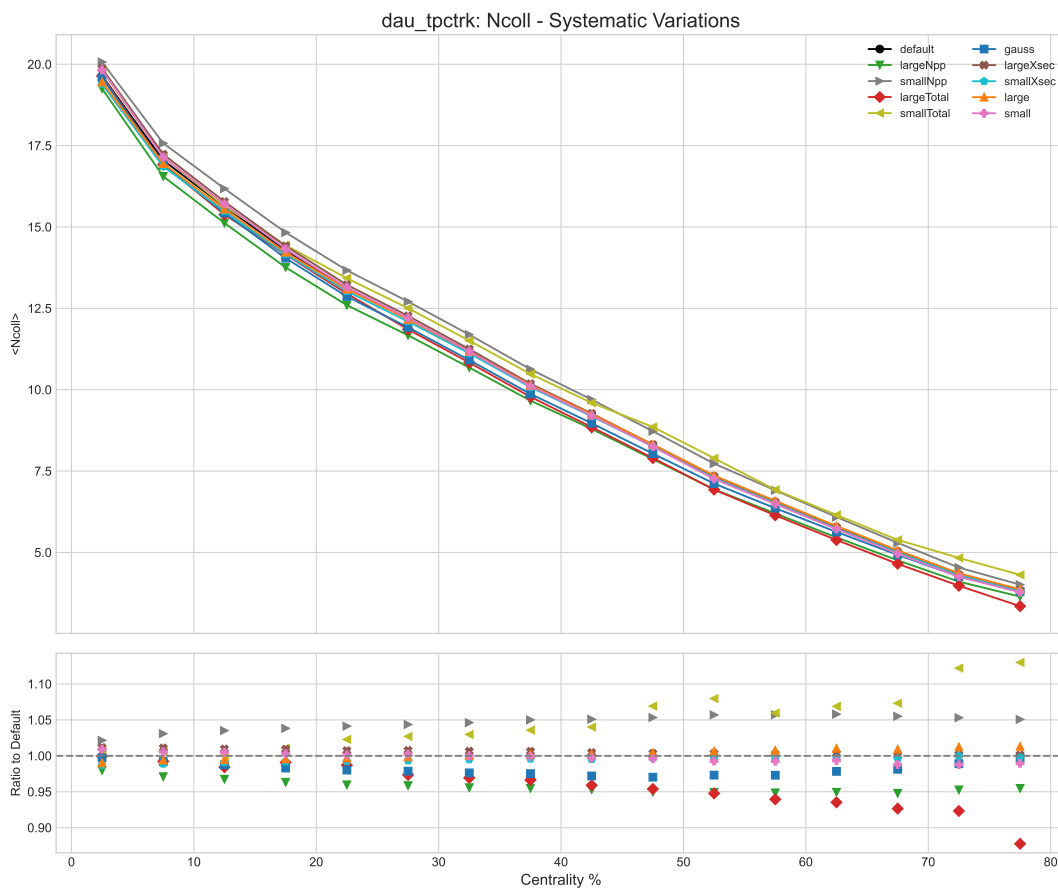
Integer Cent_Interval	Multiplicity Default	Centrality Large_Tot	Cuts Used Small_Tot
0-5%	54	54	55
5-10%	47	46	47
10-15%	42	41	42
15-20%	37	37	38
20-25%	34	33	35
25-30%	31	30	32
30-35%	28	27	29
35-40%	25	24	26
40-45%	23	22	24
45-50%	20	19	22
50-55%	18	17	19
55-60%	16	15	17
60-65%	14	13	15
65-70%	12	11	13
70-75%	10	9	12
75-80%	9	7	10

d+Au TPC - Ncoll

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

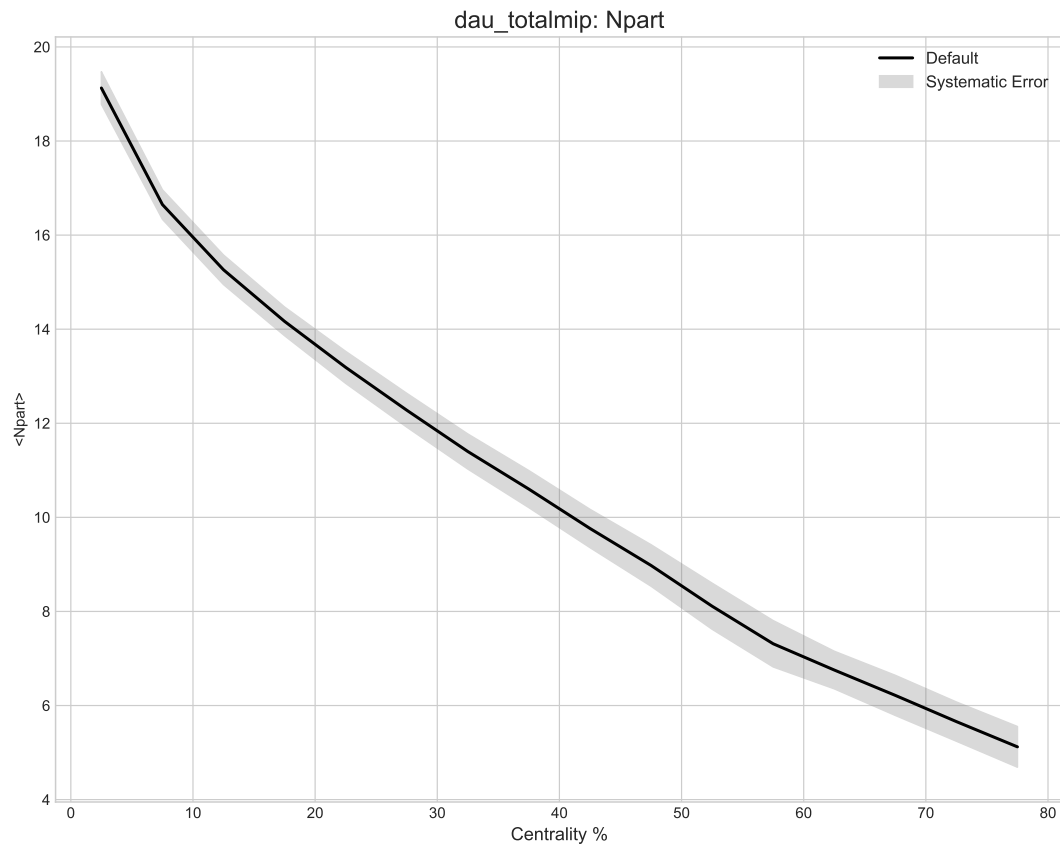
min_c	max_c	value	stat_err	syst_err	tot_err	gauss	large	largeNpp	largeTotal	largeXsec	small	smallNpp	smallTotal	smallXsec
0.0	5.0	19.6465	0.0082	0.5593	0.5593	19.5901	19.4557	19.2507	19.6385	19.8752	19.8256	20.0706	19.8123	19.3899
5.0	10.0	17.0484	0.0078	0.5995	0.5995	16.9120	16.9473	16.5539	16.9166	17.2352	17.1472	17.5744	17.1591	16.8573
10.0	15.0	15.6303	0.0076	0.6565	0.6566	15.4324	15.5509	15.1215	15.3828	15.7798	15.7085	16.1798	15.6354	15.4854
15.0	20.0	14.2845	0.0066	0.6333	0.6334	14.0416	14.2244	13.7597	14.1525	14.4156	14.3271	14.8306	14.4365	14.1657
20.0	25.0	13.1265	0.0077	0.6804	0.6805	12.8659	13.0892	12.5980	12.9535	13.2225	13.1421	13.6683	13.4264	13.0252
25.0	30.0	12.1765	0.0072	0.6836	0.6837	11.9163	12.1616	11.6720	11.8531	12.2683	12.2044	12.7078	12.5054	12.0957
30.0	35.0	11.1760	0.0068	0.6776	0.6776	10.9116	11.1791	10.6809	10.8338	11.2486	11.1748	11.6926	11.5079	11.1185
35.0	40.0	10.1208	0.0063	0.6746	0.6747	9.8705	10.1421	9.6636	9.7809	10.1840	10.1132	10.6275	10.4831	10.0770
40.0	45.0	9.2277	0.0072	0.6582	0.6582	8.9696	9.2485	8.7948	8.8497	9.2695	9.2029	9.6979	9.5969	9.1829
45.0	50.0	8.2825	0.0055	0.7658	0.7659	8.0365	8.3170	7.8662	7.9002	8.3114	8.2487	8.7243	8.8558	8.2602
50.0	55.0	7.3125	0.0062	0.7456	0.7456	7.1161	7.3618	6.9392	6.9303	7.3377	7.2597	7.7294	7.8957	7.2883
55.0	60.0	6.5350	0.0057	0.5739	0.5739	6.3589	6.5851	6.1982	6.1398	6.5506	6.4860	6.9082	6.9241	6.5183
60.0	65.0	5.7541	0.0051	0.5367	0.5368	5.6299	5.8149	5.4624	5.3819	5.7728	5.7174	6.0879	6.1506	5.7362
65.0	70.0	5.0187	0.0046	0.4742	0.4743	4.9248	5.0651	4.7569	4.6503	5.0202	4.9576	5.2948	5.3857	4.9995
70.0	75.0	4.3067	0.0040	0.5778	0.5778	4.2599	4.3609	4.1027	3.9762	4.3132	4.2536	4.5353	4.8325	4.3075
75.0	80.0	3.8154	0.0049	0.5350	0.5350	3.8033	3.8655	3.6427	3.3482	3.8201	3.7753	4.0090	4.3114	3.8077

Percentage difference from default											
min_c	max_c	tot_sys_err(%)	gauss(%)	large(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	small(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)
0.0	5.0	2.85	-0.29	-0.97	-2.01	-0.04	1.16	0.91	2.16	0.84	-1.31
5.0	10.0	3.52	-0.80	-0.59	-2.90	-0.77	1.10	0.58	3.08	0.65	-1.12
10.0	15.0	4.20	-1.27	-0.51	-3.26	-1.58	0.96	0.50	3.52	0.03	-0.93
15.0	20.0	4.43	-1.70	-0.42	-3.67	-0.92	0.92	0.30	3.82	1.06	-0.83
20.0	25.0	5.18	-1.99	-0.28	-4.03	-1.32	0.73	0.12	4.13	2.28	-0.77
25.0	30.0	5.61	-2.14	-0.12	-4.14	-2.66	0.75	0.23	4.36	2.70	-0.66
30.0	35.0	6.06	-2.37	0.03	-4.43	-3.06	0.65	-0.01	4.62	2.97	-0.51
35.0	40.0	6.67	-2.47	0.21	-4.52	-3.36	0.62	-0.07	5.01	3.58	-0.43
40.0	45.0	7.13	-2.80	0.23	-4.69	-4.10	0.45	-0.27	5.10	4.00	-0.49
45.0	50.0	9.25	-2.97	0.42	-5.03	-4.61	0.35	-0.41	5.34	6.92	-0.27
50.0	55.0	10.20	-2.69	0.68	-5.11	-5.23	0.35	-0.72	5.70	7.98	-0.33
55.0	60.0	8.78	-2.70	0.77	-5.15	-6.05	0.24	-0.75	5.71	5.95	-0.26
60.0	65.0	9.33	-2.16	1.06	-5.07	-6.47	0.32	-0.64	5.80	6.89	-0.31
65.0	70.0	9.45	-1.87	0.92	-5.22	-7.34	0.03	-1.22	5.50	7.31	-0.38
70.0	75.0	13.42	-1.09	1.26	-4.74	-7.67	0.15	-1.23	5.31	12.21	0.02
75.0	80.0	14.02	-0.32	1.31	-4.53	-12.24	0.12	-1.05	5.07	13.00	-0.20

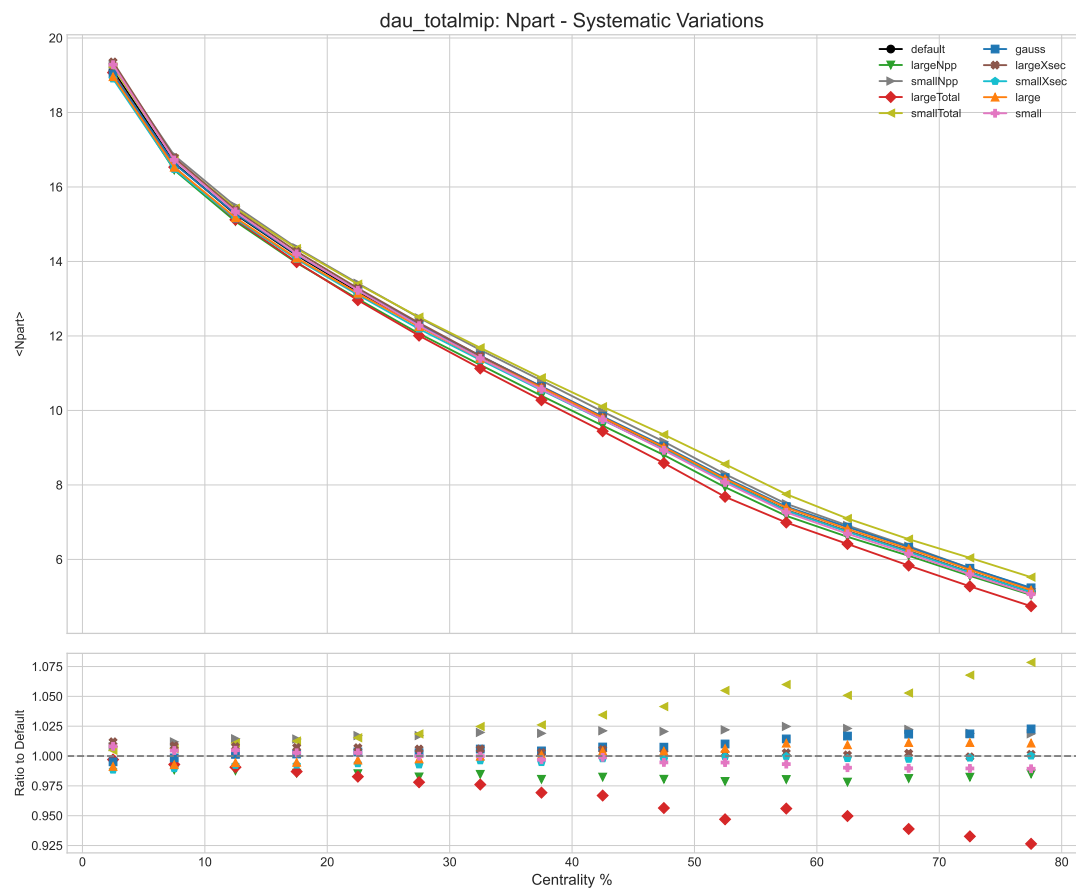
Integer Multiplicity Centrality Cuts Used			
Cent_Interval	Default	Large_Tot	Small_Tot
0-5%	54	54	55
5-10%	47	46	47
10-15%	42	41	42
15-20%	37	37	38
20-25%	34	33	35
25-30%	31	30	32
30-35%	28	27	29
35-40%	25	24	26
40-45%	23	22	24
45-50%	20	19	22
50-55%	18	17	19
55-60%	16	15	17
60-65%	14	13	15
65-70%	12	11	13
70-75%	10	9	12
75-80%	9	7	10

d+Au TotalMIP - Npart

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

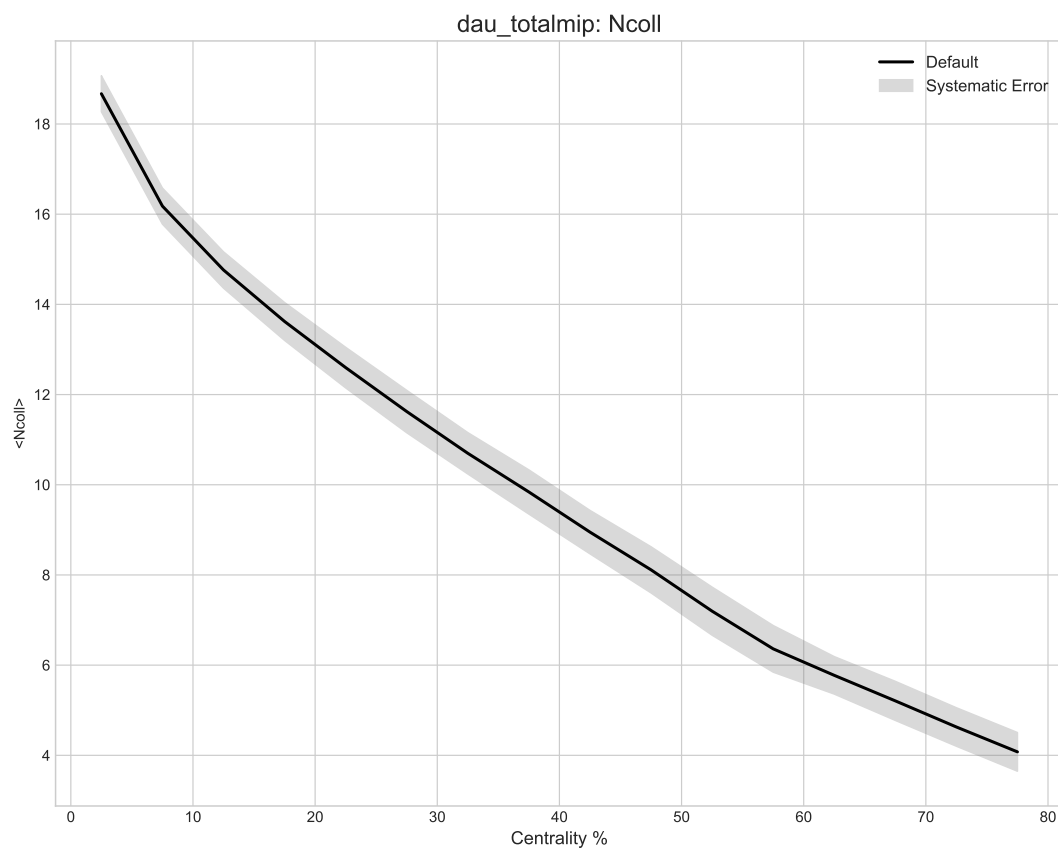
min_c	max_c	value	stat_err	syst_err	tot_err	gauss	large	largeNpp	largeTotal	largeXsec	small	smallNpp	smallTotal	smallXsec
0.0	5.0	19.1275	0.0078	0.3457	0.3458	19.0346	18.9608	18.9966	19.0709	19.3561	19.2829	19.2773	19.2189	18.9049
5.0	10.0	16.6488	0.0075	0.3136	0.3136	16.6229	16.5276	16.4556	16.5312	16.7905	16.7255	16.8467	16.7535	16.4758
10.0	15.0	15.2627	0.0075	0.3139	0.3139	15.2838	15.1783	15.0754	15.1193	15.3824	15.3366	15.4842	15.4286	15.1464
15.0	20.0	14.1657	0.0075	0.3044	0.3045	14.1899	14.0904	13.9625	13.9812	14.2676	14.2044	14.3680	14.3438	14.0602
20.0	25.0	13.1888	0.0075	0.3399	0.3400	13.2341	13.1447	12.9938	12.9607	13.2809	13.2244	13.4147	13.3907	13.1059
25.0	30.0	12.2781	0.0074	0.3590	0.3591	12.3125	12.2484	12.0633	12.0085	12.3497	12.2763	12.4882	12.5055	12.1886
30.0	35.0	11.3998	0.0075	0.3730	0.3731	11.4667	11.3978	11.2252	11.1273	11.4664	11.3987	11.6250	11.6819	11.3561
35.0	40.0	10.6002	0.0073	0.3935	0.3936	10.6457	10.6099	10.3935	10.2748	10.6316	10.5677	10.8016	10.8768	10.5439
40.0	45.0	9.7644	0.0069	0.4070	0.4071	9.8383	9.8056	9.5920	9.4411	9.8184	9.7538	9.9708	10.1005	9.7437
45.0	50.0	8.9778	0.0070	0.4410	0.4410	9.0446	9.0145	8.8034	8.5866	8.9976	8.9300	9.1626	9.3500	8.9552
50.0	55.0	8.1116	0.0066	0.4898	0.4898	8.1935	8.1629	7.9391	7.6815	8.1290	8.0674	8.2898	8.5570	8.1079
55.0	60.0	7.3131	0.0065	0.4925	0.4925	7.4179	7.3917	7.1694	6.9911	7.3326	7.2635	7.4943	7.7514	7.3109
60.0	65.0	6.7549	0.0064	0.3995	0.3996	6.8680	6.8185	6.6075	6.4151	6.7610	6.6888	6.9110	7.0983	6.7418
65.0	70.0	6.2156	0.0056	0.4263	0.4264	6.3297	6.2859	6.0987	5.8357	6.2297	6.1517	6.3543	6.5437	6.1998
70.0	75.0	5.6582	0.0051	0.4170	0.4170	5.7633	5.7222	5.5580	5.2773	5.6543	5.5994	5.7662	6.0417	5.6498
75.0	80.0	5.1213	0.0048	0.4327	0.4327	5.2377	5.1767	5.0448	4.7445	5.1287	5.0668	5.2167	5.5231	5.1222

Percentage difference from default											
min_c	max_c	tot_sys_err(%)	gauss(%)	large(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	small(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)
0.0	5.0	1.81	-0.49	-0.87	-0.68	-0.30	1.20	0.81	0.78	0.48	-1.16
5.0	10.0	1.88	-0.16	-0.73	-1.16	-0.71	0.85	0.46	1.19	0.63	-1.04
10.0	15.0	2.06	0.14	-0.55	-1.23	-0.94	0.78	0.48	1.45	1.09	-0.76
15.0	20.0	2.15	0.17	-0.53	-1.43	-1.30	0.72	0.27	1.43	1.26	-0.74
20.0	25.0	2.58	0.34	-0.33	-1.48	-1.73	0.70	0.27	1.71	1.53	-0.63
25.0	30.0	2.92	0.28	-0.24	-1.75	-2.20	0.58	-0.01	1.71	1.85	-0.73
30.0	35.0	3.27	0.59	-0.02	-1.53	-2.39	0.58	-0.01	1.98	2.47	-0.38
35.0	40.0	3.71	0.43	0.09	-1.95	-3.07	0.30	-0.31	1.90	2.61	-0.53
40.0	45.0	4.17	0.76	0.42	-1.77	-3.31	0.55	-0.11	2.11	3.44	-0.21
45.0	50.0	4.91	0.74	0.41	-1.94	-4.36	0.22	-0.53	2.06	4.14	-0.25
50.0	55.0	6.04	1.01	0.63	-2.13	-5.30	0.21	-0.54	2.20	5.49	-0.04
55.0	60.0	6.73	1.43	1.08	-1.96	-4.40	0.27	-0.68	2.48	5.99	-0.03
60.0	65.0	5.91	1.67	0.94	-2.18	-5.03	0.09	-0.98	2.31	5.08	-0.19
65.0	70.0	6.86	1.84	1.13	-1.88	-6.11	0.23	-1.03	2.23	5.28	-0.25
70.0	75.0	7.37	1.86	1.13	-1.77	-6.73	-0.07	-1.04	1.91	6.78	-0.15
75.0	80.0	8.45	2.27	1.08	-1.49	-7.36	0.14	-1.06	1.86	7.85	0.02

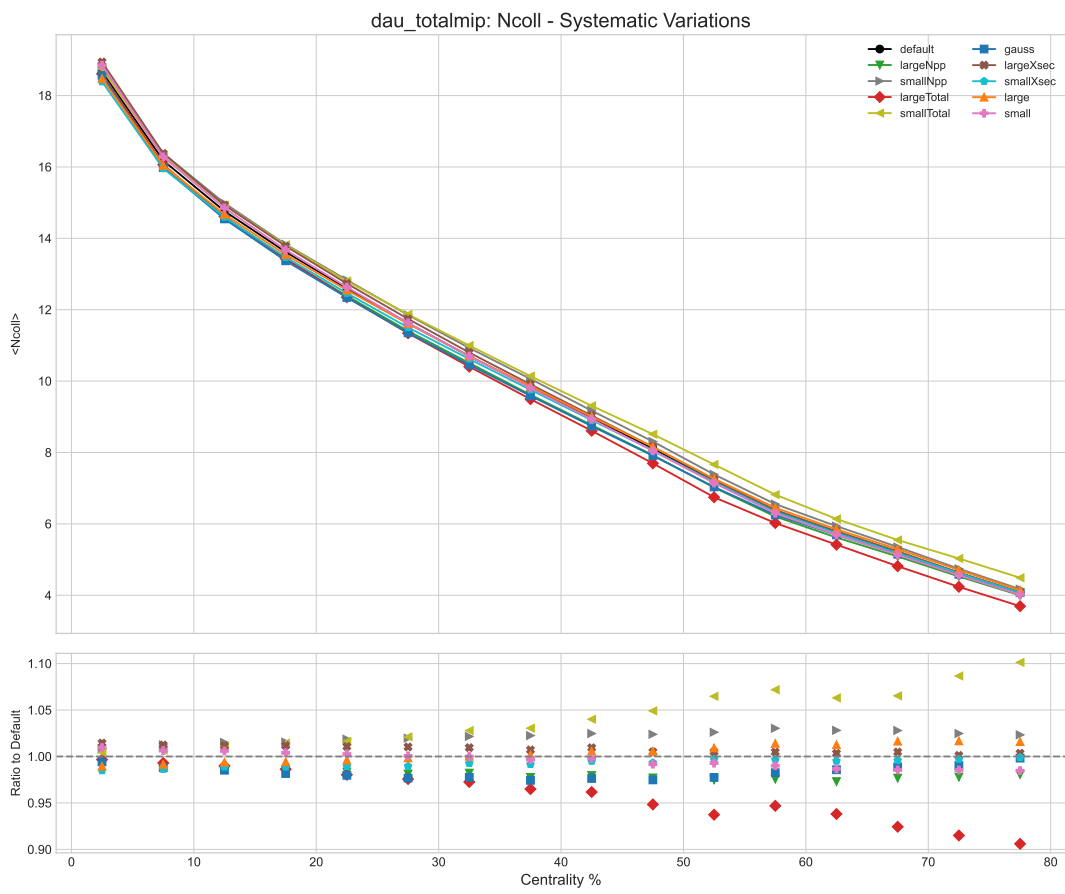
Integer Multiplicity Centrality Cuts Used			
Cent_Interval	Default	Large_Tot	Small_Tot
0-5%	285	283	288
5-10%	244	241	247
10-15%	216	212	220
15-20%	194	190	198
20-25%	175	170	179
25-30%	158	153	163
30-35%	143	138	148
35-40%	129	123	134
40-45%	115	110	121
45-50%	103	97	109
50-55%	91	85	97
55-60%	80	74	87
60-65%	70	63	76
65-70%	59	51	66
70-75%	48	41	56
75-80%	38	30	46

d+Au TotalMIP - Ncoll

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

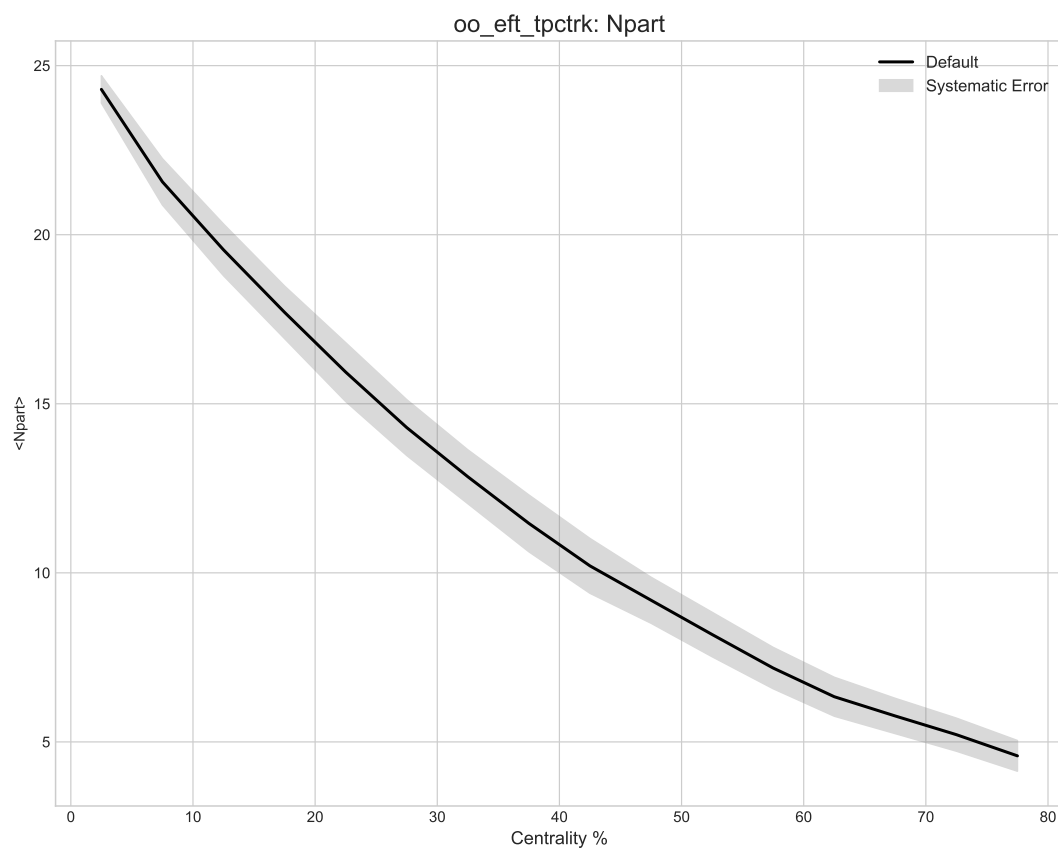
min_c	max_c	value	stat_err	syst_err	tot_err	gauss	large	largeNpp	largeTotal	largeXsec	small	smallNpp	smallTotal	smallXsec
0.0	5.0	18.6717	0.0086	0.3988	0.3989	18.5633	18.4758	18.5361	18.6093	18.9415	18.8502	18.8202	18.7550	18.3891
5.0	10.0	16.1789	0.0083	0.3997	0.3998	15.9841	16.0428	15.9879	16.0661	16.3788	16.2822	16.3820	16.2915	15.9568
10.0	15.0	14.7625	0.0083	0.4028	0.4029	14.5441	14.6722	14.5688	14.6110	14.9313	14.8515	14.9859	14.9299	14.5974
15.0	20.0	13.6209	0.0084	0.4200	0.4201	13.3710	13.5427	13.4159	13.4337	13.7803	13.6752	13.8322	13.8151	13.4690
20.0	25.0	12.5981	0.0084	0.4501	0.4502	12.3431	12.5494	12.3864	12.3502	12.7332	12.6313	12.8336	12.8030	12.4635
25.0	30.0	11.6237	0.0083	0.4701	0.4701	11.3528	11.6060	11.4043	11.3413	11.7434	11.6312	11.8534	11.8692	11.5020
30.0	35.0	10.6979	0.0084	0.4588	0.4589	10.4576	10.7002	10.5068	10.4047	10.7998	10.6969	10.9286	10.9964	10.6194
35.0	40.0	9.8397	0.0082	0.4902	0.4902	9.5869	9.8576	9.6183	9.4945	9.9098	9.8067	10.0614	10.1392	9.7564
40.0	45.0	8.9502	0.0077	0.4817	0.4818	8.7381	9.0022	8.7669	8.6078	9.0345	8.9339	9.1714	9.3084	8.9054
45.0	50.0	8.1135	0.0078	0.5117	0.5117	7.9091	8.1588	7.9265	7.6950	8.1538	8.0465	8.3072	8.5120	8.0594
50.0	55.0	7.1972	0.0073	0.5333	0.5334	7.0351	7.2635	7.0165	6.7460	7.2338	7.1458	7.3849	7.6633	7.1759
55.0	60.0	6.3615	0.0070	0.5171	0.5171	6.2506	6.4511	6.2064	6.0236	6.3921	6.2988	6.5544	6.8186	6.3401
60.0	65.0	5.7740	0.0069	0.4155	0.4156	5.6916	5.8474	5.6179	5.4167	5.7937	5.6969	5.9365	6.1383	5.7456
65.0	70.0	5.2076	0.0060	0.4333	0.4334	5.1493	5.2933	5.0861	4.8137	5.2327	5.1327	5.3533	5.5478	5.1863
70.0	75.0	4.6285	0.0054	0.4269	0.4270	4.5836	4.7061	4.5260	4.2351	4.6338	4.5614	4.7430	5.0297	4.6156
75.0	80.0	4.0766	0.0050	0.4289	0.4289	4.0693	4.1418	3.9973	3.6934	4.0908	4.0145	4.1718	4.4893	4.0726

Percentage difference from default											
min_c	max_c	tot_sys_err(%)	gauss(%)	large(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	small(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)
0.0	5.0	2.14	-0.58	-1.05	-0.73	-0.33	1.44	0.96	0.80	0.45	-1.51
5.0	10.0	2.47	-1.20	-0.84	-1.18	-0.70	1.24	0.64	1.26	0.70	-1.37
10.0	15.0	2.73	-1.48	-0.61	-1.31	-1.03	1.14	0.60	1.51	1.13	-1.12
15.0	20.0	3.08	-1.83	-0.57	-1.50	-1.37	1.17	0.40	1.55	1.43	-1.12
20.0	25.0	3.57	-2.02	-0.39	-1.68	-1.97	1.07	0.26	1.87	1.63	-1.07
25.0	30.0	4.04	-2.33	-0.15	-1.89	-2.43	1.03	0.07	1.98	2.11	-1.05
30.0	35.0	4.29	-2.25	0.02	-1.79	-2.74	0.95	-0.01	2.16	2.79	-0.73
35.0	40.0	4.98	-2.57	0.18	-2.25	-3.51	0.71	-0.34	2.25	3.04	-0.85
40.0	45.0	5.38	-2.37	0.58	-2.05	-3.82	0.94	-0.18	2.47	4.00	-0.50
45.0	50.0	6.31	-2.52	0.56	-2.30	-5.16	0.50	-0.83	2.39	4.91	-0.67
50.0	55.0	7.41	-2.25	0.92	-2.51	-6.27	0.51	-0.71	2.61	6.48	-0.30
55.0	60.0	8.13	-1.74	1.41	-2.44	-5.31	0.48	-0.99	3.03	7.19	-0.34
60.0	65.0	7.20	-1.43	1.27	-2.70	-6.19	0.34	-1.33	2.81	6.31	-0.49
65.0	70.0	8.32	-1.12	1.65	-2.33	-7.56	0.48	-1.44	2.80	6.53	-0.41
70.0	75.0	9.22	-0.97	1.68	-2.21	-8.50	0.12	-1.45	2.47	8.67	-0.28
75.0	80.0	10.52	-0.18	1.60	-1.94	-9.40	0.35	-1.52	2.33	10.12	-0.10

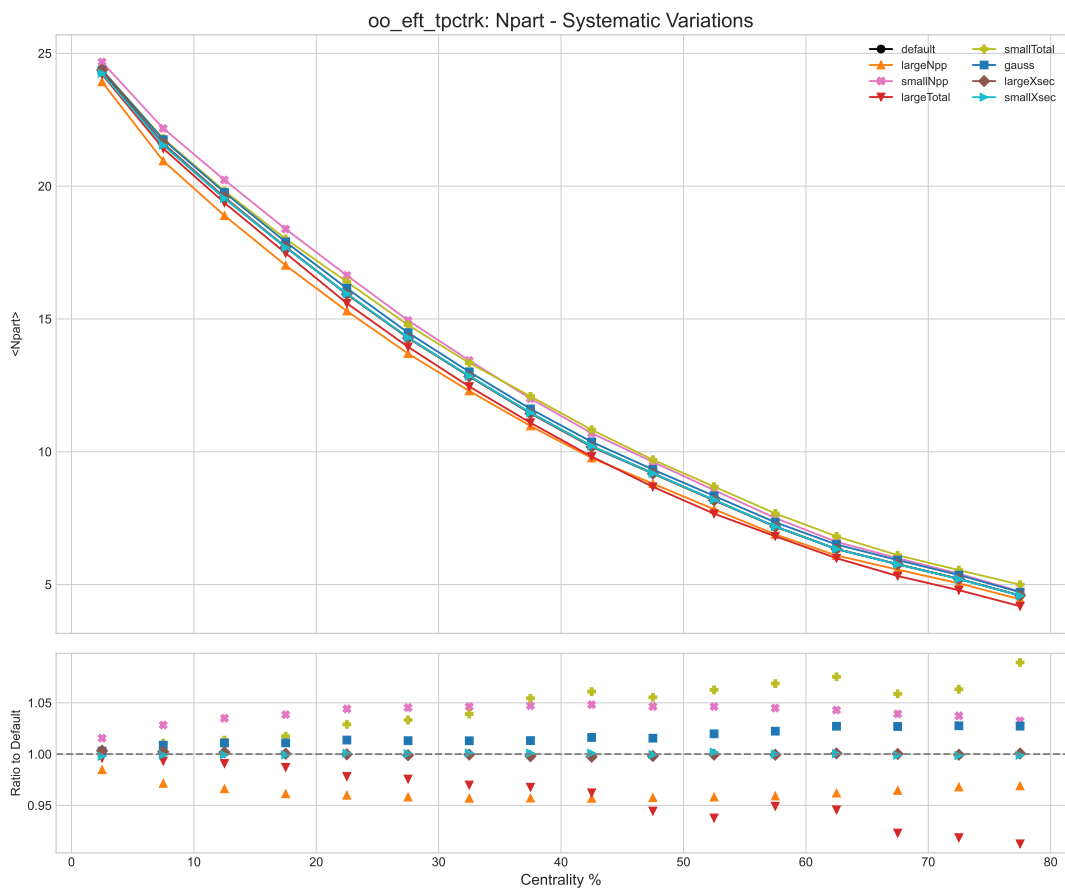
Integer Cent_Interval	Multiplicity Default	Centrality Large_Tot	Cuts Used Small_Tot
0-5%	285	283	288
5-10%	244	241	247
10-15%	216	212	220
15-20%	194	190	198
20-25%	175	170	179
25-30%	158	153	163
30-35%	143	138	148
35-40%	129	123	134
40-45%	115	110	121
45-50%	103	97	109
50-55%	91	85	97
55-60%	80	74	87
60-65%	70	63	76
65-70%	59	51	66
70-75%	48	41	56
75-80%	38	30	46

O+O EFT TPC - Npart

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

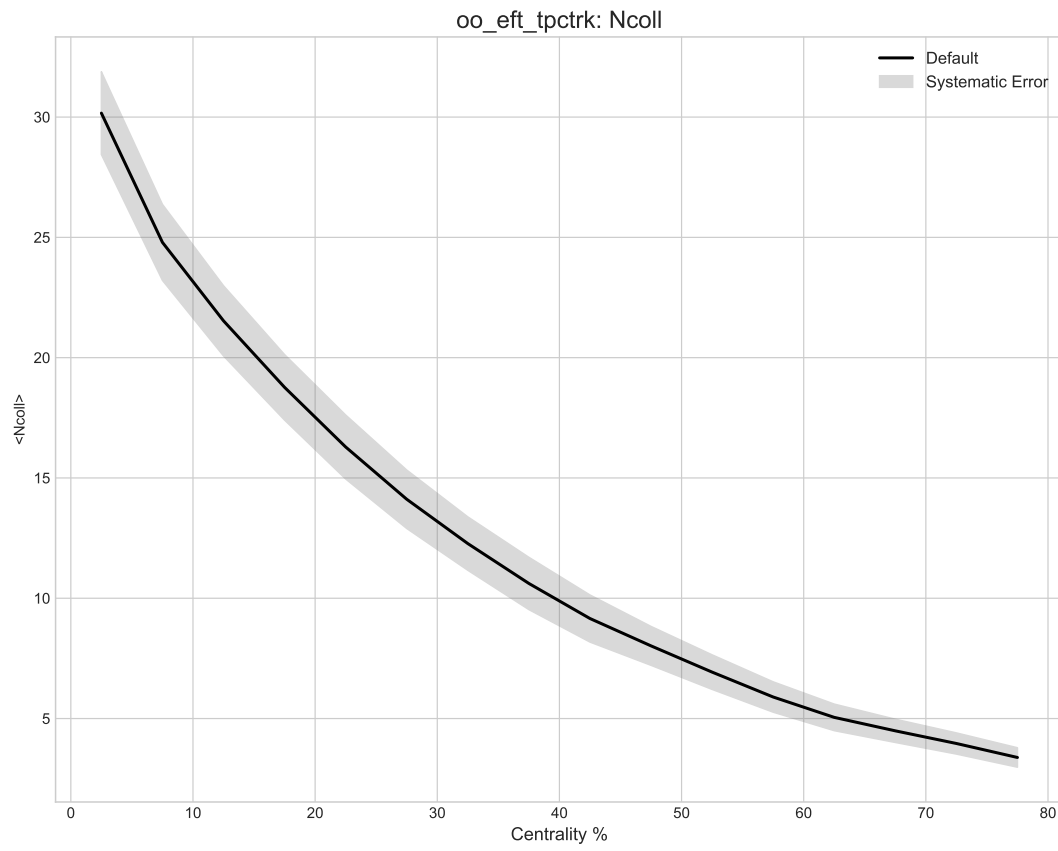
min_c	max_c	value	stat_err	syst_err	tot_err	gauss	largeNpp	largeTotal	largeXsec	smallNpp	smallTotal	smallXsec
0.0	5.0	24.2980	0.0067	0.4037	0.4037	24.3719	23.9275	24.2082	24.3784	24.6761	24.3769	24.2474
5.0	10.0	21.5627	0.0070	0.6843	0.6843	21.7514	20.9490	21.4169	21.6130	22.1745	21.7939	21.5446
10.0	15.0	19.5521	0.0075	0.7642	0.7642	19.7653	18.8917	19.3736	19.5882	20.2350	19.8184	19.5404
15.0	20.0	17.7039	0.0072	0.7767	0.7767	17.8987	17.0188	17.4763	17.7106	18.3856	18.0133	17.6930
20.0	25.0	15.9375	0.0075	0.8694	0.8695	16.1563	15.2990	15.5899	15.9348	16.6402	16.4002	15.9504
25.0	30.0	14.2961	0.0068	0.8258	0.8259	14.4836	13.6982	13.9498	14.2809	14.9445	14.7716	14.3017
30.0	35.0	12.8422	0.0073	0.7962	0.7962	13.0095	12.2894	12.4561	12.8365	13.4370	13.3442	12.8559
35.0	40.0	11.4629	0.0061	0.8401	0.8401	11.6139	10.9713	11.0930	11.4395	12.0040	12.0872	11.4741
40.0	45.0	10.2061	0.0067	0.8106	0.8107	10.3720	9.7663	9.8212	10.1795	10.6985	10.8278	10.2150
45.0	50.0	9.1905	0.0060	0.6805	0.6805	9.3333	8.8009	8.6804	9.1737	9.6173	9.6996	9.1867
50.0	55.0	8.1768	0.0054	0.6572	0.6572	8.3385	7.8345	7.6684	8.1729	8.5556	8.6888	8.1928
55.0	60.0	7.1843	0.0047	0.6113	0.6113	7.3449	6.8909	6.8195	7.1801	7.5068	7.6781	7.1852
60.0	65.0	6.3366	0.0047	0.5759	0.5759	6.5088	6.0959	5.9926	6.3412	6.6084	6.8142	6.3399
65.0	70.0	5.7661	0.0052	0.5216	0.5216	5.9214	5.5620	5.3224	5.7673	5.9921	6.1048	5.7598
70.0	75.0	5.2144	0.0037	0.4888	0.4888	5.3585	5.0472	4.7899	5.2113	5.4091	5.5442	5.2073
75.0	80.0	4.5872	0.0032	0.4539	0.4539	4.7122	4.4452	4.1862	4.5901	4.7363	4.9972	4.5820

Percentage difference from default										
min_c	max_c	tot_sys_err(%)	gauss(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)	
0.0	5.0	1.66	0.30	-1.52	-0.37	0.33	1.56	0.32	-0.21	
5.0	10.0	3.17	0.87	-2.85	-0.68	0.23	2.84	1.07	-0.08	
10.0	15.0	3.91	1.09	-3.38	-0.91	0.18	3.49	1.36	-0.06	
15.0	20.0	4.39	1.10	-3.87	-1.29	0.04	3.85	1.75	-0.06	
20.0	25.0	5.46	1.37	-4.01	-2.18	-0.02	4.41	2.90	0.08	
25.0	30.0	5.78	1.31	-4.18	-2.42	-0.11	4.54	3.33	0.04	
30.0	35.0	6.20	1.30	-4.30	-3.01	-0.04	4.63	3.91	0.11	
35.0	40.0	7.33	1.32	-4.29	-3.23	-0.20	4.72	5.45	0.10	
40.0	45.0	7.94	1.63	-4.31	-3.77	-0.26	4.82	6.09	0.09	
45.0	50.0	7.40	1.55	-4.24	-5.55	-0.18	4.64	5.54	-0.04	
50.0	55.0	8.04	1.98	-4.19	-6.22	-0.05	4.63	6.26	0.20	
55.0	60.0	8.51	2.24	-4.08	-5.08	-0.06	4.49	6.87	0.01	
60.0	65.0	9.09	2.72	-3.80	-5.43	0.07	4.29	7.54	0.05	
65.0	70.0	9.05	2.69	-3.54	-7.69	0.02	3.92	5.87	-0.11	
70.0	75.0	9.37	2.76	-3.21	-8.14	-0.06	3.73	6.33	-0.14	
75.0	80.0	9.89	2.73	-3.10	-8.74	0.06	3.25	8.94	-0.11	

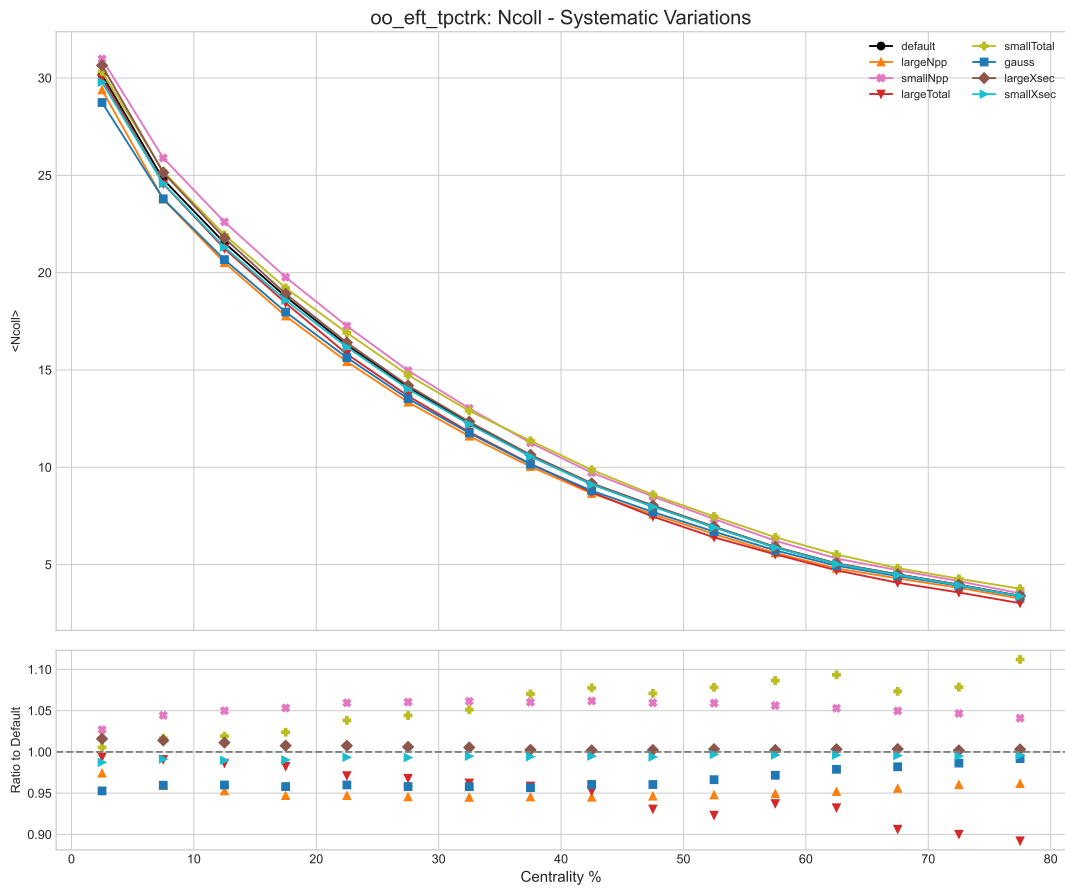
Integer Multiplicity Centrality Cuts Used			
Cent_Interval	Default	Large_Tot	Small_Tot
0-5%	97	96	98
5-10%	82	81	84
10-15%	72	71	73
15-20%	63	62	65
20-25%	56	54	58
25-30%	49	48	51
30-35%	44	42	46
35-40%	38	37	41
40-45%	34	32	36
45-50%	30	28	32
50-55%	26	24	28
55-60%	22	21	24
60-65%	19	17	21
65-70%	17	15	18
70-75%	14	12	16
75-80%	11	9	13

O+O EFT TPC - Ncoll

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

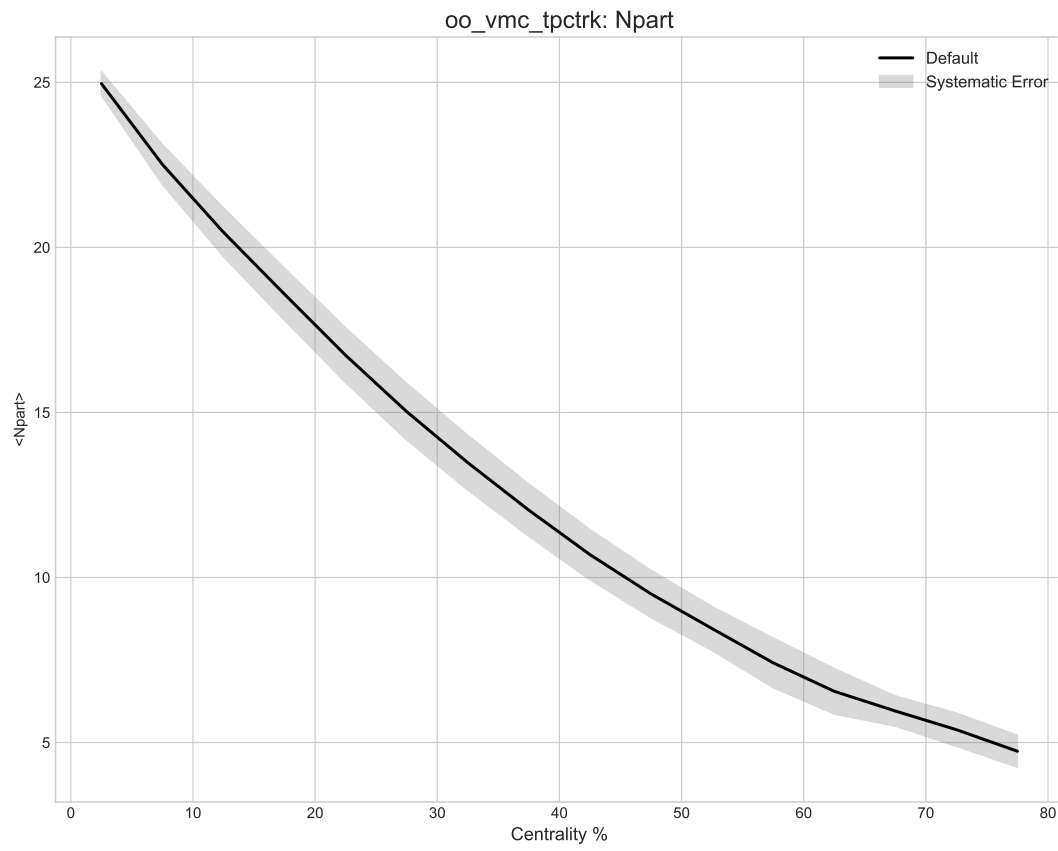
min_c	max_c	value	stat_err	syst_err	tot_err	gauss	largeNpp	largeTotal	largeXsec	smallNpp	smallTotal	smallXsec
0.0	5.0	30.1652	0.0155	1.7200	1.7201	28.7399	29.3875	29.9798	30.6459	30.9785	30.3309	29.7837
5.0	10.0	24.7916	0.0132	1.5795	1.5795	23.7912	23.7772	24.5622	25.1378	25.8898	25.2015	24.5758
10.0	15.0	21.5309	0.0128	1.4585	1.4586	20.6684	20.5134	21.2358	21.7742	22.6057	21.9418	21.3108
15.0	20.0	18.7660	0.0114	1.3617	1.3617	17.9776	17.7718	18.4380	18.9083	19.7657	19.2135	18.5845
20.0	25.0	16.2872	0.0111	1.3294	1.3294	15.6358	15.4256	15.8183	16.4088	17.2567	16.9102	16.1812
25.0	30.0	14.1144	0.0096	1.2148	1.2149	13.5222	13.3461	13.6637	14.2001	14.9677	14.7371	14.0182
30.0	35.0	12.2725	0.0098	1.1106	1.1106	11.7559	11.5962	11.8108	12.3388	13.0256	12.9009	12.2080
35.0	40.0	10.6149	0.0078	1.0872	1.0873	10.1547	10.0347	10.1762	10.6417	11.2548	11.3613	10.5542
40.0	45.0	9.1575	0.0081	0.9791	0.9792	8.7963	8.6542	8.7215	9.1752	9.7234	9.8687	9.1103
45.0	50.0	8.0204	0.0071	0.8086	0.8087	7.7036	7.5901	7.4657	8.0390	8.4979	8.5887	7.9713
50.0	55.0	6.9298	0.0061	0.7188	0.7188	6.6963	6.5683	6.3985	6.9527	7.3393	7.4720	6.9087
55.0	60.0	5.8949	0.0051	0.6312	0.6312	5.7282	5.5965	5.5256	5.9089	6.2267	6.4049	5.8722
60.0	65.0	5.0484	0.0049	0.5526	0.5527	4.9422	4.8060	4.7068	5.0649	5.3153	5.5201	5.0291
65.0	70.0	4.4886	0.0052	0.4833	0.4834	4.4078	4.2903	4.0680	4.5044	4.7118	4.8184	4.4691
70.0	75.0	3.9631	0.0037	0.4404	0.4404	3.9094	3.8054	3.5676	3.9703	4.1480	4.2744	3.9429
75.0	80.0	3.3814	0.0030	0.4046	0.4046	3.3540	3.2517	3.0170	3.3918	3.5196	3.7604	3.3674

Percentage difference from default										
min_c	max_c	tot_sys_err(%)	gauss(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)	
0.0	5.0	5.70	-4.72	-2.58	-0.61	1.59	2.70	0.55	-1.26	
5.0	10.0	6.37	-4.04	-4.09	-0.93	1.40	4.43	1.65	-0.87	
10.0	15.0	6.77	-4.01	-4.73	-1.37	1.13	4.99	1.91	-1.02	
15.0	20.0	7.26	-4.20	-5.30	-1.75	0.76	5.33	2.38	-0.97	
20.0	25.0	8.16	-4.00	-5.29	-2.88	0.75	5.95	3.82	-0.65	
25.0	30.0	8.61	-4.20	-5.44	-3.19	0.61	6.05	4.41	-0.68	
30.0	35.0	9.05	-4.21	-5.51	-3.76	0.54	6.14	5.12	-0.53	
35.0	40.0	10.24	-4.33	-5.47	-4.13	0.25	6.03	7.03	-0.57	
40.0	45.0	10.69	-3.94	-5.50	-4.76	0.19	6.18	7.77	-0.52	
45.0	50.0	10.08	-3.95	-5.36	-6.92	0.23	5.95	7.09	-0.61	
50.0	55.0	10.37	-3.37	-5.22	-7.67	0.33	5.91	7.82	-0.30	
55.0	60.0	10.71	-2.83	-5.06	-6.27	0.24	5.63	8.65	-0.39	
60.0	65.0	10.95	-2.10	-4.80	-6.77	0.33	5.29	9.34	-0.38	
65.0	70.0	10.77	-1.80	-4.42	-9.37	0.35	4.97	7.35	-0.43	
70.0	75.0	11.11	-1.36	-3.98	-9.98	0.18	4.66	7.85	-0.51	
75.0	80.0	11.97	-0.81	-3.84	-10.78	0.31	4.09	11.21	-0.41	

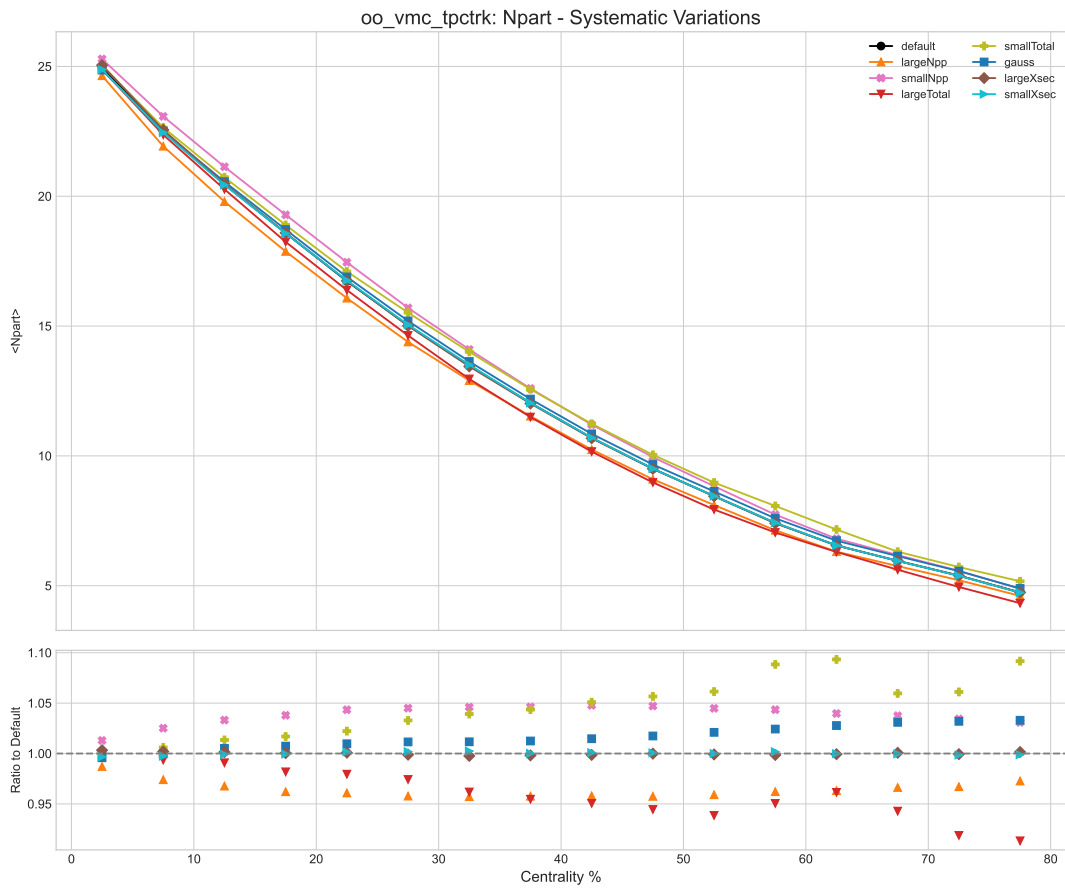
Integer Multiplicity Centrality Cuts Used			
Cent_Interval	Default	Large_Tot	Small_Tot
0-5%	97	96	98
5-10%	82	81	84
10-15%	72	71	73
15-20%	63	62	65
20-25%	56	54	58
25-30%	49	48	51
30-35%	44	42	46
35-40%	38	37	41
40-45%	34	32	36
45-50%	30	28	32
50-55%	26	24	28
55-60%	22	21	24
60-65%	19	17	21
65-70%	17	15	18
70-75%	14	12	16
75-80%	11	9	13

O+O VMC TPC - Npart

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

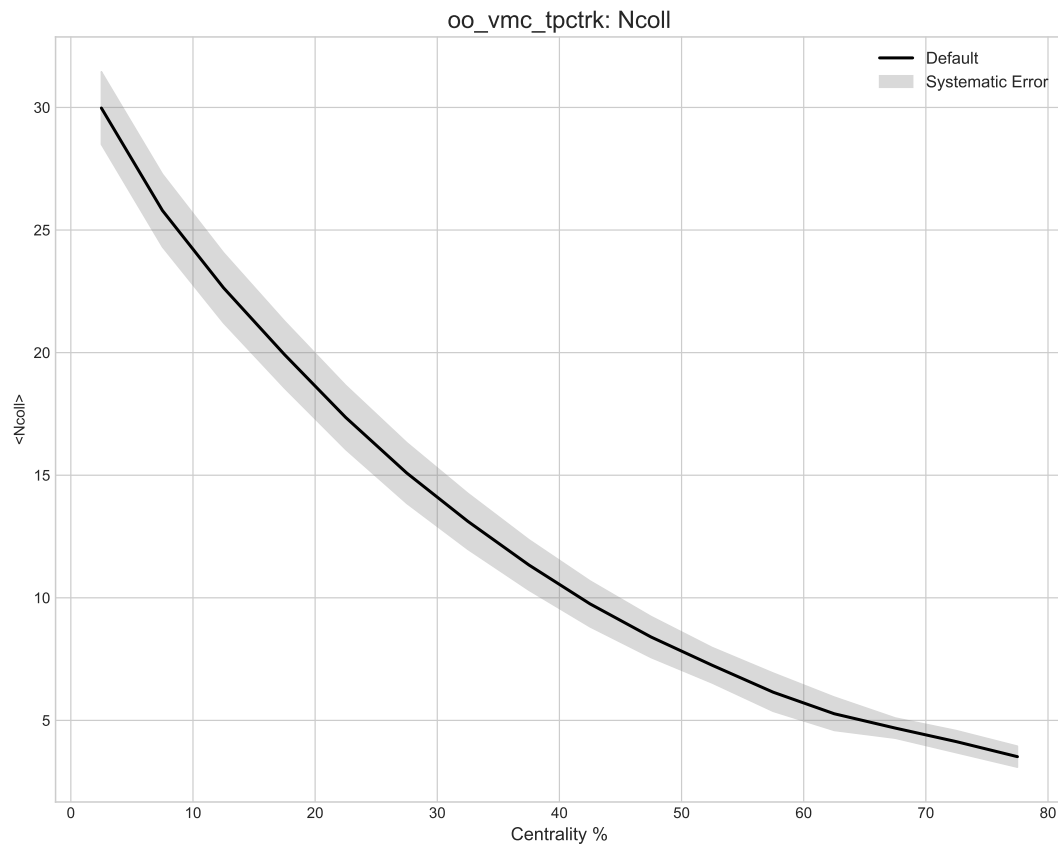
min_c	max_c	value	stat_err	syst_err	tot_err	gauss	largeNpp	largeTotal	largeXsec	smallNpp	smallTotal	smallXsec
0.0	5.0	24.9642	0.0069	0.3586	0.3587	24.8594	24.6434	24.8886	25.0465	25.2881	25.0373	24.8806
5.0	10.0	22.5122	0.0077	0.6051	0.6051	22.5058	21.9275	22.3679	22.5573	23.0773	22.6478	22.4539
10.0	15.0	20.4568	0.0081	0.7408	0.7409	20.5603	19.7959	20.2697	20.4861	21.1359	20.7326	20.4387
15.0	20.0	18.5802	0.0088	0.7932	0.7933	18.7168	17.8762	18.2426	18.5877	19.2848	18.8943	18.5718
20.0	25.0	16.7306	0.0082	0.8329	0.8329	16.8927	16.0764	16.3877	16.7448	17.4573	17.1026	16.7585
25.0	30.0	15.0244	0.0086	0.8554	0.8555	15.1978	14.3901	14.6376	15.0068	15.7005	15.5183	15.0524
30.0	35.0	13.4785	0.0079	0.8305	0.8305	13.6353	12.9015	12.9641	13.4466	14.0986	14.0071	13.5117
35.0	40.0	12.0393	0.0078	0.7940	0.7940	12.1890	11.5300	11.4935	12.0169	12.5958	12.5645	12.0409
40.0	45.0	10.6929	0.0070	0.7631	0.7631	10.8499	10.2439	10.1643	10.6782	11.2039	11.2372	10.7025
45.0	50.0	9.5035	0.0069	0.7197	0.7197	9.6690	9.1000	8.9762	9.5023	9.9521	10.0415	9.5098
50.0	55.0	8.4531	0.0062	0.6670	0.6671	8.6308	8.1075	7.9340	8.4454	8.8319	8.9725	8.4526
55.0	60.0	7.4155	0.0054	0.7527	0.7527	7.5955	7.1351	7.0484	7.4048	7.7380	8.0712	7.4290
60.0	65.0	6.5519	0.0053	0.6890	0.6891	6.7338	6.3112	6.2994	6.5475	6.8119	7.1636	6.5510
65.0	70.0	5.9558	0.0058	0.4583	0.4584	6.1402	5.7553	5.6156	5.9602	6.1792	6.3110	5.9539
70.0	75.0	5.3896	0.0042	0.5057	0.5057	5.5617	5.2122	4.9516	5.3862	5.5745	5.7189	5.3818
75.0	80.0	4.7368	0.0036	0.4840	0.4840	4.8930	4.6076	4.3266	4.7440	4.8830	5.1709	4.7311

Percentage difference from default									
min_c	max_c	tot_sys_err(%)	gauss(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)
0.0	5.0	1.44	-0.42	-1.29	-0.30	0.33	1.30	0.29	-0.33
5.0	10.0	2.69	-0.03	-2.60	-0.64	0.20	2.51	0.60	-0.26
10.0	15.0	3.62	0.51	-3.23	-0.91	0.14	3.32	1.35	-0.09
15.0	20.0	4.27	0.73	-3.79	-1.82	0.04	3.79	1.69	-0.05
20.0	25.0	4.98	0.97	-3.91	-2.05	0.09	4.34	2.22	0.17
25.0	30.0	5.69	1.15	-4.22	-2.57	-0.12	4.50	3.29	0.19
30.0	35.0	6.16	1.16	-4.28	-3.82	-0.24	4.60	3.92	0.25
35.0	40.0	6.60	1.24	-4.23	-4.53	-0.19	4.62	4.36	0.01
40.0	45.0	7.14	1.47	-4.20	-4.94	-0.14	4.78	5.09	0.09
45.0	50.0	7.57	1.74	-4.25	-5.55	-0.01	4.72	5.66	0.07
50.0	55.0	7.89	2.10	-4.09	-6.14	-0.09	4.48	6.14	-0.01
55.0	60.0	10.15	2.43	-3.78	-4.95	-0.15	4.35	8.84	0.18
60.0	65.0	10.52	2.78	-3.67	-3.85	-0.07	3.97	9.34	-0.01
65.0	70.0	7.70	3.09	-3.37	-5.71	0.07	3.75	5.96	-0.03
70.0	75.0	9.38	3.19	-3.29	-8.13	-0.06	3.43	6.11	-0.14
75.0	80.0	10.22	3.30	-2.73	-8.66	0.15	3.08	9.16	-0.12

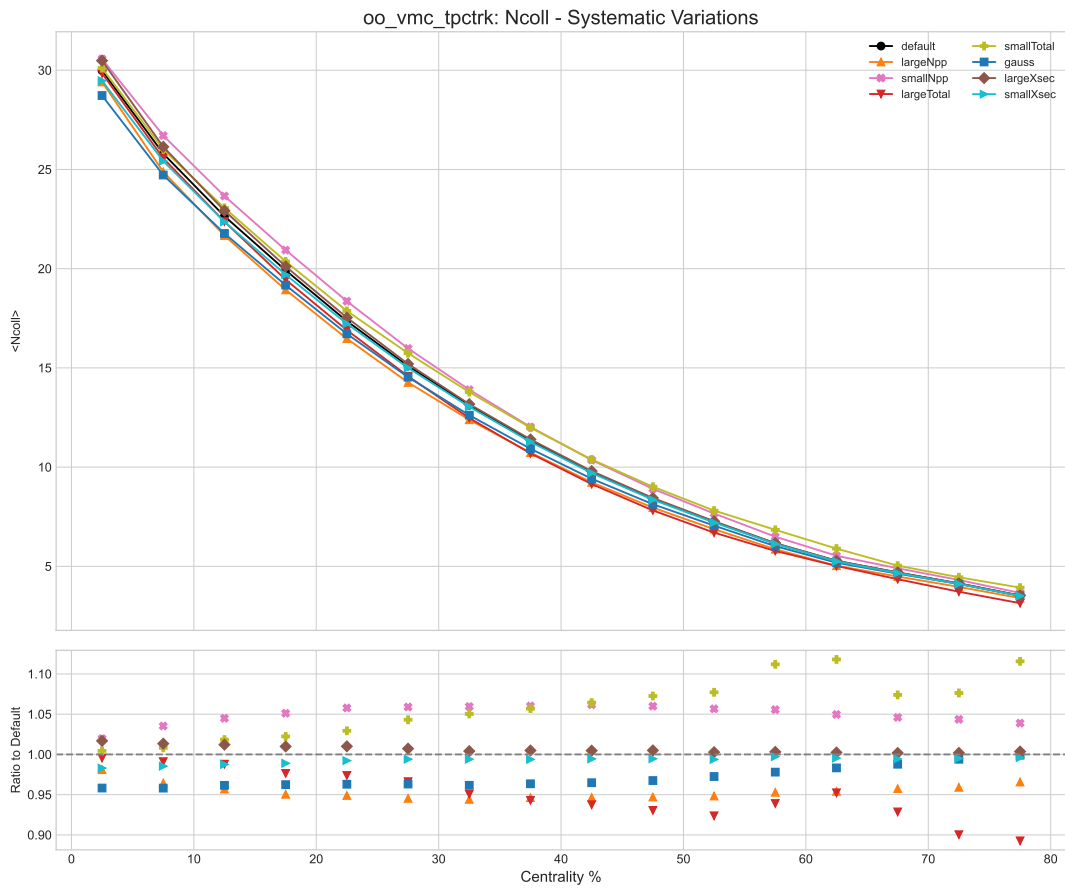
Integer Cent_Interval	Multiplicity Default	Centrality Large_Tot	Cuts Used Small_Tot
0-5%	97	96	98
5-10%	82	81	83
10-15%	71	70	73
15-20%	63	61	64
20-25%	55	54	57
25-30%	49	47	51
30-35%	43	41	45
35-40%	38	36	40
40-45%	33	31	35
45-50%	29	27	31
50-55%	25	23	27
55-60%	21	20	24
60-65%	18	17	20
65-70%	16	14	17
70-75%	13	11	15
75-80%	10	8	12

O+O VMC TPC - Ncoll

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

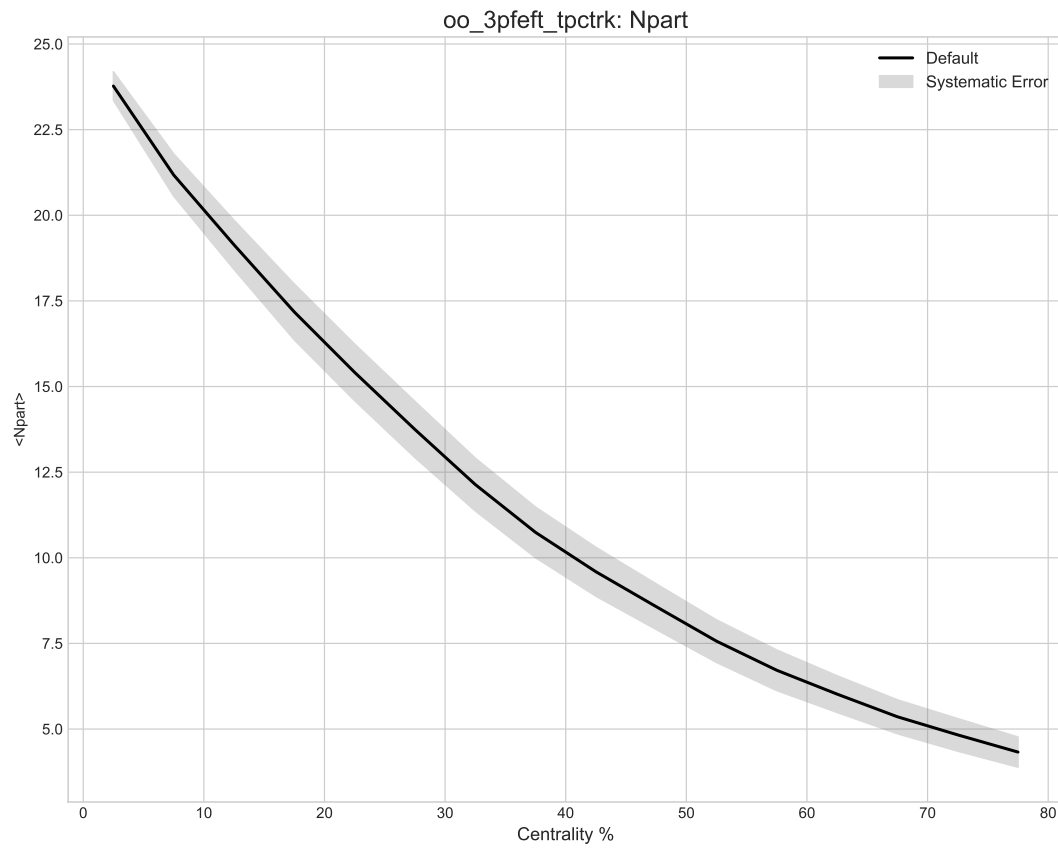
min_c	max_c	value	stat_err	syst_err	tot_err	gauss	largeNpp	largeTotal	largeXsec	smallNpp	smallTotal	smallXsec
0.0	5.0	29.9773	0.0137	1.4802	1.4803	28.7254	29.4082	29.8440	30.4817	30.5671	30.1101	29.4693
5.0	10.0	25.7969	0.0133	1.4836	1.4836	24.7172	24.8806	25.5651	26.1427	26.7073	26.0074	25.4201
10.0	15.0	22.6439	0.0130	1.4302	1.4303	21.7734	21.6700	22.3701	22.9187	23.6611	23.0613	22.3633
15.0	20.0	19.9170	0.0133	1.3664	1.3665	19.1694	18.9302	19.4469	20.1138	20.9363	20.3618	19.6975
20.0	25.0	17.3585	0.0118	1.3076	1.3077	16.7154	16.4738	16.9044	17.5332	18.3614	17.8683	17.2251
25.0	30.0	15.0906	0.0120	1.2384	1.2385	14.5355	14.2670	14.5801	15.1989	15.9799	15.7410	15.0005
30.0	35.0	13.1204	0.0105	1.1424	1.1424	12.6173	12.3881	12.4736	13.1743	13.9015	13.7805	13.0421
35.0	40.0	11.3401	0.0099	1.0299	1.0299	10.9265	10.7315	10.6934	11.3958	12.0231	11.9857	11.2706
40.0	45.0	9.7543	0.0086	0.9369	0.9370	9.4115	9.2326	9.1445	9.8003	10.3569	10.3823	9.7009
45.0	50.0	8.4020	0.0082	0.8376	0.8376	8.1292	7.9575	7.8175	8.4444	8.9057	9.0116	8.3571
50.0	55.0	7.2506	0.0070	0.7238	0.7238	7.0517	6.8760	6.6977	7.2712	7.6616	7.8103	7.2057
55.0	60.0	6.1504	0.0058	0.7810	0.7811	6.0152	5.8608	5.7752	6.1699	6.4927	6.8390	6.1338
60.0	65.0	5.2711	0.0056	0.6812	0.6812	5.1831	5.0289	5.0198	5.2838	5.5329	5.8933	5.2470
65.0	70.0	4.6860	0.0060	0.4131	0.4131	4.6289	4.4870	4.3513	4.6945	4.9016	5.0327	4.6596
70.0	75.0	4.1333	0.0042	0.4511	0.4511	4.1085	3.9644	3.7210	4.1413	4.3134	4.4489	4.1126
75.0	80.0	3.5203	0.0034	0.4298	0.4298	3.5170	3.3997	3.1424	3.5330	3.6571	3.9275	3.5065

Percentage difference from default									
min_c	max_c	tot_sys_err(%)	gauss(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)
0.0	5.0	4.94	-4.18	-1.90	-0.44	1.68	1.97	0.44	-1.69
5.0	10.0	5.75	-4.19	-3.55	-0.90	1.34	3.53	0.82	-1.46
10.0	15.0	6.32	-3.84	-4.30	-1.21	1.21	4.49	1.84	-1.24
15.0	20.0	6.86	-3.75	-4.95	-2.36	0.99	5.12	2.23	-1.10
20.0	25.0	7.53	-3.70	-5.10	-2.62	1.01	5.78	2.94	-0.77
25.0	30.0	8.21	-3.68	-5.46	-3.38	0.72	5.89	4.31	-0.60
30.0	35.0	8.71	-3.83	-5.58	-4.93	0.41	5.95	5.03	-0.60
35.0	40.0	9.08	-3.65	-5.37	-5.70	0.49	6.02	5.69	-0.61
40.0	45.0	9.61	-3.51	-5.35	-6.25	0.47	6.18	6.44	-0.55
45.0	50.0	9.97	-3.25	-5.29	-6.96	0.50	5.99	7.25	-0.53
50.0	55.0	9.98	-2.74	-5.17	-7.63	0.28	5.67	7.72	-0.62
55.0	60.0	12.70	-2.20	-4.71	-6.10	0.32	5.57	11.20	-0.27
60.0	65.0	12.92	-1.67	-4.59	-4.77	0.24	4.97	11.80	-0.46
65.0	70.0	8.82	-1.22	-4.24	-7.14	0.18	4.60	7.40	-0.56
70.0	75.0	10.91	-0.60	-4.09	-9.97	0.19	4.36	7.63	-0.50
75.0	80.0	12.21	-0.09	-3.43	-10.73	0.36	3.89	11.57	-0.39

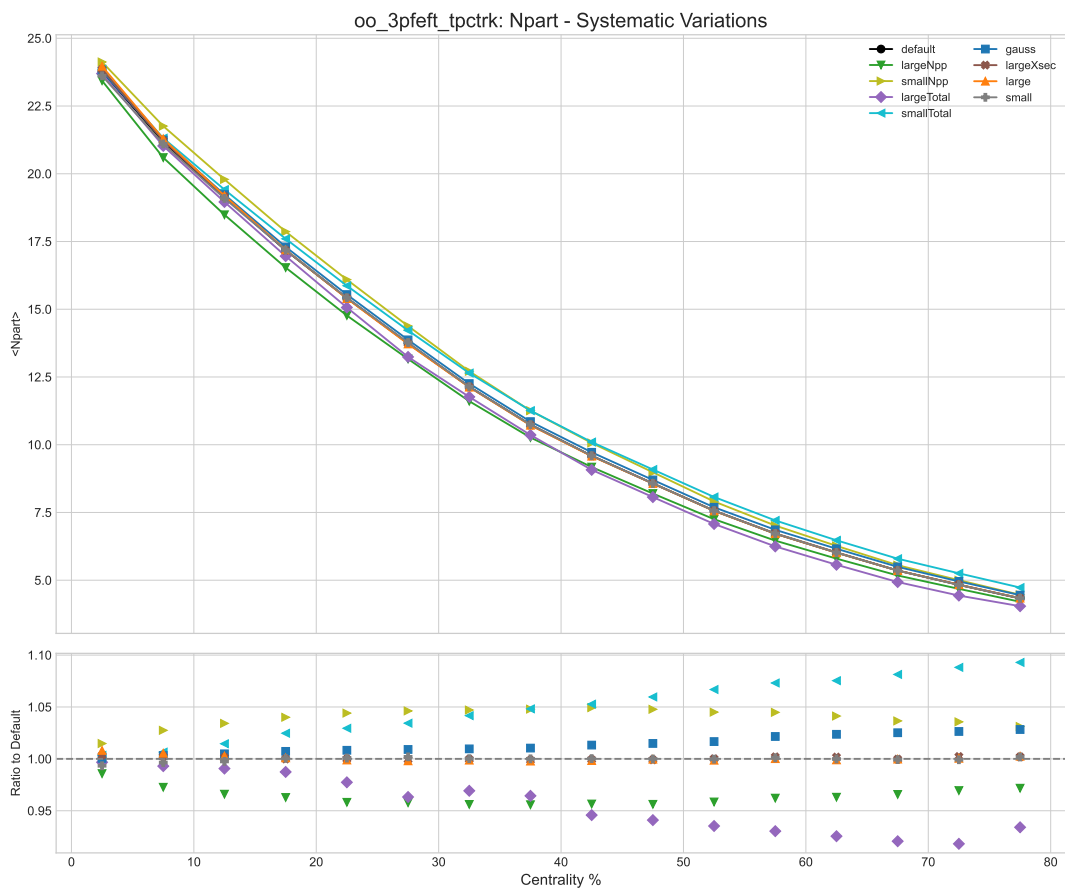
Integer Multiplicity Centrality Cuts Used			
Cent_Interval	Default	Large_Tot	Small_Tot
0-5%	97	96	98
5-10%	82	81	83
10-15%	71	70	73
15-20%	63	61	64
20-25%	55	54	57
25-30%	49	47	51
30-35%	43	41	45
35-40%	38	36	40
40-45%	33	31	35
45-50%	29	27	31
50-55%	25	23	27
55-60%	21	20	24
60-65%	18	17	20
65-70%	16	14	17
70-75%	13	11	15
75-80%	10	8	12

O+O 3PFEFT TPC - Npart

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

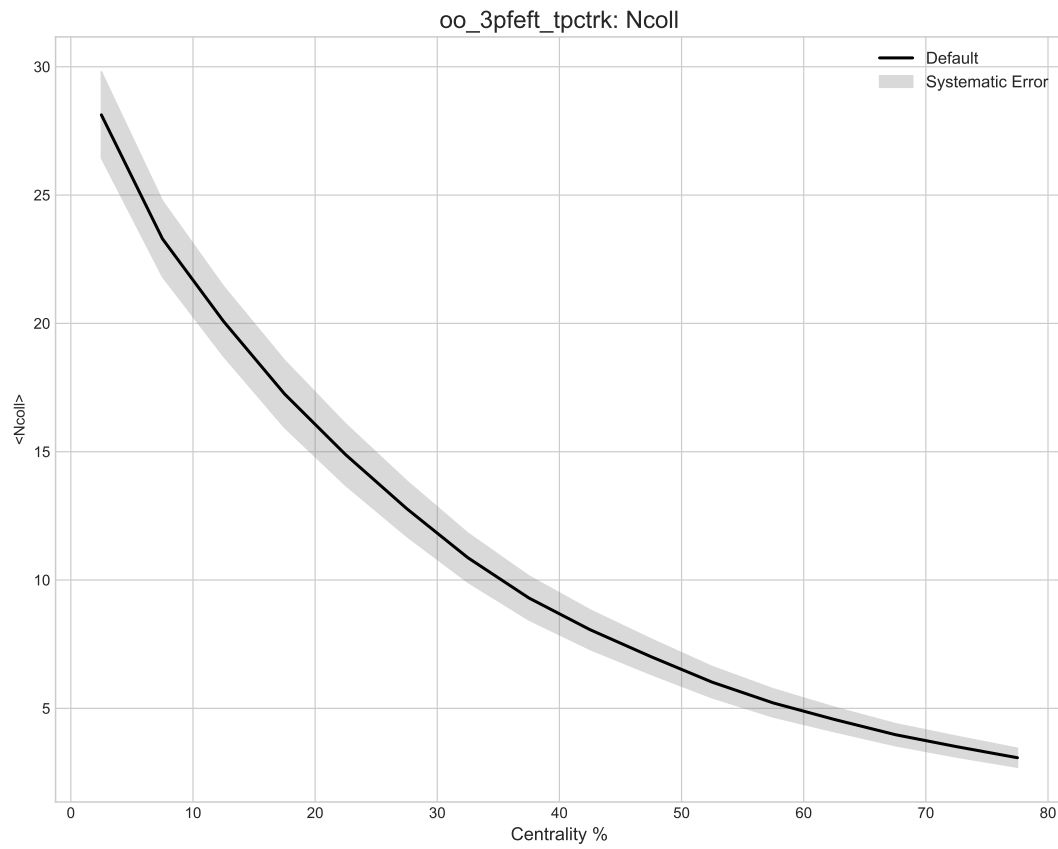
min_c	max_c	value	stat_err	syst_err	tot_err	gauss	large	largeNpp	largeTotal	largeXsec	small	smallNpp	smallTotal
0.0	5.0	23.7736	0.0067	0.4178	0.4179	23.7849	23.9670	23.4354	23.6961	23.8556	23.6159	24.1254	23.8551
5.0	10.0	21.1761	0.0071	0.6171	0.6172	21.2452	21.2907	20.5966	21.0278	21.2225	21.1005	21.7582	21.3144
10.0	15.0	19.1378	0.0074	0.7199	0.7199	19.2309	19.1913	18.4865	18.9587	19.1654	19.0908	19.7921	19.4166
15.0	20.0	17.1768	0.0075	0.8180	0.8180	17.3007	17.1936	16.5377	16.9602	17.1744	17.1851	17.8651	17.6006
20.0	25.0	15.4139	0.0078	0.8274	0.8274	15.5412	15.3966	14.7686	15.0656	15.4117	15.4259	16.0935	15.8680
25.0	30.0	13.7455	0.0071	0.8214	0.8215	13.8699	13.7192	13.1649	13.2404	13.7360	13.7720	14.3806	14.2179
30.0	35.0	12.1395	0.0069	0.7719	0.7719	12.2563	12.1243	11.6066	11.7658	12.1344	12.1416	12.7108	12.6449
35.0	40.0	10.7413	0.0067	0.7391	0.7391	10.8529	10.7180	10.2666	10.3586	10.7273	10.7417	11.2561	11.2590
40.0	45.0	9.5928	0.0067	0.7149	0.7150	9.7201	9.5770	9.1755	9.0727	9.5855	9.5950	10.0662	10.0985
45.0	50.0	8.5736	0.0060	0.6674	0.6675	8.7010	8.5706	8.1980	8.0675	8.5655	8.5726	8.9830	9.0850
50.0	55.0	7.5658	0.0053	0.6226	0.6226	7.6918	7.5556	7.2515	7.0762	7.5668	7.5637	7.9063	8.0714
55.0	60.0	6.7155	0.0053	0.5940	0.5941	6.8603	6.7168	6.4615	6.2477	6.7287	6.7223	7.0165	7.2066
60.0	65.0	6.0193	0.0046	0.5366	0.5366	6.1617	6.0132	5.7968	5.5705	6.0298	6.0186	6.2676	6.4729
65.0	70.0	5.3580	0.0040	0.4966	0.4966	5.4930	5.3567	5.1744	4.9327	5.3566	5.3542	5.5540	5.7939
70.0	75.0	4.8291	0.0042	0.4768	0.4768	4.9566	4.8316	4.6816	4.4337	4.8401	4.8251	5.0014	5.2549
75.0	80.0	4.3270	0.0030	0.4417	0.4417	4.4494	4.3379	4.2046	4.0416	4.3354	4.3348	4.4619	4.7291

Percentage difference from default													
min_c	max_c	tot_sys_err(%)	gauss(%)	large(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	small(%)	smallNpp(%)	smallTotal(%)			
0.0	5.0	1.76	0.05	0.81	-1.42	-0.33	0.34	-0.66	1.48	0.34			
5.0	10.0	2.91	0.33	0.54	-2.74	-0.70	0.22	-0.36	2.75	0.65			
10.0	15.0	3.76	0.49	0.28	-3.40	-0.94	0.14	-0.25	3.42	1.46			
15.0	20.0	4.76	0.72	0.10	-3.72	-1.26	-0.01	0.05	4.01	2.47			
20.0	25.0	5.37	0.83	-0.11	-4.19	-2.26	-0.01	0.08	4.41	2.95			
25.0	30.0	5.98	0.91	-0.19	-4.22	-3.67	-0.07	0.19	4.62	3.44			
30.0	35.0	6.36	0.96	-0.13	-4.39	-3.08	-0.04	0.02	4.71	4.16			
35.0	40.0	6.88	1.04	-0.22	-4.42	-3.56	-0.13	0.00	4.79	4.82			
40.0	45.0	7.45	1.33	-0.16	-4.35	-5.42	-0.08	0.02	4.94	5.27			
45.0	50.0	7.78	1.49	-0.03	-4.38	-5.90	-0.09	-0.01	4.78	5.96			
50.0	55.0	8.23	1.67	-0.13	-4.15	-6.47	0.01	-0.03	4.50	6.68			
55.0	60.0	8.85	2.16	0.02	-3.78	-6.97	0.20	0.10	4.48	7.31			
60.0	65.0	8.91	2.37	-0.10	-3.70	-7.45	0.18	-0.01	4.13	7.54			
65.0	70.0	9.27	2.52	-0.03	-3.43	-7.94	-0.03	-0.07	3.66	8.14			
70.0	75.0	9.87	2.64	0.05	-3.06	-8.19	0.23	-0.08	3.57	8.82			
75.0	80.0	10.21	2.83	0.25	-2.83	-6.59	0.20	0.18	3.12	9.29			

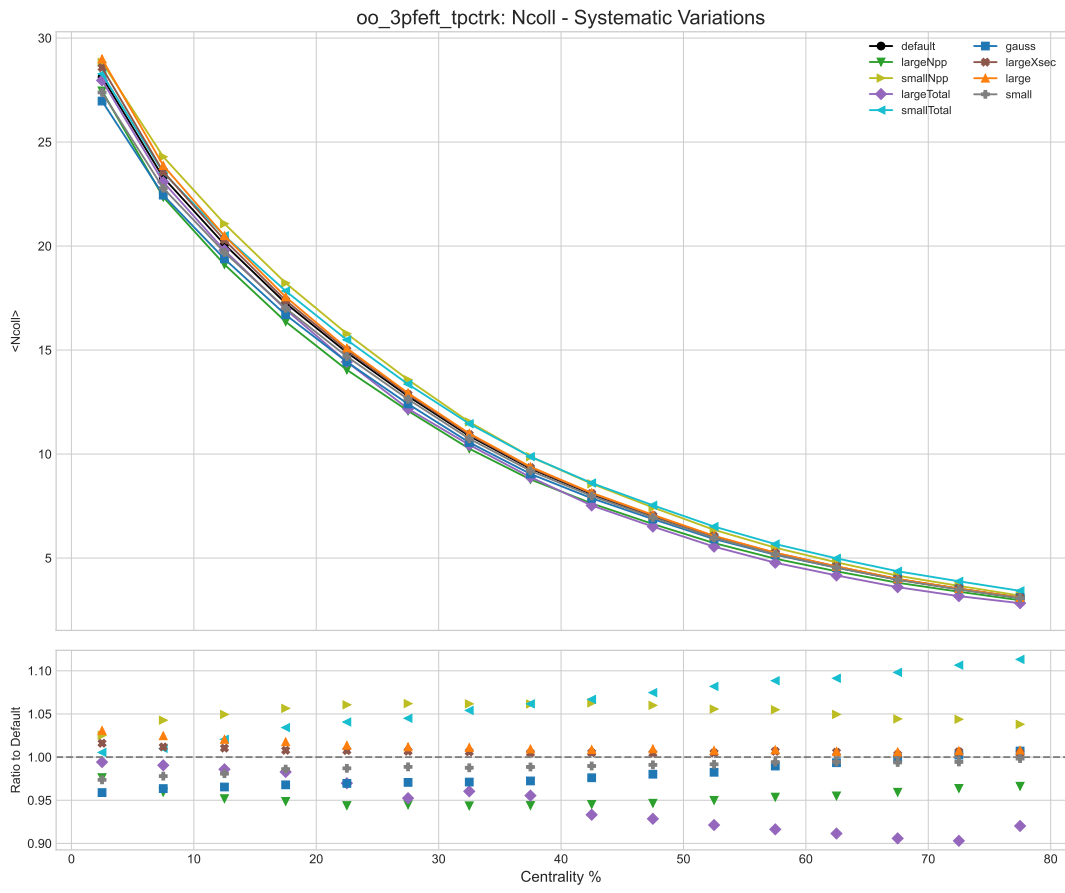
Integer Multiplicity Centrality Cuts Used			
Cent_Interval	Default	Large_Tot	Small_Tot
0-5%	96	95	97
5-10%	81	80	82
10-15%	70	69	72
15-20%	61	60	63
20-25%	54	52	56
25-30%	47	45	49
30-35%	41	40	43
35-40%	36	34	38
40-45%	32	30	34
45-50%	28	26	30
50-55%	24	22	26
55-60%	21	19	23
60-65%	18	16	20
65-70%	15	13	17
70-75%	13	11	15
75-80%	10	9	12

O+O 3PFEFT TPC - Ncoll

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

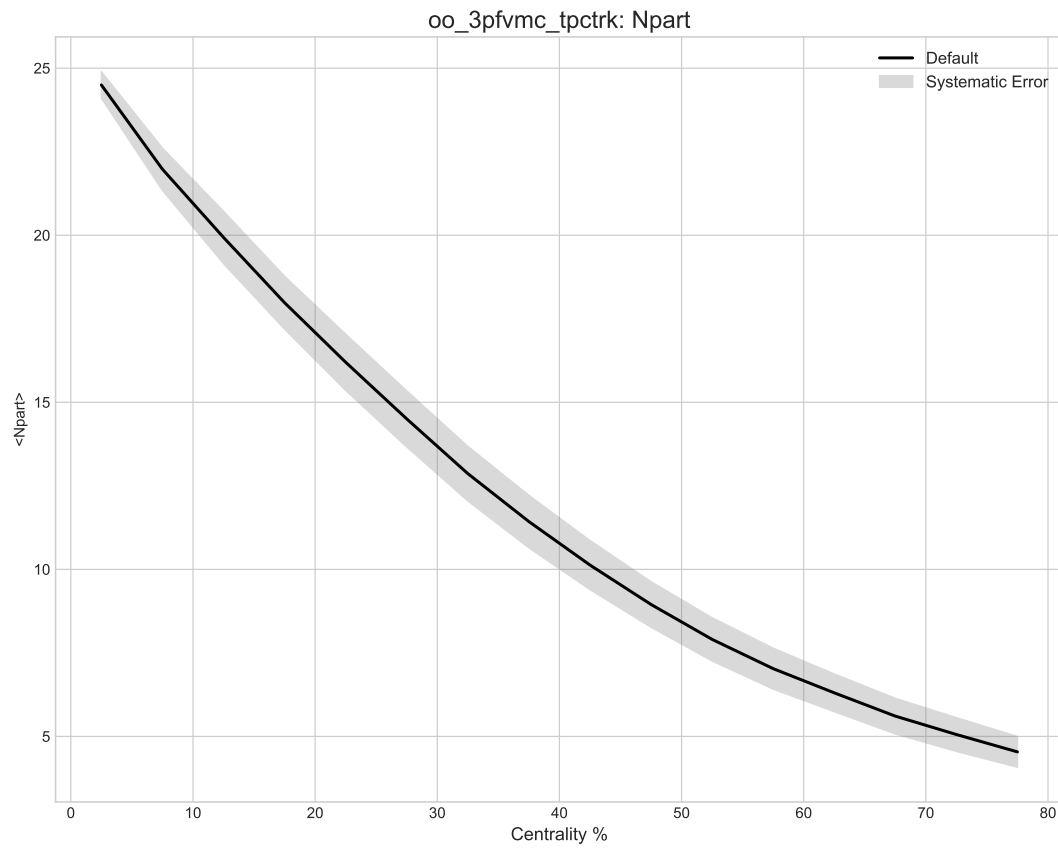
min_c	max_c	value	stat_err	syst_err	tot_err	gauss	large	largeNpp	largeTotal	largeXsec	small	smallNpp	smallTotal
0.0	5.0	28.1265	0.0147	1.6802	1.6803	26.9700	28.9943	27.4589	27.9647	28.5793	27.3865	28.8345	28.2793
5.0	10.0	23.2972	0.0130	1.4757	1.4758	22.4473	23.8728	22.3524	23.0772	23.5724	22.7837	24.2935	23.5338
10.0	15.0	20.0814	0.0121	1.3612	1.3612	19.3863	20.4934	19.1148	19.7977	20.2905	19.6963	21.0741	20.4944
15.0	20.0	17.2500	0.0112	1.3102	1.3103	16.6960	17.5561	16.3655	16.9530	17.3859	17.0099	18.2240	17.8407
20.0	25.0	14.8844	0.0109	1.1998	1.1999	14.4313	15.0864	14.0490	14.4345	14.9978	14.6912	15.7868	15.4897
25.0	30.0	12.7819	0.0094	1.0812	1.0812	12.4076	12.9325	12.0763	12.1735	12.8686	12.6376	13.5747	13.3569
30.0	35.0	10.8718	0.0085	0.9579	0.9579	10.5575	10.9929	10.2591	10.4406	10.9382	10.7376	11.5432	11.4597
35.0	40.0	9.3014	0.0079	0.8582	0.8583	9.0446	9.3887	8.7803	8.8867	9.3456	9.1952	9.8739	9.8755
40.0	45.0	8.0669	0.0075	0.7705	0.7705	7.8740	8.1376	7.6247	7.5280	8.1079	7.9840	8.5734	8.6067
45.0	50.0	7.0134	0.0064	0.6903	0.6903	6.8743	7.0796	6.6386	6.5120	7.0439	6.9504	7.4341	7.5375
50.0	55.0	6.0215	0.0054	0.6085	0.6085	5.9158	6.0678	5.7201	5.5477	6.0509	5.9715	6.3567	6.5148
55.0	60.0	5.2115	0.0052	0.5485	0.5485	5.1575	5.2554	4.9700	4.7753	5.2463	5.1813	5.4978	5.6728
60.0	65.0	4.5690	0.0044	0.4772	0.4772	4.5393	4.5989	4.3636	4.1644	4.5947	4.5437	4.7950	4.9863
65.0	70.0	3.9750	0.0037	0.4290	0.4290	3.9688	3.9989	3.8133	3.6007	3.9863	3.9504	4.1509	4.3653
70.0	75.0	3.5092	0.0038	0.4056	0.4056	3.5173	3.5344	3.3820	3.1688	3.5308	3.4897	3.6630	3.8830
75.0	80.0	3.0757	0.0027	0.3690	0.3690	3.0974	3.1001	2.9720	2.8303	3.0905	3.0704	3.1926	3.4239

Percentage difference from default													
min_c	max_c	tot_sys_err(%)	gauss(%)	large(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	small(%)	smallNpp(%)	smallTotal(%)			
0.0	5.0	5.97	-4.11	3.09	-2.37	-0.58	1.61	-2.63	2.52	0.54			
5.0	10.0	6.33	-3.65	2.47	-4.06	-0.94	1.18	-2.20	4.28	1.02			
10.0	15.0	6.78	-3.46	2.05	-4.81	-1.41	1.04	-1.92	4.94	2.06			
15.0	20.0	7.60	-3.21	1.77	-5.13	-1.72	0.79	-1.39	5.65	3.42			
20.0	25.0	8.06	-3.04	1.36	-5.61	-3.02	0.76	-1.30	6.06	4.07			
25.0	30.0	8.46	-2.93	1.18	-5.52	-4.76	0.68	-1.13	6.20	4.50			
30.0	35.0	8.81	-2.89	1.11	-5.64	-3.97	0.61	-1.23	6.18	5.41			
35.0	40.0	9.23	-2.76	0.94	-5.60	-4.46	0.48	-1.14	6.16	6.17			
40.0	45.0	9.55	-2.39	0.88	-5.48	-6.68	0.51	-1.03	6.28	6.69			
45.0	50.0	9.84	-1.98	0.95	-5.34	-7.15	0.44	-0.90	6.00	7.47			
50.0	55.0	10.11	-1.76	0.77	-5.01	-7.87	0.49	-0.83	5.57	8.19			
55.0	60.0	10.52	-1.04	0.84	-4.63	-8.37	0.67	-0.58	5.49	8.85			
60.0	65.0	10.44	-0.65	0.66	-4.49	-8.86	0.56	-0.55	4.95	9.13			
65.0	70.0	10.79	-0.16	0.60	-4.07	-9.42	0.28	-0.62	4.42	9.82			
70.0	75.0	11.56	0.23	0.72	-3.62	-9.70	0.62	-0.56	4.38	10.65			
75.0	80.0	12.00	0.70	0.79	-3.37	-7.98	0.48	-0.17	3.80	11.32			

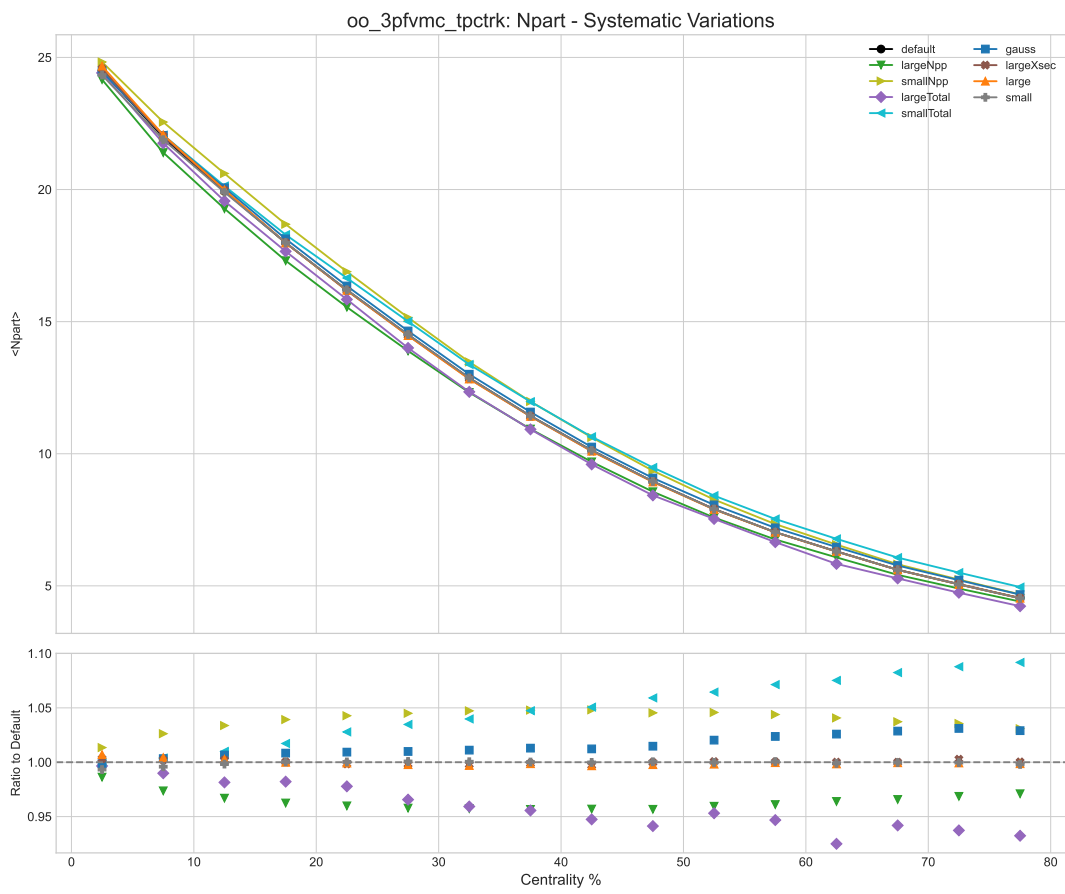
Integer Multiplicity Centrality Cuts Used			
Cent_Interval	Default	Large_Tot	Small_Tot
0-5%	96	95	97
5-10%	81	80	82
10-15%	70	69	72
15-20%	61	60	63
20-25%	54	52	56
25-30%	47	45	49
30-35%	41	40	43
35-40%	36	34	38
40-45%	32	30	34
45-50%	28	26	30
50-55%	24	22	26
55-60%	21	19	23
60-65%	18	16	20
65-70%	15	13	17
70-75%	13	11	15
75-80%	10	9	12

O+O 3PFVMC TPC - Npart

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

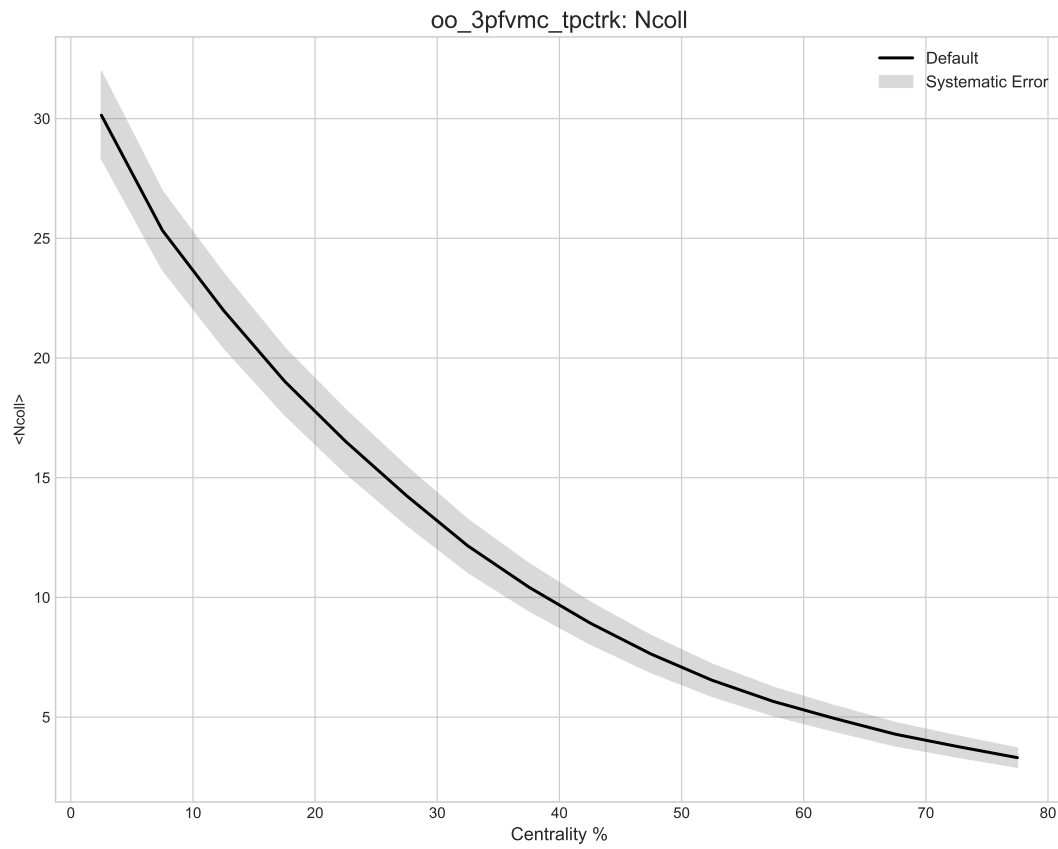
min_c	max_c	value	stat_err	syst_err	tot_err	gauss	large	largeNpp	largeTotal	largeXsec	small	smallNpp	smallTotal
0.0	5.0	24.4975	0.0067	0.3965	0.3966	24.4649	24.6709	24.1615	24.4134	24.5758	24.3323	24.8279	24.4975
5.0	10.0	21.9761	0.0074	0.6326	0.6327	22.0490	22.0701	21.3979	21.7533	22.0226	21.8856	22.5530	22.0611
10.0	15.0	19.9347	0.0077	0.7821	0.7821	20.0667	19.9906	19.2761	19.5653	19.9748	19.9008	20.6078	20.1338
15.0	20.0	17.9806	0.0078	0.7899	0.7899	18.1315	17.9792	17.3063	17.6595	17.9863	17.9870	18.6862	18.2906
20.0	25.0	16.1987	0.0082	0.8412	0.8412	16.3493	16.1860	15.5487	15.8396	16.1736	16.2029	16.8916	16.6504
25.0	30.0	14.5032	0.0075	0.8379	0.8380	14.6469	14.4704	13.8914	14.0043	14.4774	14.5134	15.1556	15.0073
30.0	35.0	12.8659	0.0073	0.8169	0.8170	13.0092	12.8271	12.3174	12.3422	12.8389	12.8719	13.4744	13.3780
35.0	40.0	11.4304	0.0073	0.7848	0.7849	11.5786	11.4147	10.9347	10.9232	11.4169	11.4324	11.9782	11.9720
40.0	45.0	10.1294	0.0065	0.7342	0.7342	10.2536	10.0960	9.6930	9.5969	10.0996	10.1274	10.6173	10.6425
45.0	50.0	8.9498	0.0064	0.6808	0.6808	9.0817	8.9297	8.5624	8.4243	8.9474	8.9536	9.3565	9.4793
50.0	55.0	7.9043	0.0056	0.6457	0.6458	8.0650	7.8902	7.5845	7.5336	7.9093	7.9030	8.2663	8.4142
55.0	60.0	7.0286	0.0057	0.6123	0.6123	7.1953	7.0258	6.7550	6.6547	7.0295	7.0329	7.3370	7.5305
60.0	65.0	6.3046	0.0050	0.5638	0.5638	6.4678	6.2946	6.0773	5.8315	6.3052	6.2980	6.5615	6.7791
65.0	70.0	5.6076	0.0043	0.5323	0.5324	5.7683	5.6049	5.4156	5.2817	5.6076	5.6062	5.8166	6.0701
70.0	75.0	5.0572	0.0046	0.5048	0.5048	5.2144	5.0537	4.8985	4.7402	5.0711	5.0555	5.2373	5.5015
75.0	80.0	4.5372	0.0033	0.4588	0.4588	4.6689	4.5309	4.4057	4.2304	4.5385	4.5281	4.6770	4.9537

Percentage difference from default													
min_c	max_c	tot_sys_err(%)	gauss(%)	large(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	small(%)	smallNpp(%)	smallTotal(%)			
0.0	5.0	1.62	-0.13	0.71	-1.37	-0.34	0.32	-0.67	1.35	0.00			
5.0	10.0	2.88	0.33	0.43	-2.63	-1.01	0.21	-0.41	2.63	0.39			
10.0	15.0	3.92	0.66	0.28	-3.30	-1.85	0.20	-0.17	3.38	1.00			
15.0	20.0	4.39	0.84	-0.01	-3.75	-1.79	0.03	0.04	3.92	1.72			
20.0	25.0	5.19	0.93	-0.08	-4.01	-2.22	-0.16	0.03	4.28	2.79			
25.0	30.0	5.78	0.99	-0.23	-4.22	-3.44	-0.18	0.07	4.50	3.48			
30.0	35.0	6.35	1.11	-0.30	-4.26	-4.07	-0.21	0.05	4.73	3.98			
35.0	40.0	6.87	1.30	-0.14	-4.34	-4.44	-0.12	0.02	4.79	4.74			
40.0	45.0	7.25	1.23	-0.33	-4.31	-5.26	-0.29	-0.02	4.82	5.07			
45.0	50.0	7.61	1.47	-0.22	-4.33	-5.87	-0.03	0.04	4.54	5.92			
50.0	55.0	8.17	2.03	-0.18	-4.05	-4.69	0.06	-0.02	4.58	6.45			
55.0	60.0	8.71	2.37	-0.04	-3.89	-5.32	0.01	0.06	4.39	7.14			
60.0	65.0	8.94	2.59	-0.16	-3.61	-7.50	0.01	-0.11	4.07	7.53			
65.0	70.0	9.49	2.87	-0.05	-3.42	-5.81	-0.00	-0.03	3.73	8.25			
70.0	75.0	9.98	3.11	-0.07	-3.14	-6.27	0.28	-0.03	3.56	8.79			
75.0	80.0	10.11	2.90	-0.14	-2.90	-6.76	0.03	-0.20	3.08	9.18			

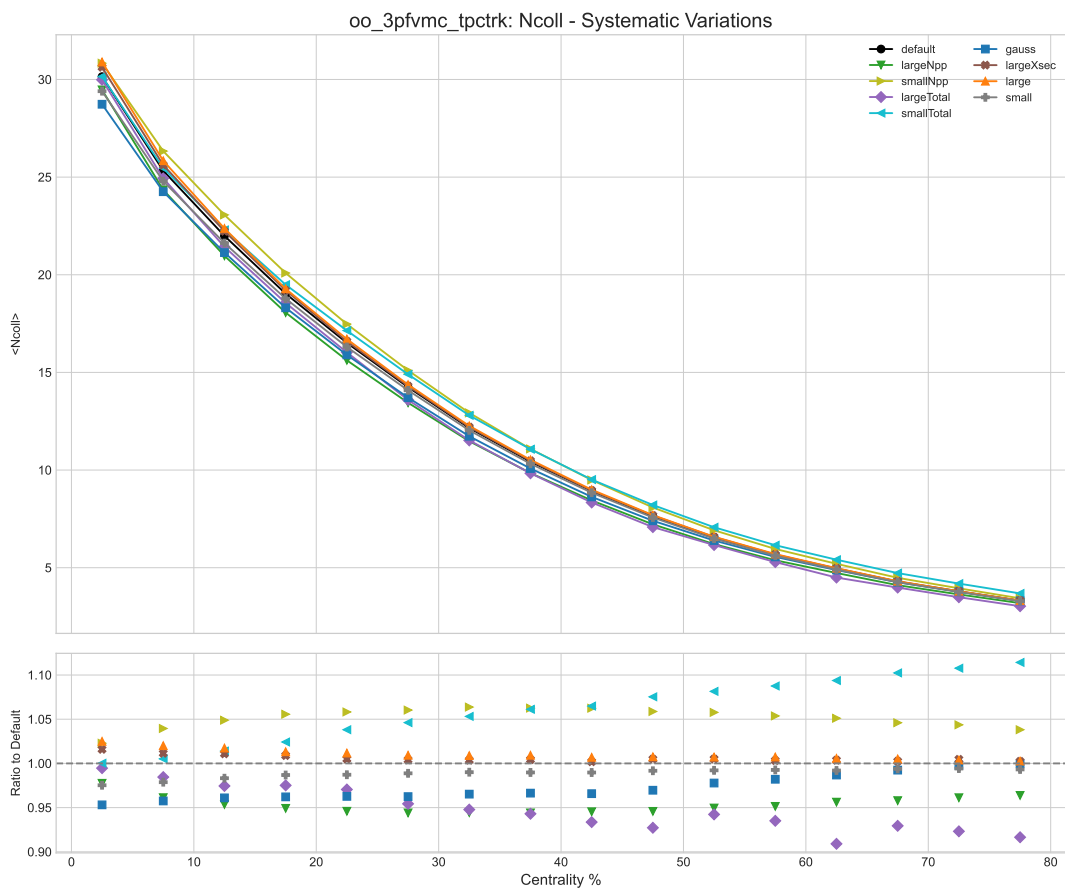
Integer Multiplicity Centrality Cuts Used			
Cent_Interval	Default	Large_Tot	Small_Tot
0-5%	97	96	97
5-10%	82	80	83
10-15%	71	69	72
15-20%	62	61	64
20-25%	55	53	57
25-30%	48	46	50
30-35%	42	40	44
35-40%	37	35	39
40-45%	32	30	34
45-50%	28	26	30
50-55%	24	23	26
55-60%	21	19	23
60-65%	18	16	20
65-70%	15	14	17
70-75%	13	11	15
75-80%	10	9	12

O+O 3PFVMC TPC - Ncoll

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

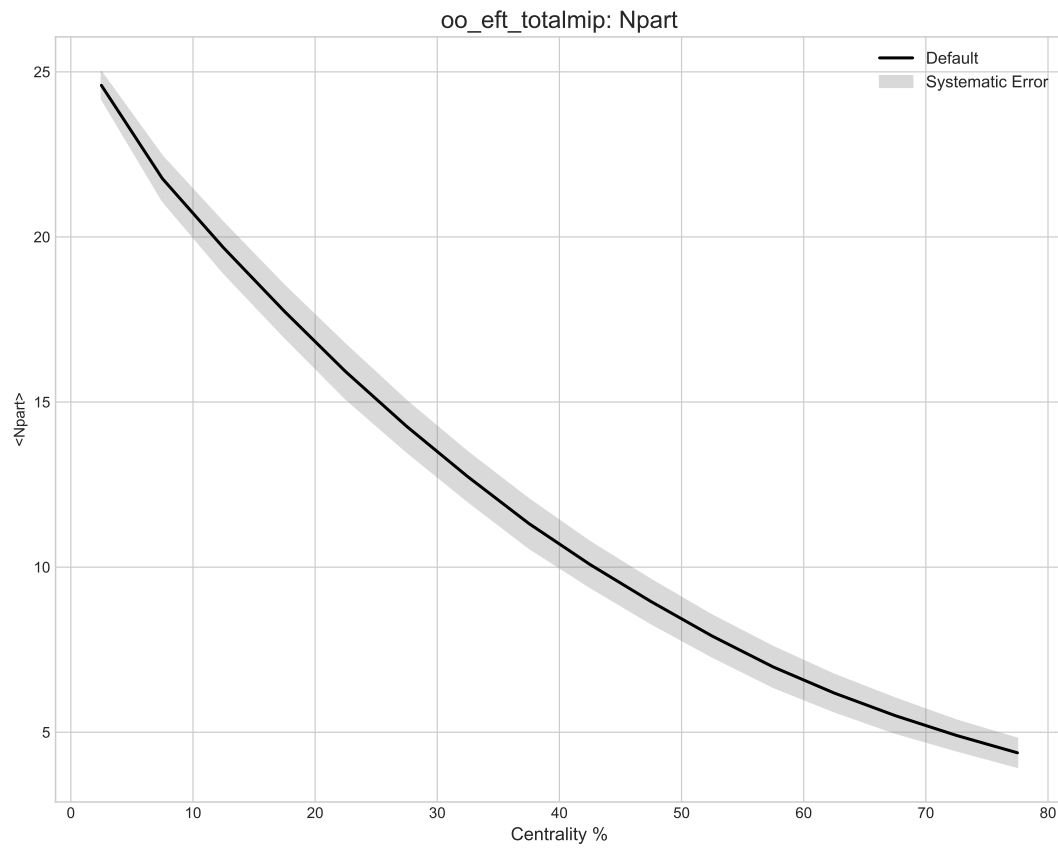
min_c	max_c	value	stat_err	syst_err	tot_err	gauss	large	largeNpp	largeTotal	largeXsec	small	smallNpp	smallTotal
0.0	5.0	30.1444	0.0152	1.8147	1.8147	28.7311	30.8960	29.4666	29.9832	30.6238	29.3995	30.8334	30.1511
5.0	10.0	25.3299	0.0138	1.6387	1.6387	24.2544	25.8380	24.3483	24.9383	25.6190	24.7922	26.3311	25.4597
10.0	15.0	21.9910	0.0131	1.5516	1.5517	21.1339	22.3726	20.9701	21.4298	22.2291	21.6229	23.0660	22.2991
15.0	20.0	19.0304	0.0123	1.3984	1.3985	18.3102	19.2813	18.0632	18.5574	19.1994	18.7804	20.0895	19.4915
20.0	25.0	16.5136	0.0121	1.3232	1.3232	15.8979	16.7071	15.6198	16.0265	16.5952	16.3021	17.4748	17.1432
25.0	30.0	14.2434	0.0105	1.2190	1.2190	13.7084	14.3748	13.4463	13.5922	14.3003	14.0840	15.1028	14.9010
30.0	35.0	12.1540	0.0097	1.1012	1.1013	11.7317	12.2612	11.4779	11.5192	12.1986	12.0333	12.9286	12.8004
35.0	40.0	10.4294	0.0091	0.9848	0.9848	10.0789	10.5253	9.8455	9.8353	10.4744	10.3228	11.0821	11.0678
40.0	45.0	8.9329	0.0077	0.8748	0.8748	8.6272	8.9942	8.4445	8.3398	8.9497	8.8410	9.4908	9.5122
45.0	50.0	7.6351	0.0073	0.7689	0.7690	7.4034	7.6924	7.2210	7.0792	7.6720	7.5716	8.0838	8.2103
50.0	55.0	6.5388	0.0061	0.6718	0.6718	6.3937	6.5857	6.2088	6.1617	6.5715	6.4876	6.9163	7.0718
55.0	60.0	5.6563	0.0059	0.5917	0.5918	5.5549	5.6974	5.3814	5.2890	5.6746	5.6146	5.9603	6.1517
60.0	65.0	4.9462	0.0050	0.5340	0.5340	4.8813	4.9744	4.7295	4.4962	4.9642	4.9066	5.1989	5.4101
65.0	70.0	4.2858	0.0042	0.4830	0.4831	4.2534	4.3075	4.1053	3.9832	4.2963	4.2595	4.4830	4.7246
70.0	75.0	3.7798	0.0044	0.4403	0.4403	3.7712	3.7942	3.6332	3.4894	3.7973	3.7580	3.9449	4.1870
75.0	80.0	3.3058	0.0030	0.3989	0.3989	3.3045	3.3144	3.1861	3.0297	3.3135	3.2833	3.4319	3.6835

Percentage difference from default													
min_c	max_c	tot_sys_err(%)	gauss(%)	large(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	small(%)	smallNpp(%)	smallTotal(%)			
0.0	5.0	6.02	-4.69	2.49	-2.25	-0.53	1.59	-2.47	2.29	0.02			
5.0	10.0	6.47	-4.25	2.01	-3.88	-1.55	1.14	-2.12	3.95	0.51			
10.0	15.0	7.06	-3.90	1.74	-4.64	-2.55	1.08	-1.67	4.89	1.40			
15.0	20.0	7.35	-3.78	1.32	-5.08	-2.49	0.89	-1.31	5.57	2.42			
20.0	25.0	8.01	-3.73	1.17	-5.41	-2.95	0.49	-1.28	5.82	3.81			
25.0	30.0	8.56	-3.76	0.92	-5.60	-4.57	0.40	-1.12	6.03	4.62			
30.0	35.0	9.06	-3.47	0.88	-5.56	-5.22	0.37	-0.99	6.37	5.32			
35.0	40.0	9.44	-3.36	0.92	-5.60	-5.70	0.43	-1.02	6.26	6.12			
40.0	45.0	9.79	-3.42	0.69	-5.47	-6.64	0.19	-1.03	6.25	6.49			
45.0	50.0	10.07	-3.03	0.75	-5.42	-7.28	0.48	-0.83	5.88	7.53			
50.0	55.0	10.27	-2.22	0.72	-5.05	-5.77	0.50	-0.78	5.77	8.15			
55.0	60.0	10.46	-1.79	0.73	-4.86	-6.49	0.32	-0.74	5.37	8.76			
60.0	65.0	10.80	-1.31	0.57	-4.38	-9.10	0.36	-0.80	5.11	9.38			
65.0	70.0	11.27	-0.76	0.51	-4.21	-7.06	0.25	-0.61	4.60	10.24			
70.0	75.0	11.65	-0.23	0.38	-3.88	-7.68	0.46	-0.58	4.37	10.77			
75.0	80.0	12.07	-0.04	0.26	-3.62	-8.35	0.23	-0.68	3.81	11.43			

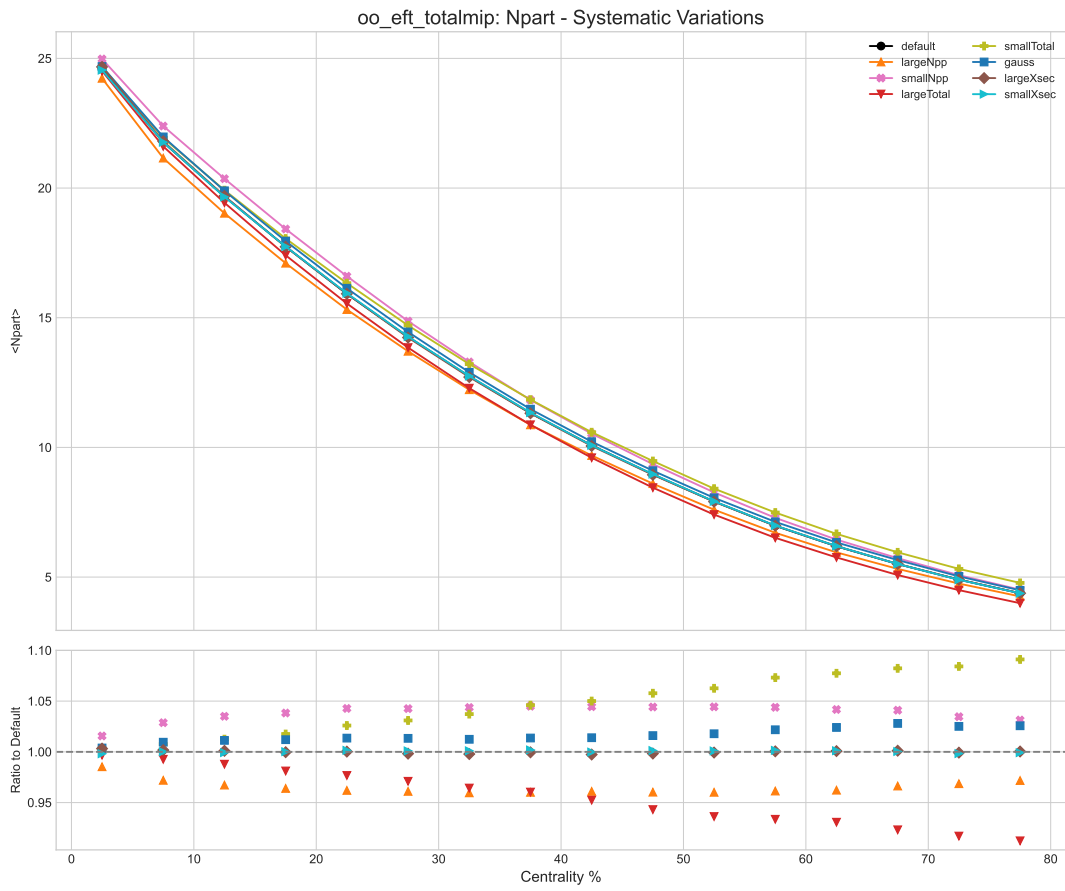
Integer Multiplicity Centrality Cuts Used			
Cent_Interval	Default	Large_Tot	Small_Tot
0-5%	97	96	97
5-10%	82	80	83
10-15%	71	69	72
15-20%	62	61	64
20-25%	55	53	57
25-30%	48	46	50
30-35%	42	40	44
35-40%	37	35	39
40-45%	32	30	34
45-50%	28	26	30
50-55%	24	23	26
55-60%	21	19	23
60-65%	18	16	20
65-70%	15	14	17
70-75%	13	11	15
75-80%	10	9	12

O+O EFT TotalMIP - Npart

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

min_c	max_c	value	stat_err	syst_err	tot_err	gauss	largeNpp	largeTotal	largeXsec	smallNpp	smallTotal	smallXsec
0.0	5.0	24.5926	0.0061	0.4114	0.4114	24.6857	24.2314	24.5128	24.6712	24.9774	24.6710	24.5471
5.0	10.0	21.7660	0.0064	0.6803	0.6803	21.9726	21.1550	21.6063	21.8034	22.3912	21.9327	21.7677
10.0	15.0	19.6724	0.0065	0.7648	0.7648	19.8948	19.0279	19.4332	19.6925	20.3613	19.9184	19.6585
15.0	20.0	17.7423	0.0065	0.7862	0.7862	17.9549	17.1025	17.4078	17.7361	18.4212	18.0538	17.7457
20.0	25.0	15.9202	0.0064	0.8251	0.8251	16.1356	15.3151	15.5493	15.9236	16.6009	16.3329	15.9438
25.0	30.0	14.2617	0.0062	0.7748	0.7749	14.4515	13.7061	13.8481	14.2360	14.8692	14.7029	14.2751
30.0	35.0	12.7389	0.0058	0.7484	0.7484	12.8964	12.2241	12.2843	12.7111	13.2957	13.2127	12.7520
35.0	40.0	11.3213	0.0055	0.7446	0.7446	11.4756	10.8695	10.8724	11.3156	11.8300	11.8422	11.3426
40.0	45.0	10.0810	0.0054	0.6899	0.6899	10.2213	9.6878	9.6011	10.0542	10.5294	10.5855	10.0790
45.0	50.0	8.9579	0.0050	0.6672	0.6673	9.1012	8.6018	8.4481	8.9428	9.3538	9.4753	8.9694
50.0	55.0	7.9142	0.0046	0.6308	0.6309	8.0553	7.5985	7.4096	7.9085	8.2655	8.4098	7.9213
55.0	60.0	6.9779	0.0043	0.6143	0.6143	7.1297	6.7083	6.5142	6.9813	7.2839	7.4883	6.9881
60.0	65.0	6.1870	0.0042	0.5642	0.5642	6.3358	5.9532	5.7580	6.1915	6.4451	6.6659	6.1939
65.0	70.0	5.5020	0.0036	0.5289	0.5289	5.6562	5.3166	5.0793	5.5081	5.7276	5.9548	5.5038
70.0	75.0	4.9041	0.0033	0.4632	0.4632	5.0272	4.7501	4.4971	4.9003	5.0740	5.3170	4.8959
75.0	80.0	4.3773	0.0030	0.4364	0.4364	4.4897	4.2535	3.9935	4.3788	4.5145	4.7759	4.3738

Percentage difference from default

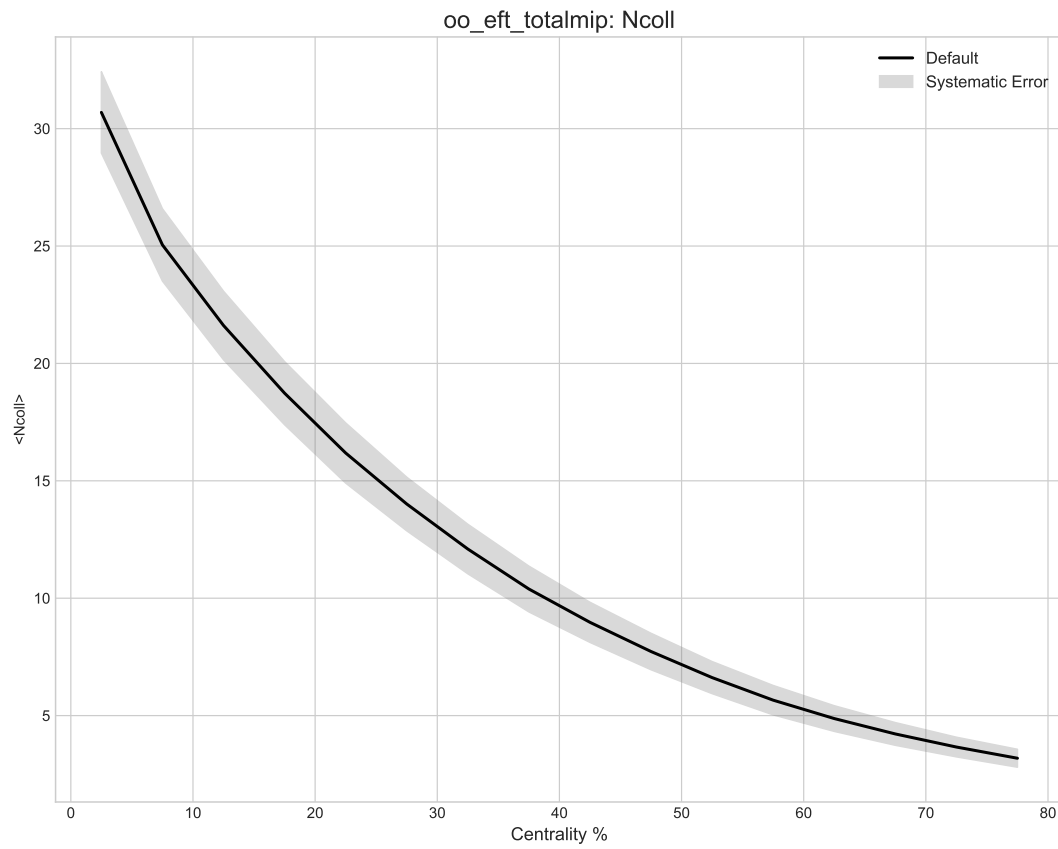
min_c	max_c	tot_sys_err(%)	gauss(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)
0.0	5.0	1.67	0.38	-1.47	-0.32	0.32	1.56	0.32	-0.19
5.0	10.0	3.13	0.95	-2.81	-0.73	0.17	2.87	0.77	0.01
10.0	15.0	3.89	1.13	-3.28	-1.22	0.10	3.50	1.25	-0.07
15.0	20.0	4.43	1.20	-3.61	-1.89	-0.03	3.83	1.76	0.02
20.0	25.0	5.18	1.35	-3.80	-2.33	0.02	4.28	2.59	0.15
25.0	30.0	5.43	1.33	-3.90	-2.90	-0.18	4.26	3.09	0.09
30.0	35.0	5.87	1.24	-4.04	-3.57	-0.22	4.37	3.72	0.10
35.0	40.0	6.58	1.36	-3.99	-3.96	-0.05	4.49	4.60	0.19
40.0	45.0	6.84	1.39	-3.90	-4.76	-0.27	4.45	5.00	-0.02
45.0	50.0	7.45	1.60	-3.98	-5.69	-0.17	4.42	5.78	0.13
50.0	55.0	7.97	1.78	-3.99	-6.38	-0.07	4.44	6.26	0.09
55.0	60.0	8.80	2.18	-3.86	-6.64	0.05	4.39	7.32	0.15
60.0	65.0	9.12	2.41	-3.78	-6.93	0.07	4.17	7.74	0.11
65.0	70.0	9.61	2.80	-3.37	-7.68	0.11	4.10	8.23	0.03
70.0	75.0	9.45	2.51	-3.14	-8.30	-0.08	3.46	8.42	-0.17
75.0	80.0	9.97	2.57	-2.83	-8.77	0.04	3.14	9.11	-0.08

Integer Multiplicity Centrality Cuts Used

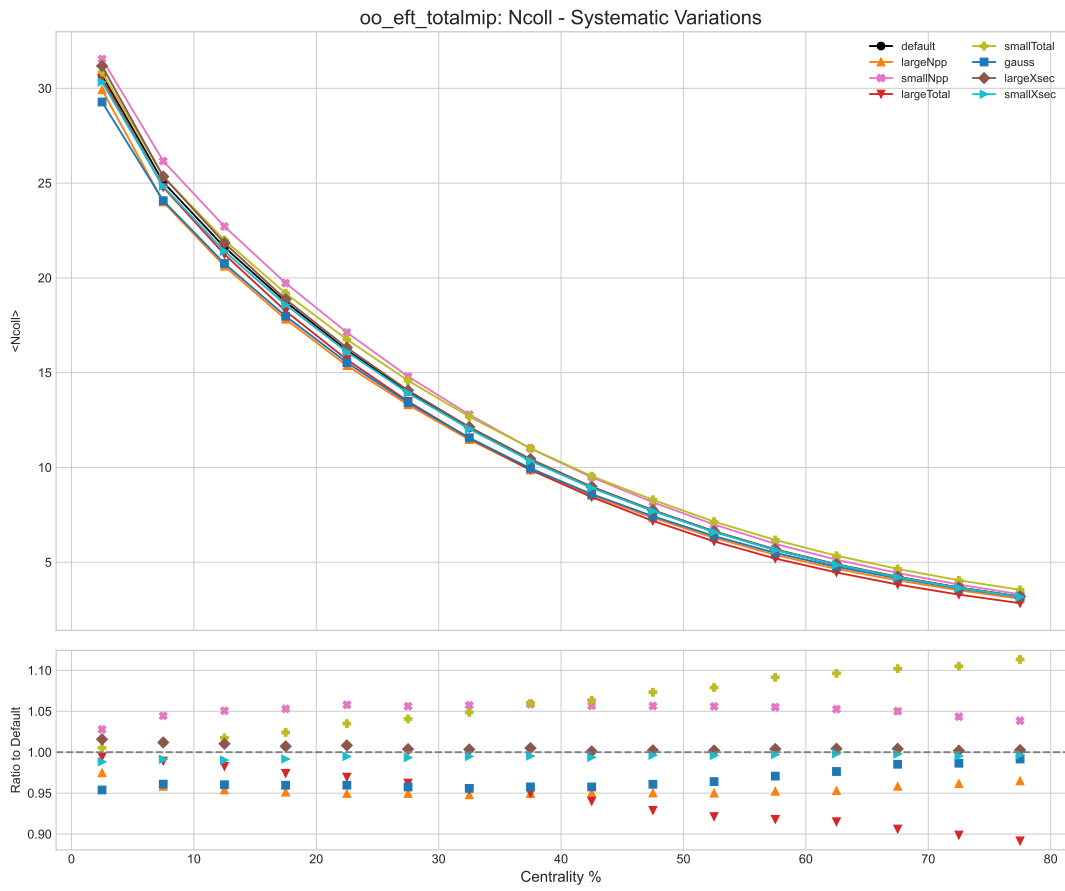
Cent_Interval	Default	Large_Tot	Small_Tot
0-5%	374	371	377
5-10%	325	321	329
10-15%	288	283	293
15-20%	256	250	262
20-25%	228	222	235
25-30%	203	196	210
30-35%	180	173	188
35-40%	159	152	167
40-45%	141	133	149
45-50%	124	116	132
50-55%	108	100	116
55-60%	94	87	102
60-65%	82	74	90
65-70%	70	62	78
70-75%	59	51	67
75-80%	49	41	57

O+O EFT TotalMIP - Ncoll

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

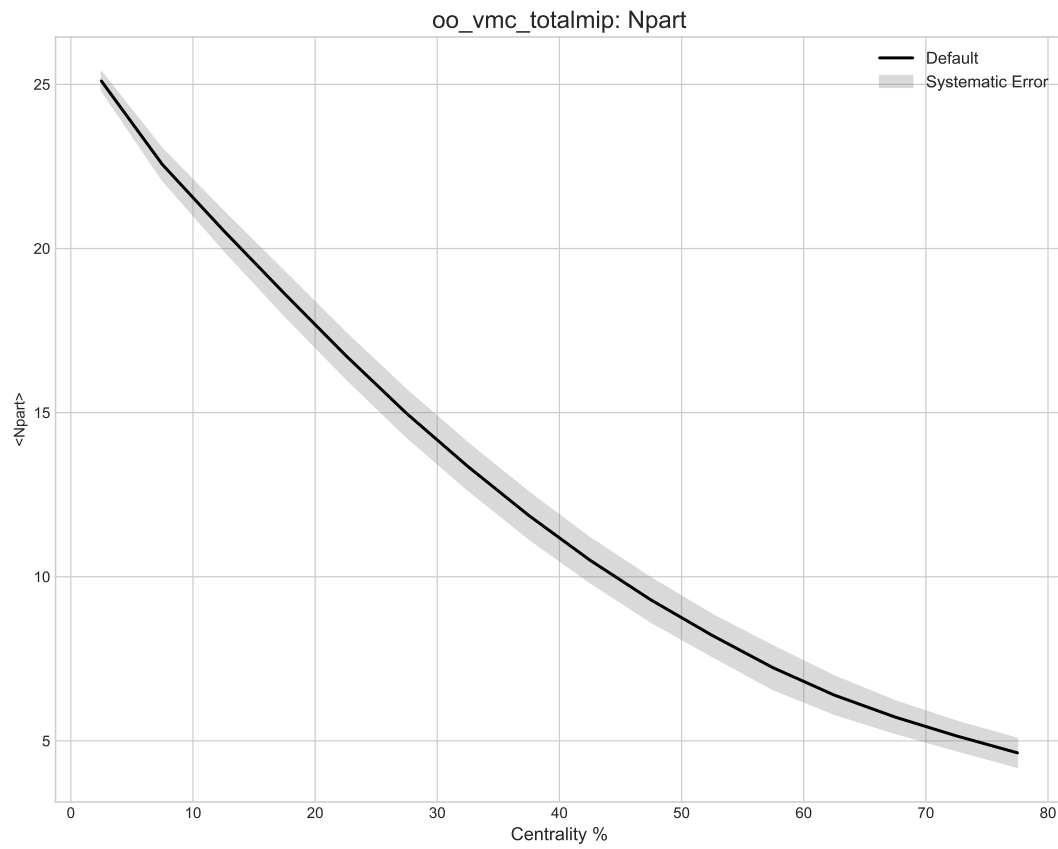
min_c	max_c	value	stat_err	syst_err	tot_err	gauss	largeNpp	largeTotal	largeXsec	smallNpp	smallTotal	smallXsec
0.0	5.0	30.6936	0.0147	1.7313	1.7314	29.2782	29.9225	30.5220	31.1779	31.5481	30.8575	30.3338
5.0	10.0	25.0478	0.0124	1.5400	1.5400	24.0707	24.0043	24.7777	25.3473	26.1621	25.3404	24.8282
10.0	15.0	21.6167	0.0114	1.4596	1.4597	20.7585	20.6198	21.2428	21.8391	22.7124	21.9962	21.4078
15.0	20.0	18.7329	0.0104	1.3450	1.3450	17.9772	17.8186	18.2512	18.8677	19.7238	19.1843	18.5779
20.0	25.0	16.1851	0.0096	1.2808	1.2808	15.5336	15.3706	15.6925	16.3203	17.1224	16.7502	16.1011
25.0	30.0	14.0127	0.0088	1.1425	1.1425	13.4172	13.3061	13.4841	14.0658	14.7989	14.5819	13.9205
30.0	35.0	12.0952	0.0079	1.0556	1.0556	11.5612	11.4660	11.5440	12.1363	12.7876	12.6832	12.0324
35.0	40.0	10.3917	0.0071	0.9757	0.9757	9.9515	9.8676	9.8715	10.4441	11.0012	11.0115	10.3423
40.0	45.0	8.9729	0.0066	0.8550	0.8550	8.5924	8.5249	8.4362	8.9821	9.4818	9.5422	8.9165
45.0	50.0	7.7294	0.0059	0.7766	0.7766	7.4269	7.3441	7.1806	7.7472	8.1656	8.2955	7.7007
50.0	55.0	6.6190	0.0052	0.6837	0.6837	6.3812	6.2878	6.0985	6.6329	6.9891	7.1417	6.5927
55.0	60.0	5.6574	0.0047	0.6259	0.6259	5.4923	5.3875	5.1937	5.6795	5.9683	6.1745	5.6410
60.0	65.0	4.8741	0.0043	0.5474	0.5474	4.7594	4.6459	4.4604	4.8950	5.1302	5.3437	4.8653
65.0	70.0	4.2184	0.0036	0.4839	0.4839	4.1563	4.0427	3.8226	4.2362	4.4297	4.6488	4.2086
70.0	75.0	3.6609	0.0032	0.4192	0.4193	3.6115	3.5207	3.2908	3.6679	3.8199	4.0453	3.6435
75.0	80.0	3.1834	0.0028	0.3821	0.3821	3.1569	3.0716	2.8388	3.1919	3.3059	3.5441	3.1727

Percentage difference from default										
min_c	max_c	tot_sys_err(%)	gauss(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)	
0.0	5.0	5.64	-4.61	-2.51	-0.56	1.58	2.78	0.53	-1.17	
5.0	10.0	6.15	-3.90	-4.17	-1.08	1.20	4.45	1.17	-0.88	
10.0	15.0	6.75	-3.97	-4.61	-1.73	1.03	5.07	1.76	-0.97	
15.0	20.0	7.18	-4.03	-4.88	-2.57	0.72	5.29	2.41	-0.83	
20.0	25.0	7.91	-4.03	-5.03	-3.04	0.83	5.79	3.49	-0.52	
25.0	30.0	8.15	-4.25	-5.04	-3.77	0.38	5.61	4.06	-0.66	
30.0	35.0	8.73	-4.41	-5.20	-4.56	0.34	5.72	4.86	-0.52	
35.0	40.0	9.39	-4.24	-5.04	-5.01	0.50	5.86	5.96	-0.48	
40.0	45.0	9.53	-4.24	-4.99	-5.98	0.10	5.67	6.34	-0.63	
45.0	50.0	10.05	-3.91	-4.99	-7.10	0.23	5.64	7.32	-0.37	
50.0	55.0	10.33	-3.59	-5.00	-7.86	0.21	5.59	7.90	-0.40	
55.0	60.0	11.06	-2.92	-4.77	-8.20	0.39	5.49	9.14	-0.29	
60.0	65.0	11.23	-2.35	-4.68	-8.49	0.43	5.25	9.63	-0.18	
65.0	70.0	11.47	-1.47	-4.16	-9.38	0.42	5.01	10.20	-0.23	
70.0	75.0	11.45	-1.35	-3.83	-10.11	0.19	4.34	10.50	-0.48	
75.0	80.0	12.00	-0.83	-3.51	-10.82	0.27	3.85	11.33	-0.34	

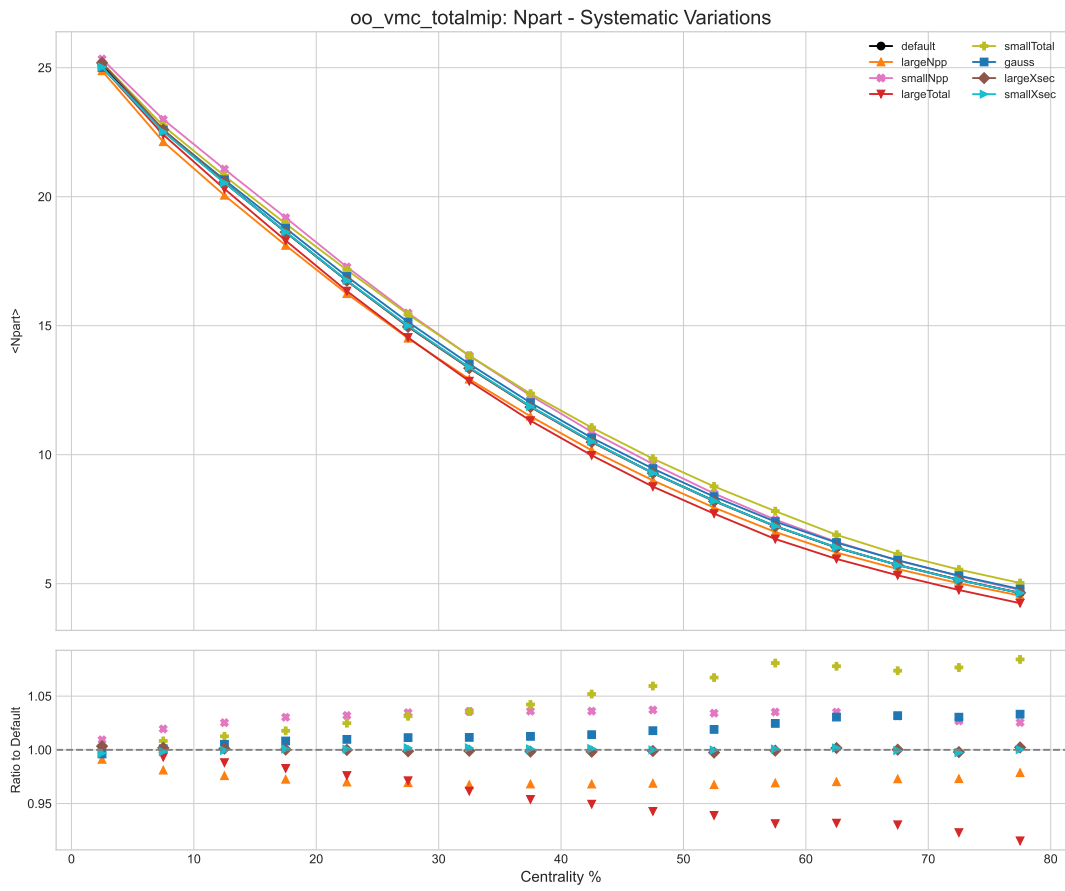
Integer Multiplicity Centrality Cuts Used			
Cent_Interval	Default	Large_Tot	Small_Tot
0-5%	374	371	377
5-10%	325	321	329
10-15%	288	283	293
15-20%	256	250	262
20-25%	228	222	235
25-30%	203	196	210
30-35%	180	173	188
35-40%	159	152	167
40-45%	141	133	149
45-50%	124	116	132
50-55%	108	100	116
55-60%	94	87	102
60-65%	82	74	90
65-70%	70	62	78
70-75%	59	51	67
75-80%	49	41	57

O+O VMC TotalMIP - Npart

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

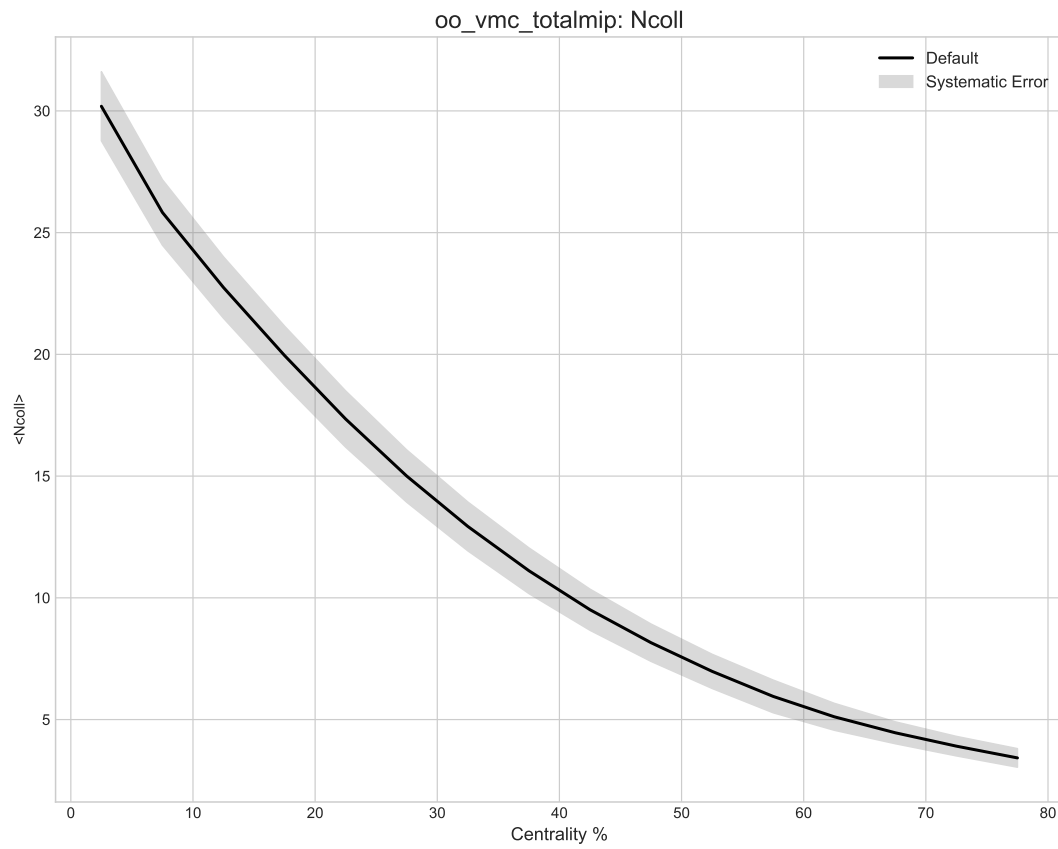
min_c	max_c	value	stat_err	syst_err	tot_err	gauss	largeNpp	largeTotal	largeXsec	smallNpp	smallTotal	smallXsec
0.0	5.0	25.1029	0.0065	0.2826	0.2826	25.0059	24.8802	25.0297	25.1880	25.3380	25.1917	25.0230
5.0	10.0	22.5563	0.0075	0.4805	0.4806	22.5803	22.1300	22.4042	22.5986	22.9961	22.7436	22.5175
10.0	15.0	20.5490	0.0080	0.5903	0.5903	20.6547	20.0543	20.3039	20.5735	21.0686	20.8072	20.5389
15.0	20.0	18.6227	0.0081	0.6723	0.6723	18.7764	18.1124	18.3007	18.6272	19.1874	18.9531	18.6362
20.0	25.0	16.7433	0.0079	0.6971	0.6971	16.9066	16.2413	16.3423	16.7413	17.2785	17.1588	16.7549
25.0	30.0	14.9723	0.0079	0.7193	0.7193	15.1424	14.5143	14.5418	14.9525	15.4906	15.4402	15.0000
30.0	35.0	13.3614	0.0075	0.7181	0.7181	13.5164	12.9276	12.8485	13.3499	13.8389	13.8381	13.3862
35.0	40.0	11.8657	0.0071	0.7107	0.7108	12.0147	11.4879	11.3182	11.8460	12.2933	12.3665	11.8763
40.0	45.0	10.5055	0.0067	0.6808	0.6808	10.6538	10.1710	9.9742	10.4868	10.8839	11.0514	10.5181
45.0	50.0	9.2936	0.0063	0.6722	0.6722	9.4594	9.0037	8.7614	9.2835	9.6389	9.8459	9.2926
50.0	55.0	8.2156	0.0058	0.6395	0.6395	8.3719	7.9490	7.7143	8.1949	8.4959	8.7683	8.2137
55.0	60.0	7.2275	0.0057	0.6613	0.6613	7.4055	7.0046	6.7310	7.2221	7.4818	7.8113	7.2329
60.0	65.0	6.3961	0.0051	0.5800	0.5800	6.5912	6.2061	5.9597	6.4069	6.6207	6.8939	6.4073
65.0	70.0	5.7236	0.0046	0.4935	0.4935	5.9056	5.5688	5.3243	5.7245	5.9040	6.1452	5.7199
70.0	75.0	5.1525	0.0042	0.4490	0.4490	5.3092	5.0140	4.7555	5.1418	5.2912	5.5476	5.1394
75.0	80.0	4.6364	0.0039	0.4381	0.4381	4.7903	4.5380	4.2437	4.6470	4.7544	5.0267	4.6375

Percentage difference from default										
min_c	max_c	tot_sys_err(%)	gauss(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)	
0.0	5.0	1.13	-0.39	-0.89	-0.29	0.34	0.94	0.35	-0.32	
5.0	10.0	2.13	0.11	-1.89	-0.67	0.19	1.95	0.83	-0.17	
10.0	15.0	2.87	0.51	-2.41	-1.19	0.12	2.53	1.26	-0.05	
15.0	20.0	3.61	0.83	-2.74	-1.73	0.02	3.03	1.77	0.07	
20.0	25.0	4.16	0.98	-3.00	-2.39	-0.01	3.20	2.48	0.07	
25.0	30.0	4.80	1.14	-3.06	-2.88	-0.13	3.46	3.13	0.19	
30.0	35.0	5.37	1.16	-3.25	-3.84	-0.09	3.57	3.57	0.19	
35.0	40.0	5.99	1.26	-3.18	-4.61	-0.17	3.60	4.22	0.09	
40.0	45.0	6.48	1.41	-3.18	-5.06	-0.18	3.60	5.20	0.12	
45.0	50.0	7.23	1.78	-3.12	-5.73	-0.11	3.72	5.94	-0.01	
50.0	55.0	7.78	1.90	-3.24	-6.10	-0.25	3.41	6.73	-0.02	
55.0	60.0	9.15	2.46	-3.08	-6.87	-0.07	3.52	8.08	0.08	
60.0	65.0	9.07	3.05	-2.97	-6.82	0.17	3.51	7.78	0.18	
65.0	70.0	8.62	3.18	-2.70	-6.98	0.02	3.15	7.37	-0.06	
70.0	75.0	8.71	3.04	-2.69	-7.71	-0.21	2.69	7.67	-0.25	
75.0	80.0	9.45	3.32	-2.12	-8.47	0.23	2.54	8.42	0.02	

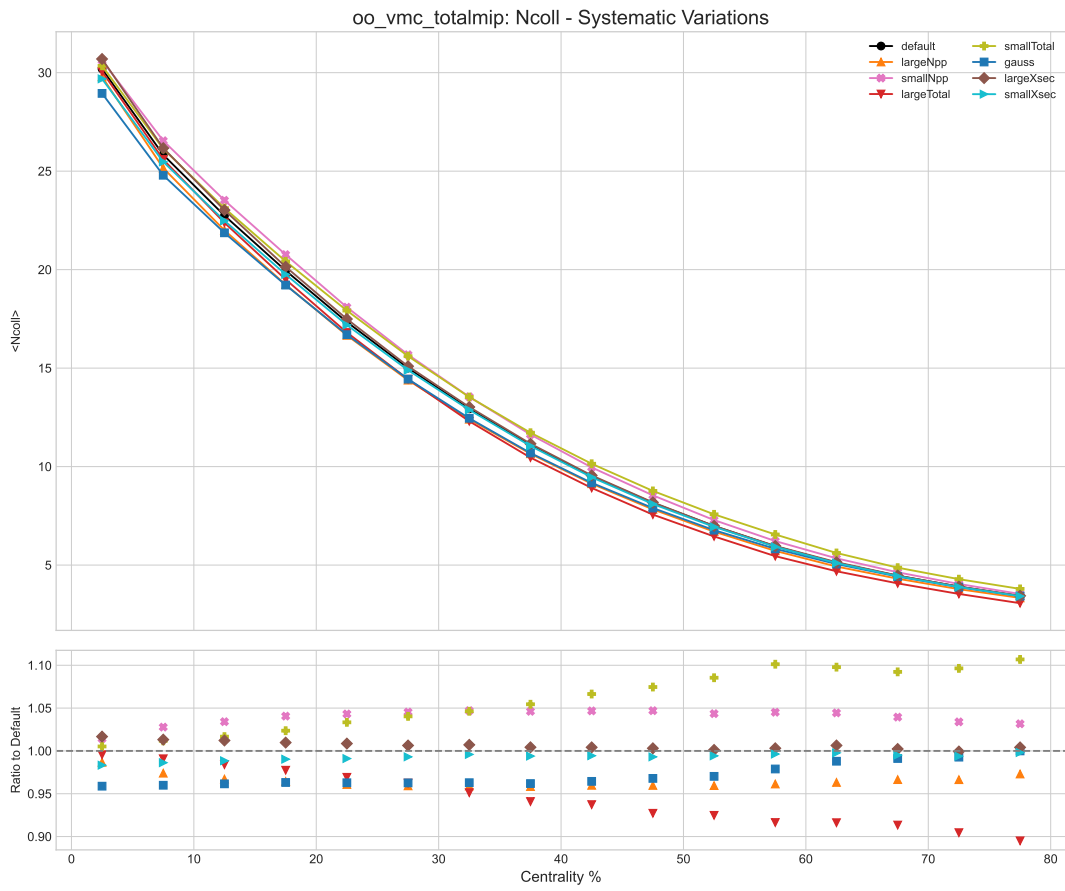
Integer Multiplicity Centrality Cuts Used			
Cent_Interval	Default	Large_Tot	Small_Tot
0-5%	372	369	376
5-10%	322	318	327
10-15%	285	280	290
15-20%	253	247	259
20-25%	224	218	231
25-30%	199	192	206
30-35%	176	168	183
35-40%	155	147	163
40-45%	136	128	144
45-50%	119	111	127
50-55%	103	96	112
55-60%	90	82	98
60-65%	77	69	85
65-70%	65	57	73
70-75%	54	46	62
75-80%	44	35	52

O+O VMC TotalMIP - Ncoll

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

min_c	max_c	value	stat_err	syst_err	tot_err	gauss	largeNpp	largeTotal	largeXsec	smallNpp	smallTotal	smallXsec
0.0	5.0	30.1911	0.0130	1.4181	1.4182	28.9478	29.7901	30.0578	30.6955	30.6226	30.3477	29.6874
5.0	10.0	25.8281	0.0130	1.3438	1.3439	24.7921	25.1558	25.5912	26.1687	26.5451	26.1322	25.4731
10.0	15.0	22.7445	0.0129	1.2598	1.2599	21.8700	21.9961	22.3801	23.0215	23.5203	23.1237	22.4826
15.0	20.0	19.9516	0.0123	1.2071	1.2072	19.2159	19.2397	19.5016	20.1457	20.7618	20.4225	19.7583
20.0	25.0	17.3435	0.0115	1.1552	1.1552	16.6989	16.6631	16.8089	17.4922	18.0917	17.9223	17.1886
25.0	30.0	15.0008	0.0110	1.0737	1.0737	14.4407	14.3885	14.4374	15.0959	15.6806	15.6059	14.8969
30.0	35.0	12.9336	0.0100	1.0031	1.0032	12.4536	12.4037	12.3028	13.0254	13.5416	13.5321	12.8805
35.0	40.0	11.1139	0.0091	0.9380	0.9380	10.6900	10.6507	10.4568	11.1610	11.6275	11.7194	11.0465
40.0	45.0	9.5127	0.0082	0.8451	0.8451	9.1743	9.1304	8.9160	9.5522	9.9572	10.1445	9.4593
45.0	50.0	8.1574	0.0075	0.7674	0.7674	7.8955	7.8287	7.5633	8.1825	8.5406	8.7659	8.1007
50.0	55.0	6.9806	0.0065	0.7018	0.7019	6.7731	6.6977	6.4553	6.9886	7.2844	7.5769	6.9404
55.0	60.0	5.9512	0.0062	0.6718	0.6718	5.8256	5.7212	5.4535	5.9695	6.2198	6.5535	5.9292
60.0	65.0	5.1146	0.0053	0.5538	0.5538	5.0530	4.9264	4.6859	5.1471	5.3418	5.6149	5.1030
65.0	70.0	4.4575	0.0047	0.4497	0.4497	4.4181	4.3080	4.0716	4.4676	4.6333	4.8689	4.4355
70.0	75.0	3.9084	0.0041	0.4011	0.4011	3.8799	3.7771	3.5352	3.9058	4.0409	4.2851	3.8843
75.0	80.0	3.4249	0.0037	0.3819	0.3819	3.4250	3.3326	3.0653	3.4391	3.5335	3.7908	3.4186

Percentage difference from default

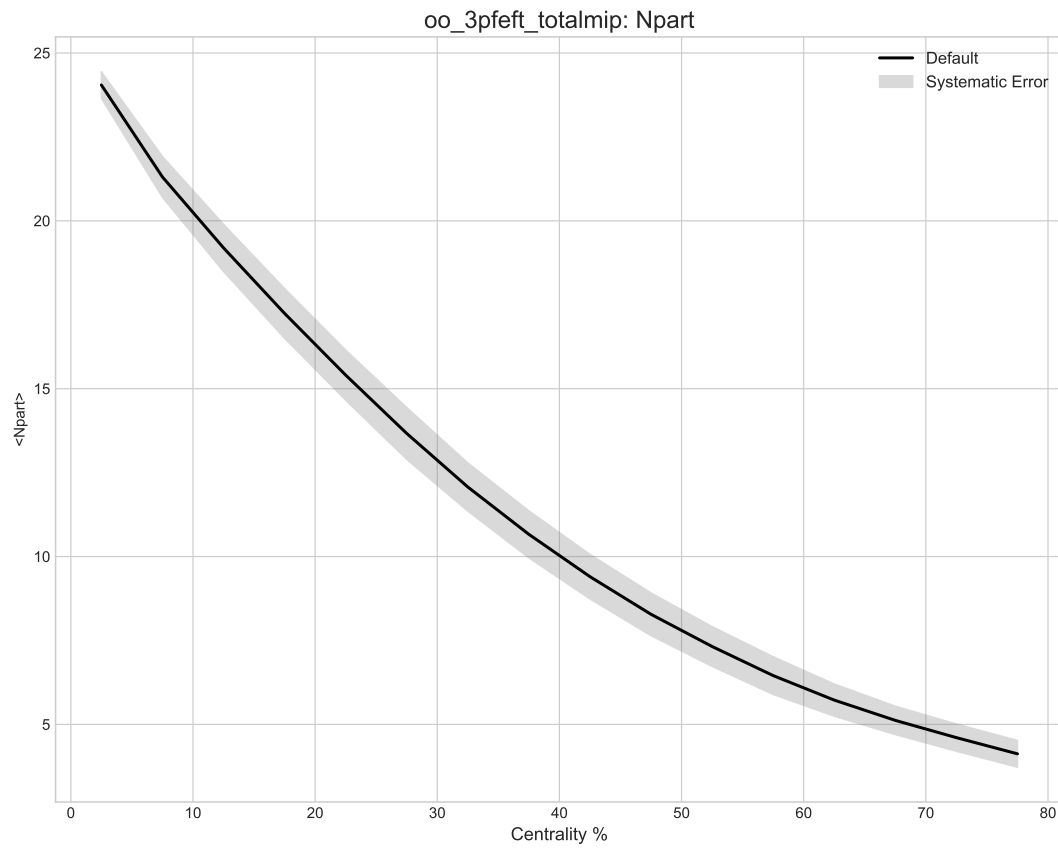
min_c	max_c	tot_sys_err(%)	gauss(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)
0.0	5.0	4.70	-4.12	-1.33	-0.44	1.67	1.43	0.52	-1.67
5.0	10.0	5.20	-4.01	-2.60	-0.92	1.32	2.78	1.18	-1.37
10.0	15.0	5.54	-3.85	-3.29	-1.60	1.22	3.41	1.67	-1.15
15.0	20.0	6.05	-3.69	-3.57	-2.26	0.97	4.06	2.36	-0.97
20.0	25.0	6.66	-3.72	-3.92	-3.08	0.86	4.31	3.34	-0.89
25.0	30.0	7.16	-3.73	-4.08	-3.76	0.63	4.53	4.03	-0.69
30.0	35.0	7.76	-3.71	-4.10	-4.88	0.71	4.70	4.63	-0.41
35.0	40.0	8.44	-3.81	-4.17	-5.91	0.42	4.62	5.45	-0.61
40.0	45.0	8.88	-3.56	-4.02	-6.27	0.42	4.67	6.64	-0.56
45.0	50.0	9.41	-3.21	-4.03	-7.28	0.31	4.70	7.46	-0.69
50.0	55.0	10.05	-2.97	-4.05	-7.52	0.11	4.35	8.54	-0.58
55.0	60.0	11.29	-2.11	-3.86	-8.36	0.31	4.51	10.12	-0.37
60.0	65.0	10.83	-1.21	-3.68	-8.38	0.64	4.44	9.78	-0.23
65.0	70.0	10.09	-0.88	-3.35	-8.66	0.23	3.94	9.23	-0.49
70.0	75.0	10.26	-0.73	-3.36	-9.55	-0.07	3.39	9.64	-0.62
75.0	80.0	11.15	0.00	-2.70	-10.50	0.41	3.17	10.68	-0.19

Integer Multiplicity Centrality Cuts Used

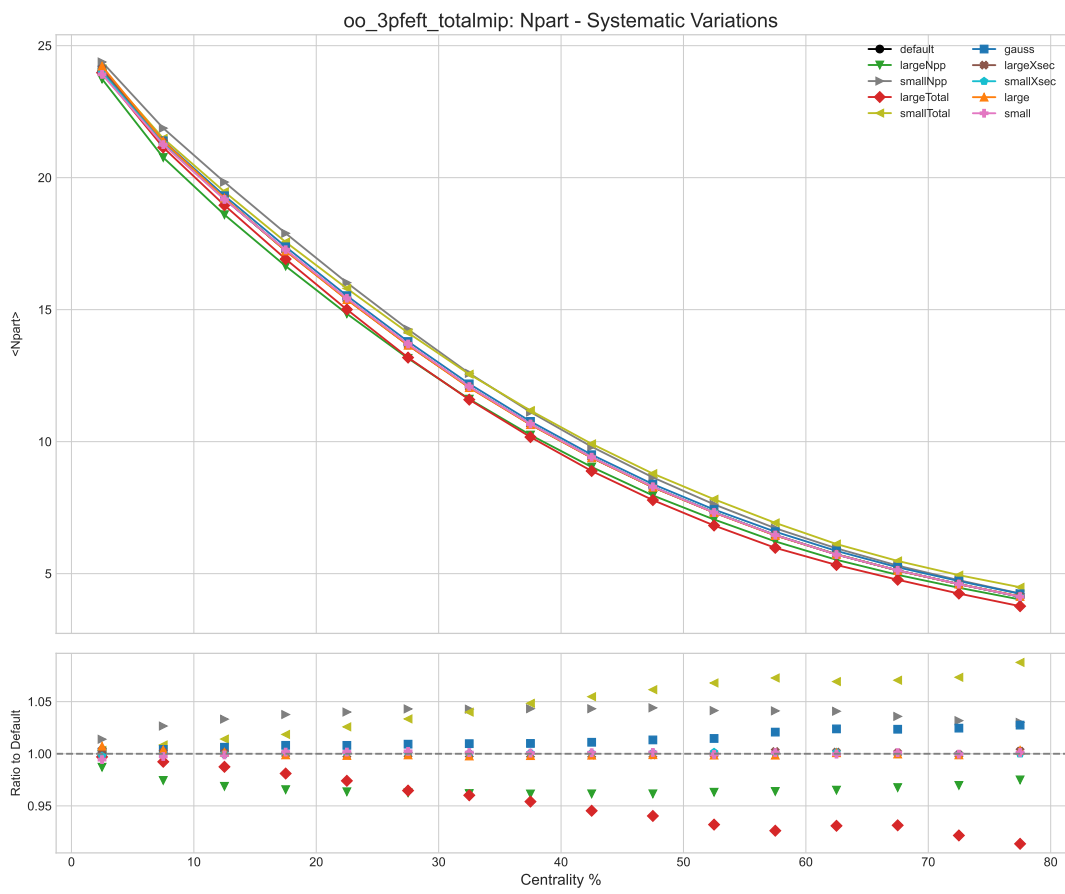
Cent_Interval	Default	Large_Tot	Small_Tot
0-5%	372	369	376
5-10%	322	318	327
10-15%	285	280	290
15-20%	253	247	259
20-25%	224	218	231
25-30%	199	192	206
30-35%	176	168	183
35-40%	155	147	163
40-45%	136	128	144
45-50%	119	111	127
50-55%	103	96	112
55-60%	90	82	98
60-65%	77	69	85
65-70%	65	57	73
70-75%	54	46	62
75-80%	44	35	52

O+O 3PFEFT TotalMIP - Npart

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

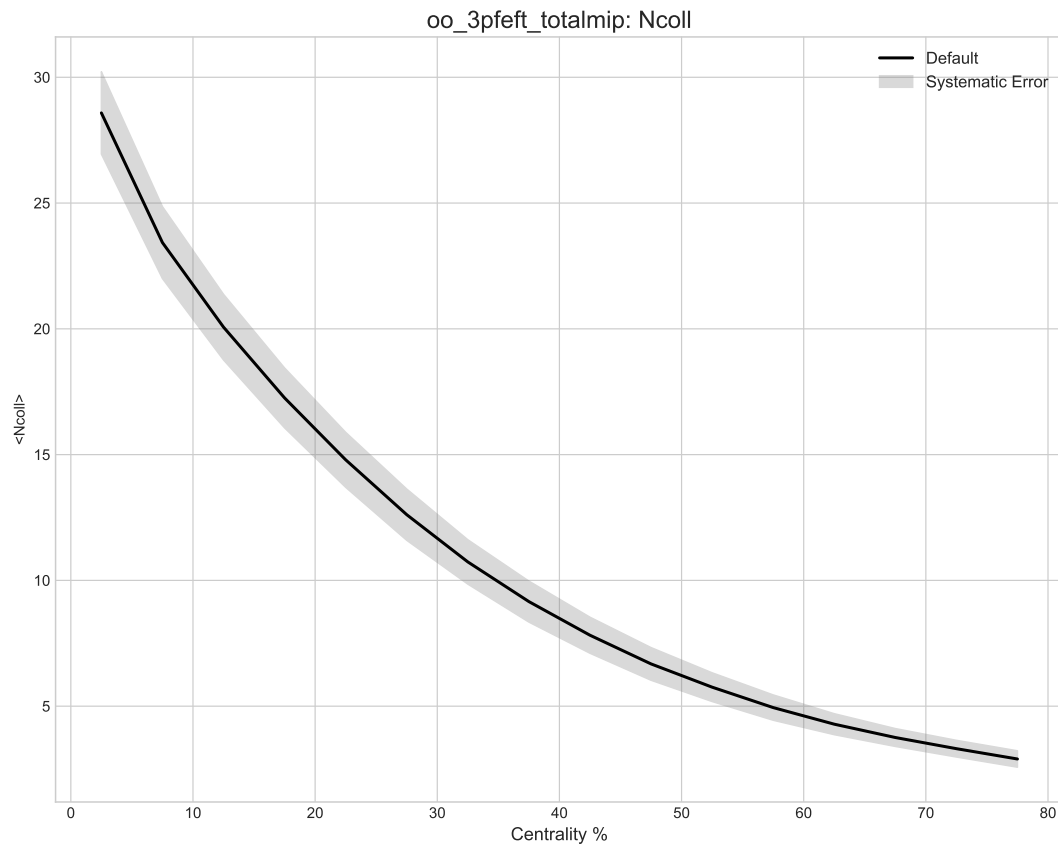
min_c	max_c	value	stat_err	syst_err	tot_err	gauss	large	largeNpp	largeTotal	largeXsec	small	smallNpp	smallTotal	smallXsec
0.0	5.0	24.0473	0.0060	0.3950	0.3951	24.0845	24.2214	23.7368	23.9749	24.1268	23.9122	24.3821	24.1247	23.9937
5.0	10.0	21.3074	0.0064	0.6072	0.6072	21.3969	21.3883	20.7628	21.1435	21.3383	21.2520	21.8738	21.4875	21.2995
10.0	15.0	19.1953	0.0067	0.7031	0.7031	19.3155	19.2240	18.5961	18.9551	19.2061	19.1742	19.8322	19.4661	19.1939
15.0	20.0	17.2426	0.0068	0.7406	0.7406	17.3822	17.2236	16.6490	16.9158	17.2370	17.2700	17.8917	17.5623	17.2527
20.0	25.0	15.4059	0.0065	0.7458	0.7459	15.5299	15.3806	14.8446	15.0061	15.3916	15.4314	16.0226	15.8038	15.4143
25.0	30.0	13.6672	0.0063	0.7718	0.7718	13.7930	13.6516	13.1562	13.1846	13.6588	13.6991	14.2553	14.1240	13.6707
30.0	35.0	12.0718	0.0059	0.7168	0.7168	12.1888	12.0455	11.6112	11.5909	12.0623	12.0858	12.5882	12.5541	12.0752
35.0	40.0	10.6619	0.0058	0.7003	0.7004	10.7675	10.6426	10.2510	10.1727	10.6511	10.6689	11.1230	11.1779	10.6698
40.0	45.0	9.4015	0.0053	0.6641	0.6641	9.5051	9.3884	9.0401	8.8874	9.3911	9.4107	9.8076	9.9162	9.4071
45.0	50.0	8.2799	0.0051	0.6360	0.6360	8.3899	8.2724	7.9619	7.7860	8.2710	8.2906	8.6448	8.7889	8.2878
50.0	55.0	7.3208	0.0047	0.5927	0.5927	7.4281	7.3112	7.0507	6.8232	7.3186	7.3163	7.6241	7.8179	7.3318
55.0	60.0	6.4542	0.0044	0.5618	0.5618	6.5882	6.4453	6.2210	5.9778	6.4679	6.4612	6.7197	6.9234	6.4551
60.0	65.0	5.7243	0.0040	0.4799	0.4800	5.8608	5.7297	5.5246	5.3287	5.7331	5.7221	5.9576	6.1207	5.7327
65.0	70.0	5.1199	0.0037	0.4225	0.4226	5.2403	5.1185	4.9538	4.7685	5.1238	5.1255	5.3033	5.4809	5.1168
70.0	75.0	4.6060	0.0033	0.4056	0.4056	4.7192	4.6010	4.4663	4.2451	4.6023	4.6043	4.7524	4.9438	4.6030
75.0	80.0	4.1237	0.0029	0.3988	0.3988	4.2370	4.1375	4.0204	3.7675	4.1341	4.1286	4.2477	4.4850	4.1238

Percentage difference from default											
min_c	max_c	tot_sys_err(%)	gauss(%)	large(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	small(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)
0.0	5.0	1.64	0.15	0.72	-1.29	-0.30	0.33	-0.56	1.39	0.32	-0.22
5.0	10.0	2.85	0.42	0.38	-2.56	-0.77	0.14	-0.26	2.66	0.85	-0.04
10.0	15.0	3.66	0.63	0.15	-3.12	-1.25	0.06	-0.11	3.32	1.41	-0.01
15.0	20.0	4.30	0.81	-0.11	-3.44	-1.90	-0.03	0.16	3.76	1.85	0.06
20.0	25.0	4.84	0.80	-0.16	-3.64	-2.60	-0.09	0.17	4.00	2.58	0.05
25.0	30.0	5.65	0.92	-0.11	-3.74	-3.53	-0.06	0.23	4.30	3.34	0.03
30.0	35.0	5.94	0.97	-0.22	-3.82	-3.98	-0.08	0.12	4.28	4.00	0.03
35.0	40.0	6.57	0.99	-0.18	-3.85	-4.59	-0.10	0.06	4.32	4.84	0.07
40.0	45.0	7.06	1.10	-0.14	-3.84	-5.47	-0.11	0.10	4.32	5.48	0.06
45.0	50.0	7.68	1.33	-0.09	-3.84	-5.97	-0.11	0.13	4.41	6.15	0.09
50.0	55.0	8.10	1.47	-0.13	-3.69	-6.80	-0.03	-0.06	4.14	6.79	0.15
55.0	60.0	8.70	2.08	-0.14	-3.61	-7.38	0.21	0.11	4.11	7.27	0.02
60.0	65.0	8.38	2.39	0.10	-3.49	-6.91	0.15	-0.04	4.08	6.93	0.15
65.0	70.0	8.25	2.35	-0.03	-3.24	-6.86	0.08	0.11	3.58	7.05	-0.06
70.0	75.0	8.81	2.46	-0.11	-3.03	-7.84	-0.08	-0.04	3.18	7.33	-0.07
75.0	80.0	9.67	2.75	0.33	-2.51	-8.64	0.25	0.12	3.01	8.76	0.00

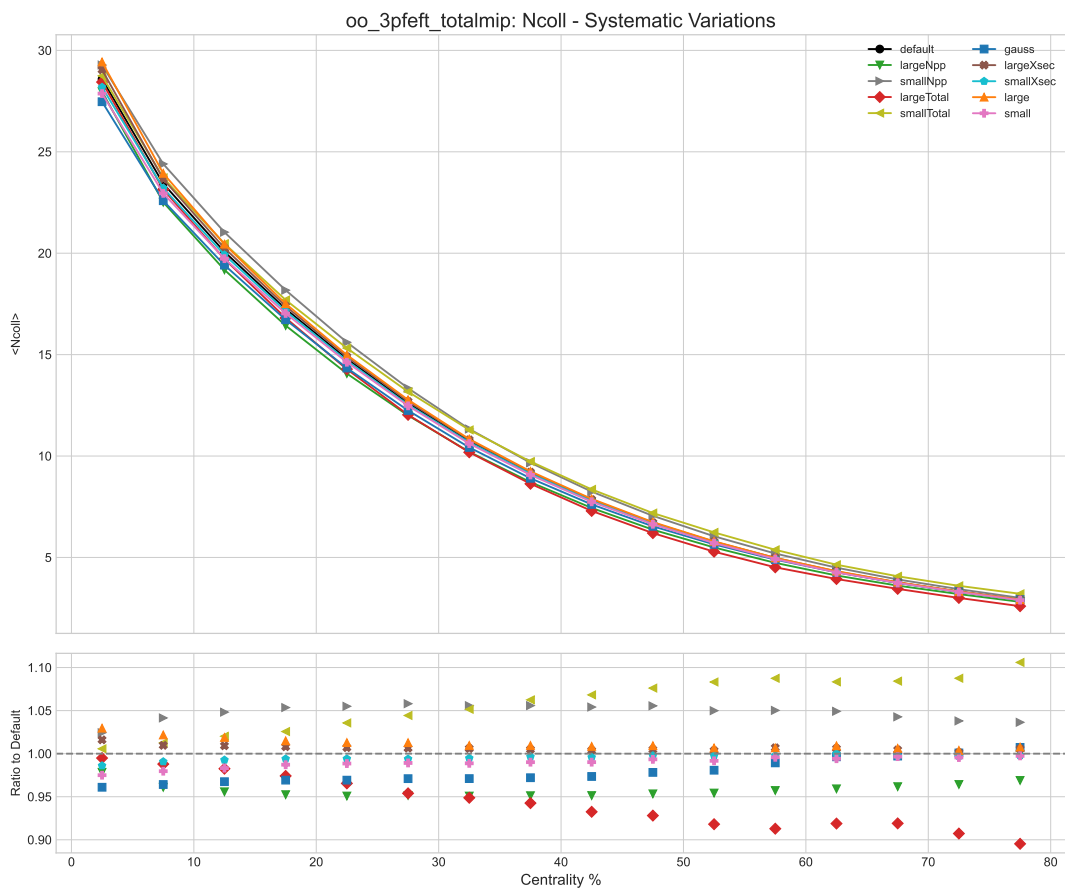
Integer Multiplicity Centrality Cuts Used			
Cent_Interval	Default	Large_Tot	Small_Tot
0-5%	371	368	374
5-10%	320	316	325
10-15%	282	277	287
15-20%	250	244	256
20-25%	221	214	228
25-30%	195	187	202
30-35%	171	164	179
35-40%	151	143	159
40-45%	132	124	140
45-50%	116	108	124
50-55%	101	93	109
55-60%	88	80	95
60-65%	76	69	83
65-70%	65	58	72
70-75%	55	47	62
75-80%	45	37	53

O+O 3PFEFT TotalMIP - Ncoll

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

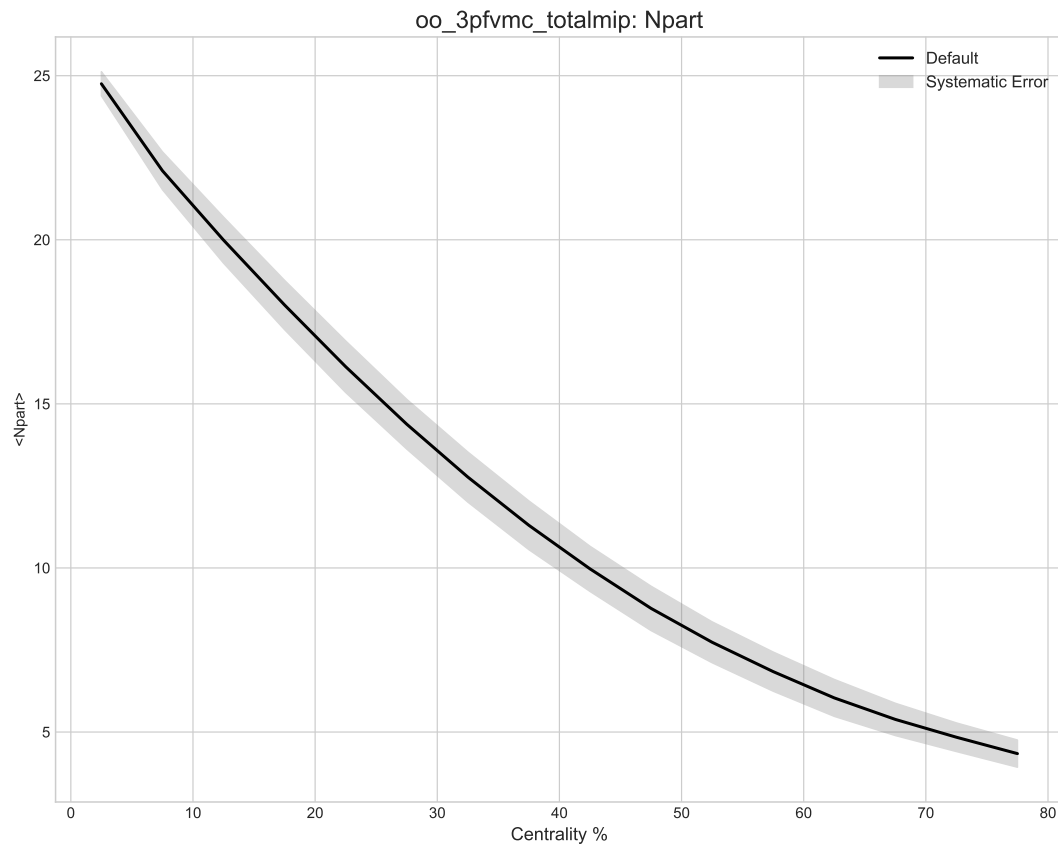
min_c	max_c	value	stat_err	syst_err	tot_err	gauss	large	largeNpp	largeTotal	largeXsec	small	smallNpp	smallTotal	smallXsec
0.0	5.0	28.5845	0.0137	1.6365	1.6366	27.4693	29.4291	27.9660	28.4385	29.0442	27.8699	29.2799	28.7465	28.1859
5.0	10.0	23.4284	0.0120	1.4299	1.4299	22.5891	23.9364	22.5142	23.1449	23.6536	22.9516	24.4008	23.7216	23.2117
10.0	15.0	20.0679	0.0112	1.3049	1.3049	19.4150	20.4454	19.1798	19.7175	20.2521	19.7312	21.0352	20.4733	19.9209
15.0	20.0	17.2532	0.0104	1.1912	1.1912	16.7228	17.5100	16.4320	16.8072	17.3920	17.0319	18.1770	17.6964	17.1495
20.0	25.0	14.7939	0.0093	1.0931	1.0931	14.3388	14.9841	14.0662	14.2831	14.8972	14.6243	15.6073	15.3222	14.7061
25.0	30.0	12.6123	0.0084	1.0193	1.0193	12.2464	12.7731	12.0013	12.0315	12.6926	12.4764	13.3440	13.1715	12.5349
30.0	35.0	10.7347	0.0074	0.8825	0.8825	10.4239	10.8369	10.2066	10.1846	10.7983	10.6144	11.3331	11.2875	10.6784
35.0	40.0	9.1577	0.0069	0.8152	0.8152	8.9019	9.2443	8.7109	8.6318	9.2022	9.0672	9.6688	9.7300	9.1254
40.0	45.0	7.8212	0.0059	0.7170	0.7171	7.6141	7.8869	7.4408	7.2932	7.8501	7.7441	8.2446	8.3551	7.7889
45.0	50.0	6.6802	0.0054	0.6495	0.6496	6.5349	6.7420	6.3687	6.1998	6.7118	6.6362	7.0511	7.1885	6.6651
50.0	55.0	5.7561	0.0047	0.5717	0.5717	5.6455	5.7943	5.4930	5.2848	5.7760	5.7083	6.0434	6.2351	5.7430
55.0	60.0	4.9444	0.0043	0.5038	0.5038	4.8916	4.9775	4.7329	4.5137	4.9782	4.9236	5.1929	5.3770	4.9333
60.0	65.0	4.2843	0.0037	0.4176	0.4176	4.2687	4.3234	4.1094	3.9365	4.3094	4.2591	4.4950	4.6414	4.2843
65.0	70.0	3.7546	0.0033	0.3557	0.3557	3.7441	3.7776	3.6107	3.4506	3.7708	3.7416	3.9150	4.0706	3.7445
70.0	75.0	3.3109	0.0029	0.3323	0.3323	3.3136	3.3234	3.1924	3.0038	3.3146	3.2961	3.4367	3.6007	3.3016
75.0	80.0	2.9003	0.0025	0.3267	0.3267	2.9214	2.9222	2.8102	2.5967	2.9162	2.8929	3.0059	3.2076	2.8942

Percentage difference from default											
min_c	max_c	tot_sys_err(%)	gauss(%)	large(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	small(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)
0.0	5.0	5.73	-3.90	2.95	-2.16	-0.51	1.61	-2.50	2.43	0.57	-1.39
5.0	10.0	6.10	-3.58	2.17	-3.90	-1.21	0.96	-2.04	4.15	1.25	-0.92
10.0	15.0	6.50	-3.25	1.88	-4.43	-1.75	0.92	-1.68	4.82	2.02	-0.73
15.0	20.0	6.90	-3.07	1.49	-4.76	-2.58	0.80	-1.28	5.35	2.57	-0.60
20.0	25.0	7.39	-3.08	1.29	-4.92	-3.45	0.70	-1.15	5.50	3.57	-0.59
25.0	30.0	8.08	-2.90	1.27	-4.84	-4.61	0.64	-1.08	5.80	4.43	-0.61
30.0	35.0	8.22	-2.90	0.95	-4.92	-5.12	0.59	-1.12	5.57	5.15	-0.53
35.0	40.0	8.90	-2.79	0.95	-4.88	-5.74	0.49	-0.99	5.58	6.25	-0.35
40.0	45.0	9.17	-2.65	0.84	-4.86	-6.75	0.37	-0.99	5.41	6.83	-0.41
45.0	50.0	9.72	-2.18	0.93	-4.66	-7.19	0.47	-0.66	5.55	7.61	-0.23
50.0	55.0	9.93	-1.92	0.66	-4.57	-8.19	0.34	-0.83	4.99	8.32	-0.23
55.0	60.0	10.19	-1.07	0.67	-4.28	-8.71	0.68	-0.42	5.02	8.75	-0.23
60.0	65.0	9.75	-0.36	0.91	-4.08	-8.12	0.59	-0.59	4.92	8.34	-0.00
65.0	70.0	9.47	-0.28	0.61	-3.83	-8.09	0.43	-0.34	4.27	8.42	-0.27
70.0	75.0	10.04	0.08	0.38	-3.58	-9.27	0.11	-0.45	3.80	8.75	-0.28
75.0	80.0	11.26	0.73	0.76	-3.11	-10.47	0.55	-0.26	3.64	10.59	-0.21

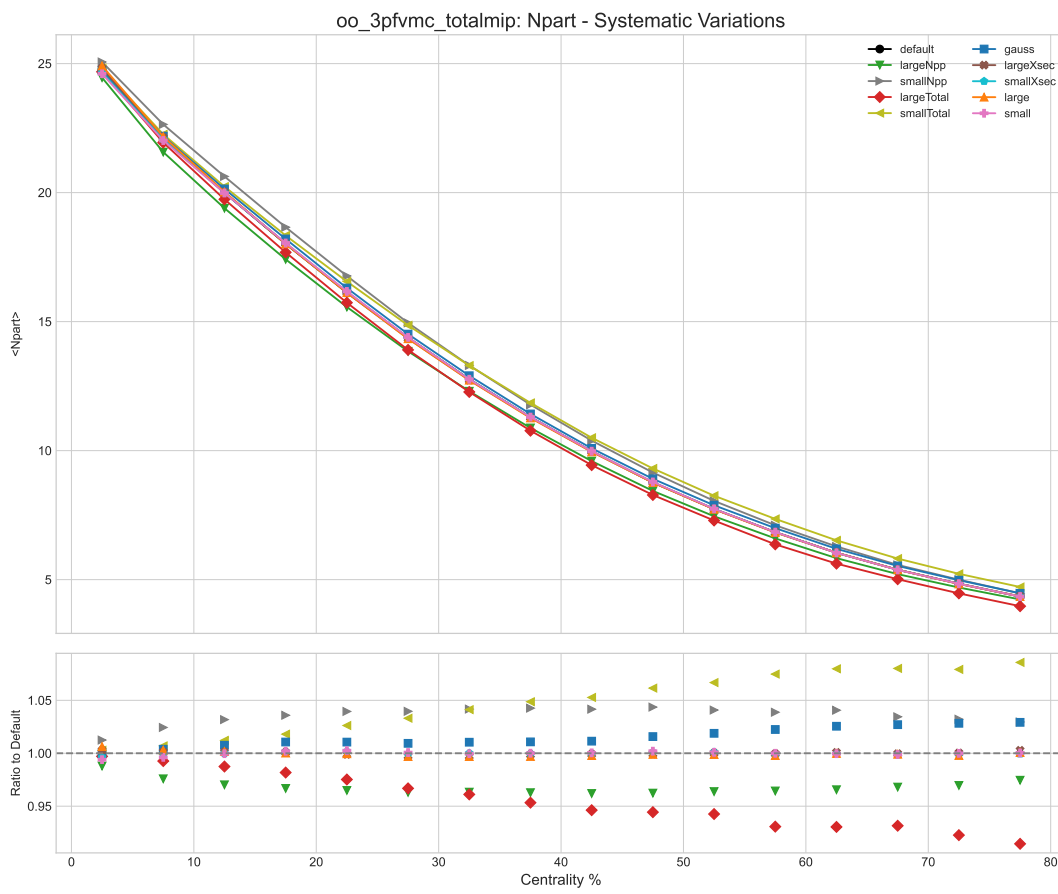
Integer Multiplicity Centrality Cuts Used			
Cent_Interval	Default	Large_Tot	Small_Tot
0-5%	371	368	374
5-10%	320	316	325
10-15%	282	277	287
15-20%	250	244	256
20-25%	221	214	228
25-30%	195	187	202
30-35%	171	164	179
35-40%	151	143	159
40-45%	132	124	140
45-50%	116	108	124
50-55%	101	93	109
55-60%	88	80	95
60-65%	76	69	83
65-70%	65	58	72
70-75%	55	47	62
75-80%	45	37	53

O+O 3PFVMC TotalMIP - Npart

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

min_c	max_c	value	stat_err	syst_err	tot_err	gauss	large	largeNpp	largeTotal	largeXsec	small	smallNpp	smallTotal	smallXsec
0.0	5.0	24.7544	0.0061	0.3667	0.3668	24.7453	24.9114	24.4567	24.6821	24.8214	24.6051	25.0633	24.8243	24.6589
5.0	10.0	22.1039	0.0067	0.5764	0.5764	22.1878	22.1711	21.5710	21.9410	22.1311	22.0186	22.6427	22.2580	22.0708
10.0	15.0	19.9912	0.0071	0.7009	0.7009	20.1426	20.0255	19.3959	19.7408	20.0129	19.9838	20.6268	20.2346	19.9936
15.0	20.0	18.0123	0.0072	0.7508	0.7509	18.2035	18.0157	17.4142	17.6842	18.0269	18.0465	18.6587	18.3369	18.0355
20.0	25.0	16.1328	0.0070	0.7850	0.7851	16.3029	16.1253	15.5665	15.7332	16.1158	16.1717	16.7701	16.5558	16.1623
25.0	30.0	14.3809	0.0067	0.7571	0.7571	14.5157	14.3360	13.8493	13.9042	14.3441	14.3902	14.9505	14.8566	14.3668
30.0	35.0	12.7700	0.0065	0.7626	0.7627	12.9033	12.7300	12.3014	12.2736	12.7358	12.7653	13.3030	13.2963	12.7720
35.0	40.0	11.2996	0.0062	0.7421	0.7421	11.4212	11.2650	10.8791	10.7723	11.2837	11.3002	11.7810	11.8498	11.2949
40.0	45.0	9.9735	0.0058	0.6880	0.6880	10.0875	9.9516	9.5945	9.4374	9.9651	9.9772	10.3887	10.4993	9.9670
45.0	50.0	8.7683	0.0054	0.6757	0.6757	8.9065	8.7588	8.4384	8.2805	8.7627	8.7855	9.1507	9.3077	8.7767
50.0	55.0	7.7354	0.0051	0.6228	0.6228	7.8811	7.7260	7.4555	7.2914	7.7367	7.7424	8.0511	8.2519	7.7439
55.0	60.0	6.8414	0.0049	0.5966	0.5966	6.9950	6.8265	6.5978	6.3674	6.8378	6.8413	7.1064	7.3530	6.8337
60.0	65.0	6.0403	0.0043	0.5624	0.5624	6.1945	6.0391	5.8323	5.6200	6.0450	6.0380	6.2858	6.5222	6.0366
65.0	70.0	5.3848	0.0040	0.4918	0.4918	5.5302	5.3796	5.2128	5.0161	5.3824	5.3784	5.5703	5.8162	5.3787
70.0	75.0	4.8419	0.0037	0.4361	0.4362	4.9789	4.8316	4.6948	4.4675	4.8431	4.8415	4.9977	5.2253	4.8367
75.0	80.0	4.3422	0.0032	0.4142	0.4142	4.4690	4.3453	4.2316	3.9707	4.3540	4.3434	4.4705	4.7148	4.3402

Percentage difference from default

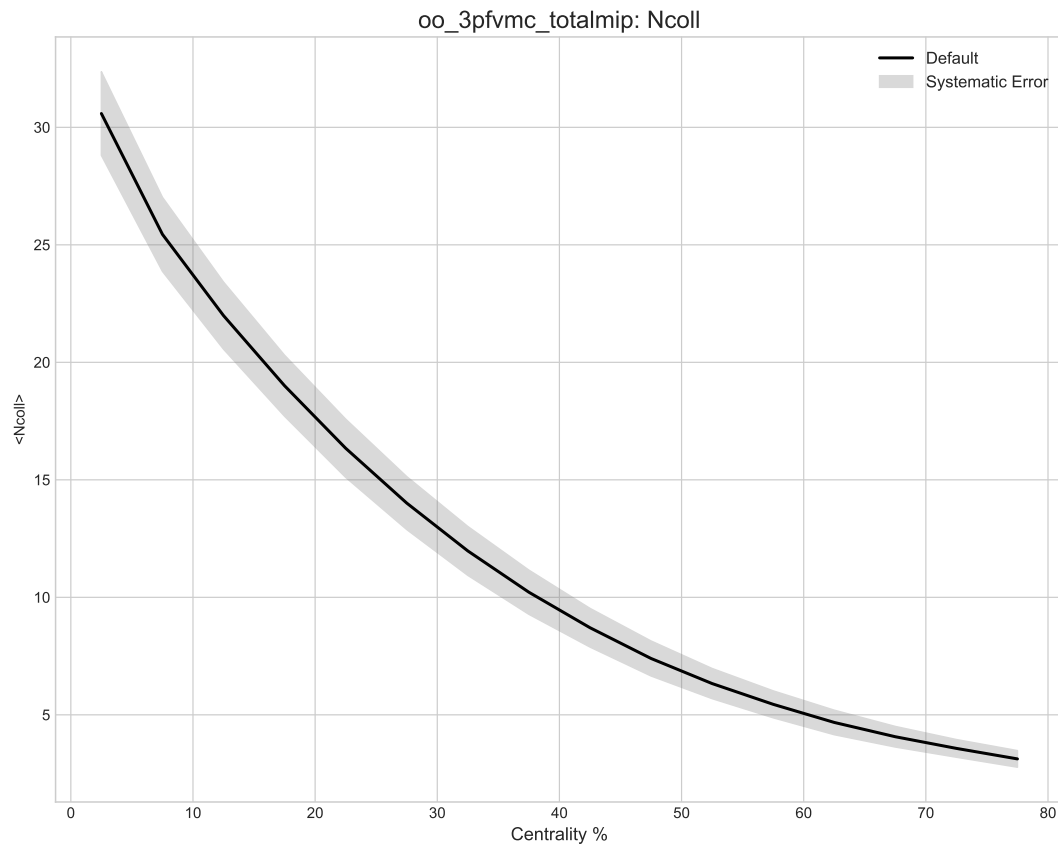
min_c	max_c	tot_sys_err(%)	gauss(%)	large(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	small(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)
0.0	5.0	1.48	-0.04	0.63	-1.20	-0.29	0.27	-0.60	1.25	0.28	-0.39
5.0	10.0	2.61	0.38	0.30	-2.41	-0.74	0.12	-0.39	2.44	0.70	-0.15
10.0	15.0	3.51	0.76	0.17	-2.98	-1.25	0.11	-0.04	3.18	1.22	0.01
15.0	20.0	4.17	1.06	0.02	-3.32	-1.82	0.08	0.19	3.59	1.80	0.13
20.0	25.0	4.87	1.05	-0.05	-3.51	-2.48	-0.11	0.24	3.95	2.62	0.18
25.0	30.0	5.26	0.94	-0.31	-3.70	-3.31	-0.26	0.07	3.96	3.31	-0.10
30.0	35.0	5.97	1.04	-0.31	-3.67	-3.89	-0.27	-0.04	4.17	4.12	0.02
35.0	40.0	6.57	1.08	-0.31	-3.72	-4.67	-0.14	0.01	4.26	4.87	-0.04
40.0	45.0	6.90	1.14	-0.22	-3.80	-5.38	-0.08	0.04	4.16	5.27	-0.06
45.0	50.0	7.71	1.58	-0.11	-3.76	-5.56	-0.06	0.20	4.36	6.15	0.10
50.0	55.0	8.05	1.88	-0.12	-3.62	-5.74	0.02	0.09	4.08	6.68	0.11
55.0	60.0	8.72	2.25	-0.22	-3.56	-6.93	-0.05	-0.00	3.87	7.48	-0.11
60.0	65.0	9.31	2.55	-0.02	-3.44	-6.96	0.08	-0.04	4.06	7.98	-0.06
65.0	70.0	9.13	2.70	-0.10	-3.19	-6.85	-0.04	-0.12	3.45	8.01	-0.11
70.0	75.0	9.01	2.83	-0.21	-3.04	-7.73	0.03	-0.01	3.22	7.92	-0.11
75.0	80.0	9.54	2.92	0.07	-2.55	-8.55	0.27	0.03	2.96	8.58	-0.04

Integer Multiplicity Centrality Cuts Used

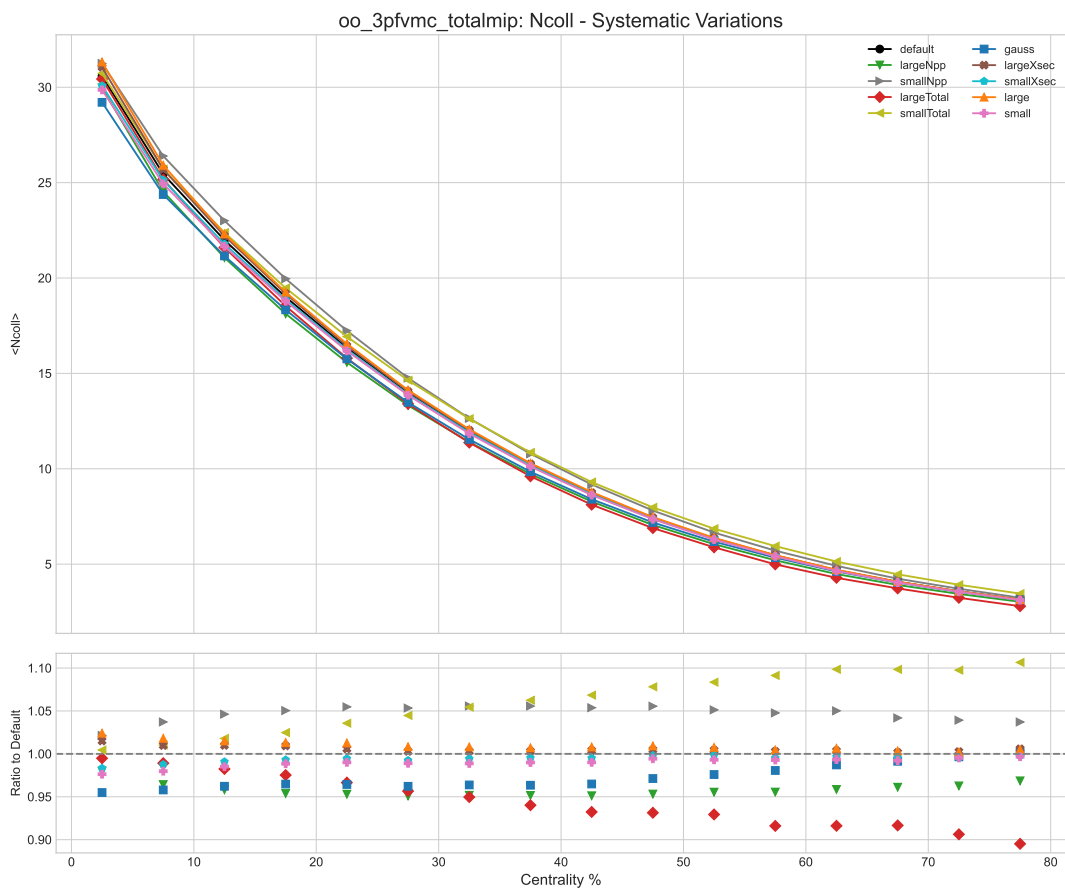
Cent_Interval	Default	Large_Tot	Small_Tot
0-5%	372	369	375
5-10%	322	318	326
10-15%	284	279	289
15-20%	252	246	258
20-25%	223	216	230
25-30%	197	190	205
30-35%	174	166	182
35-40%	153	145	161
40-45%	134	126	142
45-50%	117	110	125
50-55%	102	95	110
55-60%	89	81	97
60-65%	76	69	84
65-70%	65	58	73
70-75%	55	47	62
75-80%	45	37	53

O+O 3PFVMC TotalMIP - Ncoll

Final Result with Total Error



Systematic Variations and Ratios



Summary Data Table

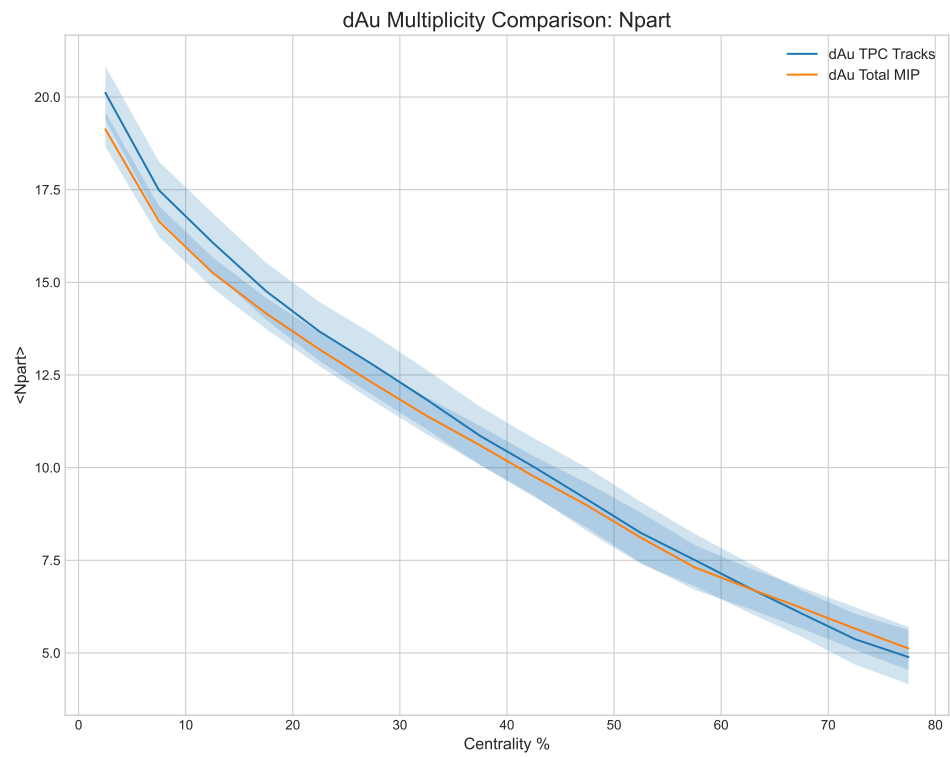
min_c	max_c	value	stat_err	syst_err	tot_err	gauss	large	largeNpp	largeTotal	largeXsec	small	smallNpp	smallTotal	smallXsec
0.0	5.0	30.5885	0.0142	1.7784	1.7784	29.2098	31.3256	29.9734	30.4319	31.0496	29.8642	31.2460	30.7269	30.0769
5.0	10.0	25.4472	0.0128	1.5744	1.5745	24.3777	25.9084	24.5368	25.1712	25.6957	24.9336	26.3933	25.7072	25.1313
10.0	15.0	21.9890	0.0122	1.4266	1.4267	21.1586	22.3203	21.0692	21.6020	22.2062	21.6614	23.0055	22.3833	21.7871
15.0	20.0	19.0034	0.0114	1.2972	1.2972	18.3342	19.2544	18.1264	18.5334	19.1791	18.7852	19.9618	19.4748	18.8710
20.0	25.0	16.3446	0.0104	1.2384	1.2385	15.7618	16.5521	15.5770	15.7975	16.4396	16.1833	17.2397	16.9284	16.2551
25.0	30.0	14.0108	0.0095	1.1262	1.1262	13.4811	14.1272	13.3234	13.4033	14.0510	13.8569	14.7586	14.6380	13.9041
30.0	35.0	11.9750	0.0087	1.0400	1.0400	11.5414	12.0719	11.3973	11.3719	12.0112	11.8419	12.6431	12.6268	11.9071
35.0	40.0	10.2174	0.0078	0.9401	0.9401	9.8438	10.2898	9.7226	9.6060	10.2478	10.1147	10.7862	10.8560	10.1707
40.0	45.0	8.7080	0.0069	0.8228	0.8229	8.4024	8.7777	8.2833	8.1187	8.7419	8.6221	9.1764	9.3039	8.6649
45.0	50.0	7.3985	0.0062	0.7444	0.7445	7.1859	7.4668	7.0521	6.8905	7.4330	7.3565	7.8095	7.9765	7.3844
50.0	55.0	6.3308	0.0055	0.6413	0.6413	6.1784	6.3784	6.0486	5.8839	6.3575	6.2872	6.6555	6.8596	6.3231
55.0	60.0	5.4473	0.0050	0.5733	0.5733	5.3418	5.4724	5.2048	4.9900	5.4626	5.4083	5.7073	5.9450	5.4216
60.0	65.0	4.6764	0.0043	0.5217	0.5217	4.6161	4.7066	4.4825	4.2843	4.6927	4.6455	4.9110	5.1370	4.6602
65.0	70.0	4.0678	0.0039	0.4381	0.4381	4.0326	4.0820	3.9097	3.7285	4.0741	4.0358	4.2382	4.4681	4.0493
70.0	75.0	3.5719	0.0034	0.3764	0.3764	3.5595	3.5793	3.4387	3.2371	3.5810	3.5551	3.7120	3.9203	3.5578
75.0	80.0	3.1242	0.0029	0.3534	0.3535	3.1317	3.1401	3.0260	2.7967	3.1424	3.1139	3.2402	3.4571	3.1188

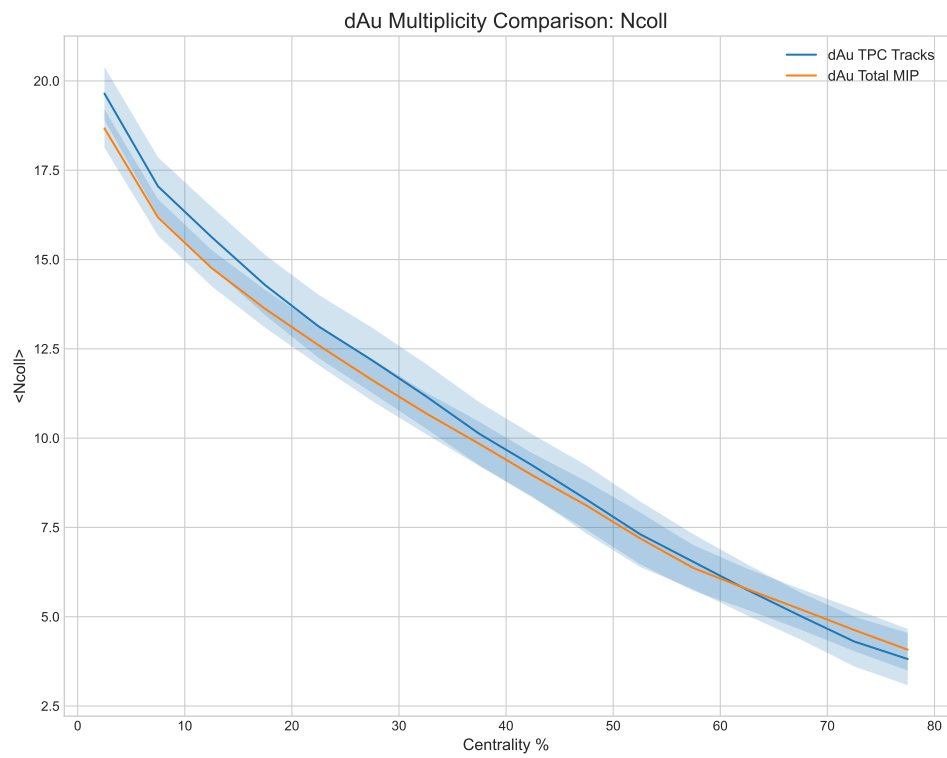
Percentage difference from default											
min_c	max_c	tot_sys_err(%)	gauss(%)	large(%)	largeNpp(%)	largeTotal(%)	largeXsec(%)	small(%)	smallNpp(%)	smallTotal(%)	smallXsec(%)
0.0	5.0	5.81	-4.51	2.41	-2.01	-0.51	1.51	-2.37	2.15	0.45	-1.67
5.0	10.0	6.19	-4.20	1.81	-3.58	-1.08	0.98	-2.02	3.72	1.02	-1.24
10.0	15.0	6.49	-3.78	1.51	-4.18	-1.76	0.99	-1.49	4.62	1.79	-0.92
15.0	20.0	6.83	-3.52	1.32	-4.61	-2.47	0.92	-1.15	5.04	2.48	-0.70
20.0	25.0	7.58	-3.57	1.27	-4.70	-3.35	0.58	-0.99	5.48	3.57	-0.55
25.0	30.0	8.04	-3.78	0.83	-4.91	-4.34	0.29	-1.10	5.34	4.48	-0.76
30.0	35.0	8.68	-3.62	0.81	-4.82	-5.04	0.30	-1.11	5.58	5.44	-0.57
35.0	40.0	9.20	-3.66	0.71	-4.84	-5.98	0.30	-1.00	5.57	6.25	-0.46
40.0	45.0	9.45	-3.51	0.80	-4.88	-6.77	0.39	-0.99	5.38	6.84	-0.49
45.0	50.0	10.06	-2.87	0.92	-4.68	-6.87	0.47	-0.57	5.56	7.81	-0.19
50.0	55.0	10.13	-2.41	0.75	-4.46	-7.06	0.42	-0.69	5.13	8.35	-0.12
55.0	60.0	10.52	-1.94	0.46	-4.45	-8.40	0.28	-0.71	4.77	9.14	-0.47
60.0	65.0	11.16	-1.29	0.65	-4.15	-8.38	0.35	-0.66	5.02	9.85	-0.35
65.0	70.0	10.77	-0.87	0.35	-3.89	-8.34	0.15	-0.78	4.19	9.84	-0.45
70.0	75.0	10.54	-0.35	0.21	-3.73	-9.37	0.26	-0.47	3.92	9.75	-0.39
75.0	80.0	11.31	0.24	0.51	-3.14	-10.48	0.58	-0.33	3.71	10.66	-0.17

Integer Cent_Interval	Multiplicity Default	Centrality Large_Tot	Cuts Used Small_Tot
0-5%	372	369	375
5-10%	322	318	326
10-15%	284	279	289
15-20%	252	246	258
20-25%	223	216	230
25-30%	197	190	205
30-35%	174	166	182
35-40%	153	145	161
40-45%	134	126	142
45-50%	117	110	125
50-55%	102	95	110
55-60%	89	81	97
60-65%	76	69	84
65-70%	65	58	73
70-75%	55	47	62
75-80%	45	37	53

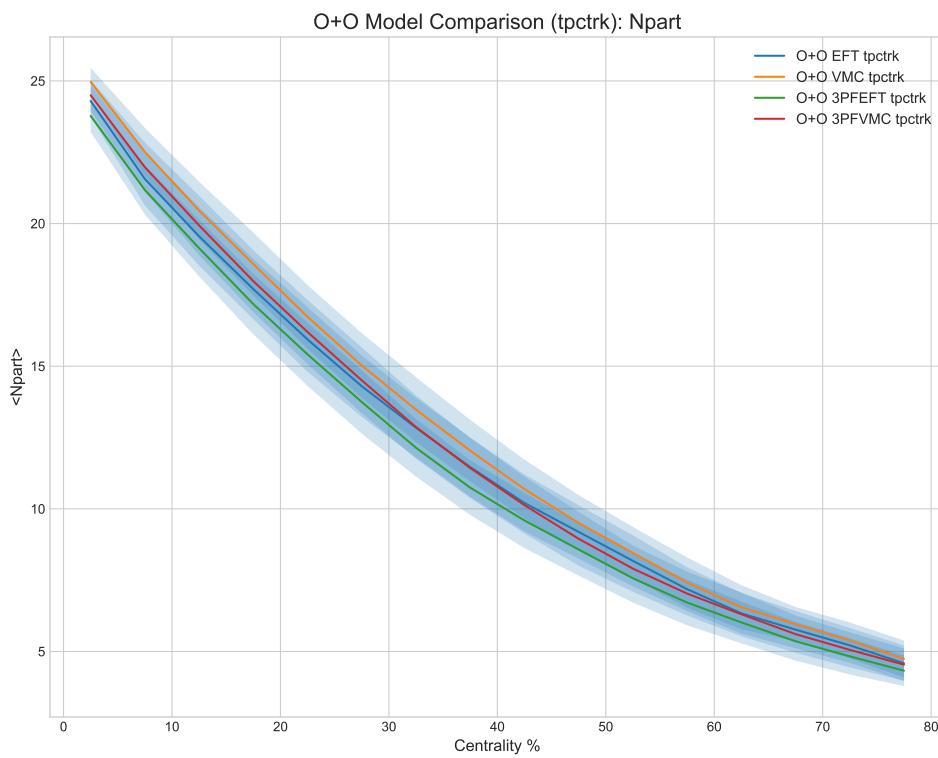
Comparison Plots

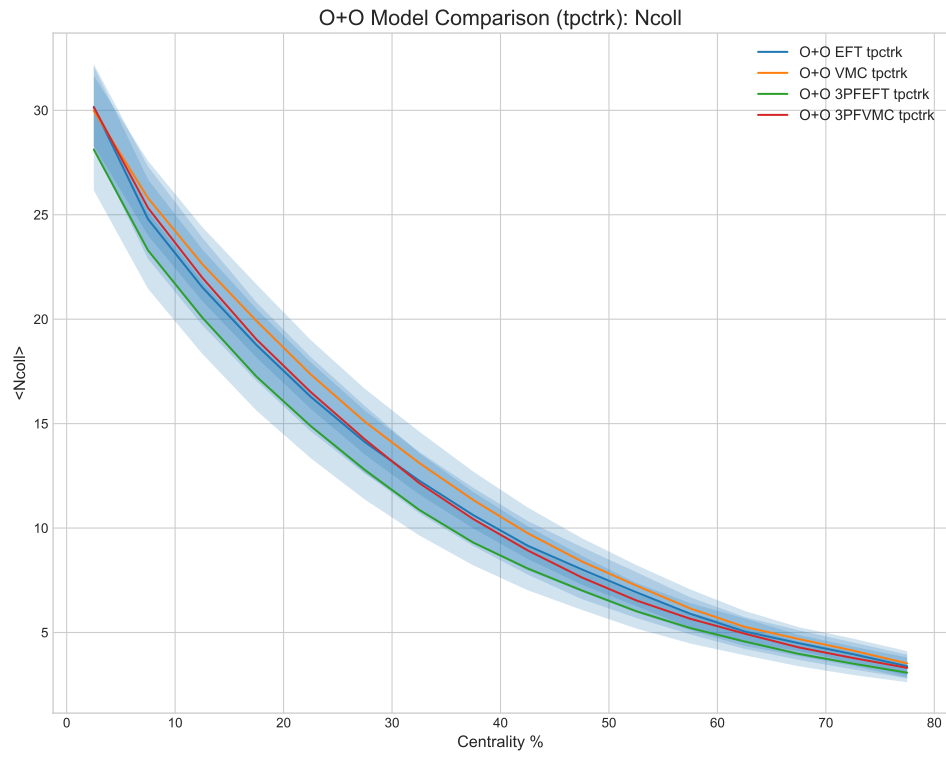
d+Au Multiplicity Comparison

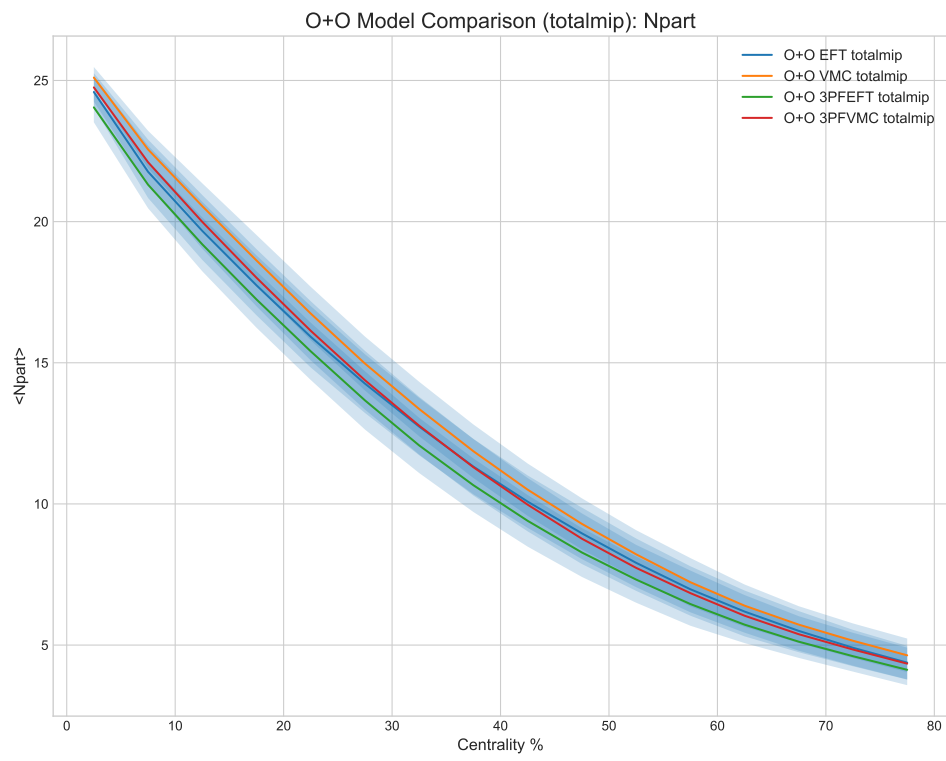


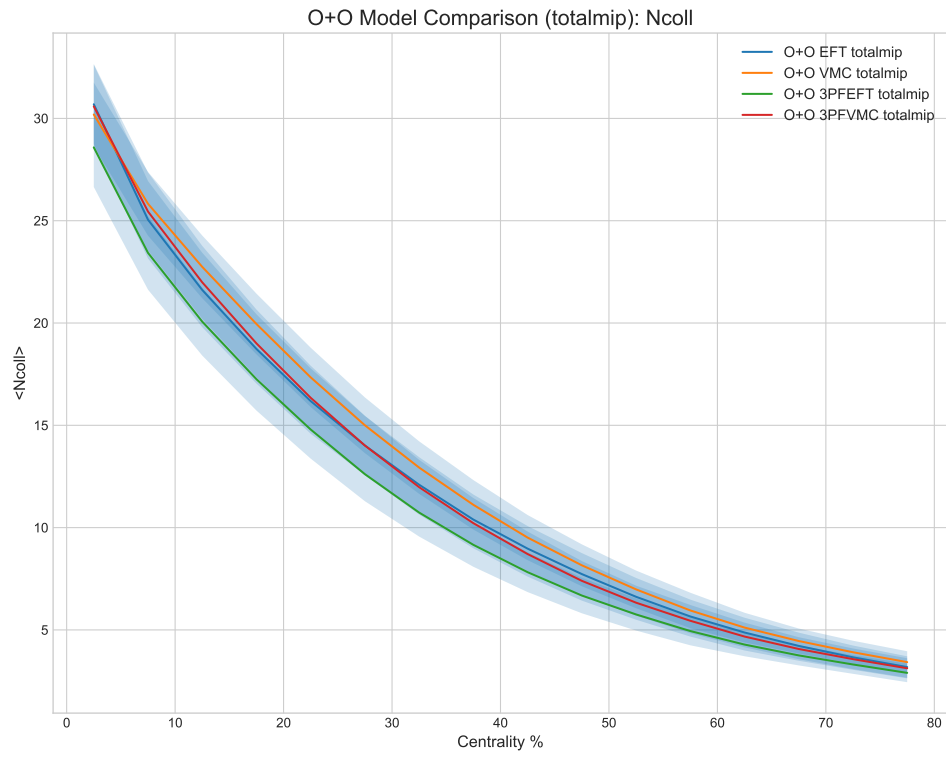


O+O Model Comparison









O+O Multiplicity Comparison

